

TỐI ƯU HÓA DANH MỤC ĐẦU TƯ CỔ PHIẾU NGÂN HÀNG VIỆT NAM

Ứng dụng Mô hình Black-Litterman kết hợp Phân cụm K-means và Tín hiệu Idiosyncratic Momentum

Black-Litterman Portfolio Optimization with K-means Clustering and Idiosyncratic Momentum Signals: Evidence from Vietnamese Banking Stocks

Dữ liệu	25 cổ phiếu ngân hàng niêm yết (HOSE/HNX), 06/2014 – 05/2026, tần suất tháng
Phương pháp	Rolling window 36 tháng · K-means (k=4) · Black-Litterman · SLSQP
Tín hiệu BL	Idiosyncratic momentum 6-1 tháng + Low-volatility composite (Blitz & van Vliet 2007)
Kết quả chính	BL_KIO Sharpe = 0.9396 BL_KIO Return = 30.1% BL_KIO MDD = -30.7%
BL gốc (no-view)	BL Sharpe = 0.7967 BL Return = 27.2% BL MDD = -36.7%
Benchmark 1/N	EW Sharpe = 0.8373 EW Return = 26.6% EW MDD = -36.6%
Giai đoạn OOS	106 tháng (08/2017 – 05/2026) · RF = VN10Y/12 (time-varying)
Kiểm định	Jobson-Korkie (1981) · Paired t-test · Jarque-Bera · Ljung-Box · Monte Carlo 2,000 sims
Tham chiếu	Black & Litterman (1992) · Jegadeesh & Titman (1993) · DeMiguel et al. (2009) · Baker et al. (2011)

1. Tóm tắt Hiệu suất – 7 Danh mục (OOS 106 tháng, annualised)

Nguồn dữ liệu: output/performance_summary_v2.csv

So sánh toàn diện 7 danh mục đầu tư trên 7 chỉ số. MKT = Market-cap weighted; TAN = Tangency (max-Sharpe lịch sử); MV = Min-Variance; BL = Black-Litterman gốc (không có views – chỉ dùng equilibrium Π , $\mu_{BL}=\Pi$); BL_KIO = BL + K-means Idiosyncratic + low-vol signal (mô hình đề xuất); EW = Equal-Weight 1/N (benchmark DeMiguel et al. 2009); RP = Risk-Parity (inverse-vol). Lãi suất phi rủi ro = VN10Y/12 (time-varying).

Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
MKT	+23.40%	+27.05%	0.7289	1.1842	-32.51%	0.7199	61.3%
TAN	+19.20%	+27.81%	0.5577	0.8727	-51.54%	0.3725	62.3%
MV	+16.24%	+25.40%	0.4942	0.8013	-38.44%	0.4224	57.5%
BL	+27.23%	+29.55%	0.7967	1.3132	-36.73%	0.7413	61.3%
BL_KIO	+30.12%	+28.13%	0.9396	1.6281	-30.66%	0.9824	61.3%
EW	+26.62%	+27.38%	0.8373	1.4530	-36.55%	0.7283	62.3%
RP	+25.67%	+27.14%	0.8097	1.3900	-35.72%	0.7186	62.3%

1b. Kiểm định t-test theo cặp (BL_KIO vs 6 danh mục còn lại)

Nguồn dữ liệu: output/ttest_results_v2.csv | output/oos_returns_v2.csv

H_0 : $E[r_{BL_KIO}] = E[r_{other}]$ (paired t-test trên lợi nhuận tháng OOS). $t > 0$: BL_KIO có lợi nhuận trung bình cao hơn. $p < 0.05$: có ý nghĩa thống kê 5%.

Pair	t_statistic	p_value	is_significant_5pct
BL_KIO vs MKT	1.1420	0.2561	False
BL_KIO vs TAN	2.0206	0.0459	True
BL_KIO vs MV	2.7305	0.007418	True
BL_KIO vs BL	0.4791	0.6329	False
BL_KIO vs EW	0.7742	0.4405	False
BL_KIO vs RP	0.9924	0.3233	False

1c. Kiểm định Jobson-Korkie (1981) – So sánh Sharpe Ratio

Nguồn dữ liệu: output/jobson_korkie_results.csv | output/oos_returns_v2.csv

H_0 : $SR_{BL_KIO} = SR_{other}$ (two-sided). $z > 0$: BL_KIO có Sharpe cao hơn. $p < 0.05$: sự khác biệt Sharpe có ý nghĩa thống kê. Lưu ý: BL_KIO vs MV có $p = 0.020$ (significant ★).

Pair	z_statistic	p_value	is_significant_5pct
BL_KIO vs MKT	0.9652	0.3345	No
BL_KIO vs TAN	1.9395	0.0524	No

Pair	z_statistic	p_value	is_significant_5pct
BL_KIO vs MV	2.3255	0.0200	★ Yes
BL_KIO vs BL	0.6977	0.4854	No
BL_KIO vs EW	0.6068	0.5440	No
BL_KIO vs RP	0.7750	0.4383	No

2. Phân tích Tính vững theo Giai đoạn (21 dòng = 7 portfolio × 3 giai đoạn)

Nguồn dữ liệu: output/subperiod_analysis.csv

Giai đoạn OOS chia làm 2 nửa bằng nhau (mỗi nửa ~53 tháng). Period 1 (08/2017–12/2021): thị trường bull, tăng trưởng mạnh. Period 2 (01/2022–05/2026): biến động cao, lãi suất tăng, hậu COVID. BL_KIO thống trị Period 2 (Sharpe 0.861 vs BL gốc 0.518, EW 0.384, TAN 0.043, MV –0.097), chứng minh K-means signal thêm giá trị thực sự (+0.343 Sharpe) so với BL gốc.

Toàn kỳ (08/2017–05/2026, N=106)

Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar
MKT	+23.40%	+27.05%	0.7289	1.1842	-32.51%	0.7199
TAN	+19.20%	+27.81%	0.5577	0.8727	-51.54%	0.3725
MV	+16.24%	+25.40%	0.4942	0.8013	-38.44%	0.4224
BL	+27.23%	+29.55%	0.7967	1.3132	-36.73%	0.7413
BL_KIO	+30.12%	+28.13%	0.9396	1.6281	-30.66%	0.9824
EW	+26.62%	+27.38%	0.8373	1.4530	-36.55%	0.7283
RP	+25.67%	+27.14%	0.8097	1.3900	-35.72%	0.7186

Period 1 – Bull market (2017-08-31 → 2021-12-31, N=53)

Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar
MKT	+34.94%	+31.19%	0.9959	1.6137	-31.07%	1.1244
TAN	+35.79%	+31.69%	1.0072	1.6457	-38.05%	0.9407
MV	+32.38%	+31.69%	0.8997	1.5005	-38.44%	0.8424
BL	+39.44%	+34.15%	1.0417	1.7697	-29.29%	1.3468
BL_KIO	+41.47%	+35.15%	1.0696	1.9065	-30.66%	1.3525
EW	+43.31%	+32.06%	1.2301	2.2541	-30.36%	1.4267
RP	+41.98%	+31.93%	1.1937	2.1417	-32.14%	1.3063

Period 2 – High volatility (2022-01-31 → 2026-05-31, N=53)

Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar
MKT	+12.78%	+22.15%	0.4190	0.6843	-32.51%	0.3932
TAN	+4.49%	+22.96%	0.0428	0.0614	-51.54%	0.0870
MV	+1.92%	+16.39%	-0.0965	-0.1390	-31.67%	0.0606
BL	+16.00%	+24.13%	0.5181	0.8090	-36.73%	0.4356

Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar
BL_KIO	+19.61%	+18.70%	0.8612	1.3701	-22.22%	0.8827
EW	+11.72%	+21.42%	0.3837	0.5933	-36.55%	0.3207
RP	+11.08%	+21.02%	0.3606	0.5599	-35.72%	0.3103

3. Phân tích Độ nhạy Tham số (One-at-a-time) – 7 Danh mục

Nguồn dữ liệu: output/sensitivity_results.csv

Thay đổi một tham số tại một thời điểm, giữ nguyên các tham số còn lại (baseline: lookback=36, max_weight=30%, k=4, rf=VN10Y dynamic). Mỗi bảng hiển thị Sharpe của tất cả 7 danh mục để so sánh trực tiếp. ★ = baseline value.

Tham số: lookback

Tham số	Baseline	MKT	TAN	MV	BL	BL_KIO	EW	RP
24		0.7393	0.7036	0.7036	0.6878	0.7029	0.8783	0.8793
36	★	0.7289	0.5577	0.4942	0.7967	0.9396	0.8373	0.8097
48		0.6367	0.7140	0.4672	0.6491	0.8854	0.8183	0.7804

Tham số: max_weight

Tham số	Baseline	MKT	TAN	MV	BL	BL_KIO	EW	RP
20%		0.7289	0.6501	0.5477	0.8388	0.8704	0.8373	0.8107
30%	★	0.7289	0.5577	0.4942	0.7967	0.9396	0.8373	0.8097
40%		0.7289	0.5113	0.4587	0.7928	0.9015	0.8373	0.8097

Tham số: k_clusters

Tham số	Baseline	MKT	TAN	MV	BL	BL_KIO	EW	RP
2		0.7289	0.5577	0.4942	0.7967	0.7237	0.8373	0.8097
3		0.7289	0.5577	0.4942	0.7967	0.8338	0.8373	0.8097
4	★	0.7289	0.5577	0.4942	0.7967	0.9396	0.8373	0.8097
5		0.7289	0.5577	0.4942	0.7967	0.9395	0.8373	0.8097
6		0.6263	0.4819	0.4124	0.7260	1.0400	0.8947	0.8554

Tham số: rf_rate

Tham số	Baseline	MKT	TAN	MV	BL	BL_KIO	EW	RP
4% fixed		0.7173	0.5483	0.4819	0.7713	0.9368	0.8259	0.7982
5% fixed		0.6804	0.5143	0.4426	0.7352	0.9084	0.7894	0.7613
6% fixed		0.6434	0.4850	0.4032	0.7173	0.8719	0.7529	0.7245
VN10Y (dynamic)	★	0.7289	0.5577	0.4942	0.7967	0.9396	0.8373	0.8097

4. Kiểm định Số cụm K tối ưu – Elbow Method + Silhouette Score

Nguồn dữ liệu: output/kmeans_k_summary.csv | output/kmeans_k_validation.csv

Elbow method: inertia (WCSS) giảm – tìm điểm 'gãy'. Silhouette score (Rousseeuw 1987): giá trị cao = cụm tách biệt tốt. k=4 được chọn dựa trên hiệu suất OOS tốt nhất (Sharpe 0.9396) và silhouette chấp nhận được (0.389, chênh lệch nhỏ so với k=2: 0.420). k=5 cho kết quả gần giống (0.9395) xác nhận k=4 không phải điểm đơn lẻ.

k	Mean_Inertia	Mean_Silhouette	Std_Silhouette
2	19.4828	0.4201	0.0892
3	11.0918	0.4057	0.0531
4	7.3568	0.3891	0.0480
5	5.0307	0.3825	0.0602
6	3.5828	0.3716	0.0666

4b. BL_KIO Sharpe theo từng giá trị k (cùng composite signal)

Nguồn dữ liệu: output/sensitivity_results.csv

Sensitivity của k trong PARAM_GRID (k=2...6, baseline k=4). Chú ý: k=6 có mẫu khác (yêu cầu ≥8 cổ phiếu active) nên EW Sharpe khác.

k	Baseline	MKT	TAN	MV	BL	BL_KIO	EW	RP
2		0.7289	0.5577	0.4942	0.7967	0.7237	0.8373	0.8097
3		0.7289	0.5577	0.4942	0.7967	0.8338	0.8373	0.8097
4	★	0.7289	0.5577	0.4942	0.7967	0.9396	0.8373	0.8097
5		0.7289	0.5577	0.4942	0.7967	0.9395	0.8373	0.8097
6		0.6263	0.4819	0.4124	0.7260	1.0400	0.8947	0.8554

5. Chất lượng Tín hiệu – Information Coefficient (IC)

Nguồn dữ liệu: output/signal_ic_analysis.csv

IC = Spearman rank correlation giữa tín hiệu idiosyncratic momentum và lợi nhuận thực tế tháng sau. IC > 0: dự báo đúng chiều. Hit rate: tỷ lệ bước mà cluster Long thực sự vượt cluster Short. Mean IC = +0.0667 (tín hiệu dự báo dương, ổn định qua 106 tháng).

Chỉ số IC	Giá trị
Mean IC (Spearman)	+0.0667
IC > 0 – % bước đúng chiều	50.0%
Hit Rate (Long > Short)	50.9%
Std IC	0.3618
t-stat IC ≠ 0	+1.898
N bước OOS	106

IC theo từng bước rebalancing (106 tháng):

Date	IC	Hit	N_stocks
2017-08-31	-0.3571	1	7
2017-09-30	-0.0714	0	7
2017-10-31	-0.1071	0	7
2017-11-30	-0.3214	0	7
2017-12-31	-0.4643	0	7
2018-01-31	0.0714	0	7
2018-02-28	0.5357	1	7
2018-03-31	0.1786	1	7
2018-04-30	0.000000	1	7
2018-05-31	-0.3571	1	7
2018-06-30	-0.2500	0	7
2018-07-31	-0.1786	0	7
2018-08-31	0.3929	1	7
2018-09-30	-0.8214	0	7
2018-10-31	0.3571	1	7
2018-11-30	0.7500	1	7
2018-12-31	0.5000	1	7

Date	IC	Hit	N_stocks
2019-01-31	0.6071	1	7
2019-02-28	0.1429	1	7
2019-03-31	0.000000	1	7
2019-04-30	-0.0180	0	7
2019-05-31	0.5357	1	7
2019-06-30	0.8929	1	7
2019-07-31	0.6429	1	7
2019-08-31	0.3214	0	7
2019-09-30	0.8214	1	7
2019-10-31	0.6786	1	7
2019-11-30	0.8571	1	7
2019-12-31	-0.0357	0	7
2020-01-31	-0.6429	0	7
2020-02-29	-0.9643	0	7
2020-03-31	0.2143	0	8
2020-04-30	-0.0714	0	8
2020-05-31	-0.1429	0	8
2020-06-30	-0.7381	0	8
2020-07-31	-0.0952	0	8
2020-08-31	-0.1500	0	9
2020-09-30	0.4667	1	9
2020-10-31	0.4061	0	10
2020-11-30	-0.2606	0	10
2020-12-31	-0.0727	0	11
2021-01-31	0.7636	1	11
2021-02-28	0.4825	1	12
2021-03-31	0.2527	1	13
2021-04-30	0.6154	1	13
2021-05-31	0.4890	1	13
2021-06-30	-0.3582	0	14

Date	IC	Hit	N_stocks
2021-07-31	0.0374	1	14
2021-08-31	-0.1429	0	15
2021-09-30	-0.3429	0	15
2021-10-31	-0.1357	0	15
2021-11-30	0.0500	0	15
2021-12-31	-0.1500	1	15
2022-01-31	-0.0107	0	15
2022-02-28	0.4286	0	15
2022-03-31	0.2107	0	15
2022-04-30	-0.1821	0	15
2022-05-31	-0.2036	1	15
2022-06-30	-0.2214	0	15
2022-07-31	-0.1500	1	15
2022-08-31	0.4286	1	15
2022-09-30	0.5647	1	16
2022-10-31	0.2294	1	16
2022-11-30	-0.0794	1	16
2022-12-31	0.2588	1	16
2023-01-31	-0.2088	0	16
2023-02-28	0.0853	0	16
2023-03-31	0.2824	1	16
2023-04-30	-0.5735	0	16
2023-05-31	-0.0441	1	16
2023-06-30	-0.2824	0	16
2023-07-31	0.2500	1	16
2023-08-31	0.0235	0	16
2023-09-30	-0.2475	1	17
2023-10-31	-0.0588	0	17
2023-11-30	0.0613	0	17
2023-12-31	-0.0917	1	19

Date	IC	Hit	N_stocks
2024-01-31	0.2772	1	19
2024-02-29	0.1135	0	22
2024-03-31	0.2506	1	23
2024-04-30	-0.1709	1	23
2024-05-31	-0.0304	0	24
2024-06-30	-0.0496	1	24
2024-07-31	0.3748	1	24
2024-08-31	0.3209	0	24
2024-09-30	-0.008462	0	25
2024-10-31	0.1662	1	25
2024-11-30	0.1354	0	25
2024-12-31	0.3738	1	25
2025-01-31	-0.0304	1	25
2025-02-28	-0.2277	0	25
2025-03-31	-0.3958	0	25
2025-04-30	-0.0262	1	25
2025-05-31	0.2585	0	25
2025-06-30	0.1608	1	25
2025-07-31	-0.2592	0	25
2025-08-31	-0.2308	0	25
2025-09-30	0.3908	1	25
2025-10-31	0.2146	1	25
2025-11-30	0.0565	1	25
2025-12-31	-0.0569	1	25
2026-01-31	-0.4147	0	25
2026-02-28	0.4489	1	25
2026-03-31	-0.0392	0	25
2026-04-30	0.1369	1	25
2026-05-31	0.0492	0	25

6. Phân tích Drawdown Chi tiết – 7 Danh mục

Nguồn dữ liệu: output/drawdown_summary.csv | output/drawdown_periods.csv

6a. Tóm tắt Drawdown

BL_KIO có Max_DD thấp nhất trong các danh mục tối ưu hóa (-30.7%). Avg_DD_Depth = -9.48% (nhỏ hơn nhiều so với TAN -24.9% và EW -15.4%). Avg_Recovery_Months BL_KIO = 3.8 (nhanh hơn BL gốc, TAN 9.6 và MV 11.8).

Portfolio	N_Drawdowns	Max_DD_pct	Avg_DD_Depth_pct	Avg_Recovery_Months	N_Months_Below_Neg20pct
MKT	9	-32.5100	-15.8300	6.9000	19
TAN	5	-51.5400	-24.9000	9.6000	51
MV	5	-38.4400	-21.3200	11.8000	26
BL	9	-36.7300	-16.2100	6.8000	22
BL_KIO	12	-30.6600	-9.4800	3.8000	21
EW	7	-36.5500	-15.3500	9.0000	34
RP	7	-35.7200	-15.6100	6.5000	34

6b. Từng sự kiện Drawdown

Liệt kê từng đợt drawdown: ngày bắt đầu, ngày đáy, ngày phục hồi, mức sụt giảm tối đa, số tháng phục hồi.

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
MKT	2018-04-30	2018-06-30	2019-12-31 00:00:00	-26.9400	2	18.0000	20
MKT	2020-02-29	2020-03-31	2020-11-30 00:00:00	-31.0700	1	8.0000	9
MKT	2021-01-31	2021-01-31	2021-02-28 00:00:00	-7.8800	0	1.0000	1
MKT	2021-07-31	2021-08-31	2022-01-31 00:00:00	-14.3800	1	5.0000	6
MKT	2022-02-28	2022-10-31	2024-02-29 00:00:00	-32.5100	8	16.0000	24
MKT	2024-04-30	2024-06-30	2024-08-31 00:00:00	-6.9900	2	2.0000	4
MKT	2024-10-31	2024-11-30	2025-01-31 00:00:00	-2.8100	1	2.0000	3
MKT	2025-04-30	2025-04-30	2025-07-31 00:00:00	-7.6900	0	3.0000	3
MKT	2025-09-30	2026-03-31	Not recovered	-12.2300	6	nan	8
TAN	2018-04-30	2020-03-31	2020-12-31 00:00:00	-38.0500	23	9.0000	32
TAN	2021-01-31	2021-01-31	2021-02-28 00:00:00	-8.0800	0	1.0000	1
TAN	2021-07-31	2021-08-31	2021-12-31 00:00:00	-15.6200	1	4.0000	5
TAN	2022-02-28	2022-12-31	2025-08-31 00:00:00	-51.5400	10	32.0000	42

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
TAN	2025-09-30	2026-03-31	2026-05-31 00:00:00	-11.1900	6	2.0000	8
MV	2018-04-30	2020-03-31	2020-12-31 00:00:00	-38.4400	23	9.0000	32
MV	2021-01-31	2021-01-31	2021-03-31 00:00:00	-8.0800	0	2.0000	2
MV	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.0300	2	5.0000	7
MV	2022-02-28	2022-12-31	2025-07-31 00:00:00	-31.6700	10	31.0000	41
MV	2025-09-30	2026-03-31	Not recovered	-15.3700	6	nan	8
BL	2018-04-30	2018-07-31	2020-01-31 00:00:00	-29.2900	3	18.0000	21
BL	2020-02-29	2020-03-31	2020-08-31 00:00:00	-28.3400	1	5.0000	6
BL	2020-10-31	2020-10-31	2020-11-30 00:00:00	-0.5000	0	1.0000	1
BL	2021-01-31	2021-01-31	2021-02-28 00:00:00	-10.1200	0	1.0000	1
BL	2021-06-30	2021-08-31	2022-01-31 00:00:00	-15.7400	2	5.0000	7
BL	2022-02-28	2022-10-31	2024-02-29 00:00:00	-36.7300	8	16.0000	24
BL	2024-04-30	2024-06-30	2025-01-31 00:00:00	-11.2600	2	7.0000	9
BL	2025-04-30	2025-04-30	2025-05-31 00:00:00	-0.6400	0	1.0000	1
BL	2025-09-30	2026-03-31	Not recovered	-13.3000	6	nan	8
BL_KIO	2017-10-31	2017-10-31	2017-11-30 00:00:00	-2.0500	0	1.0000	1
BL_KIO	2017-12-31	2017-12-31	2018-01-31 00:00:00	-1.1400	0	1.0000	1
BL_KIO	2018-04-30	2019-01-31	2020-01-31 00:00:00	-30.4700	9	12.0000	21
BL_KIO	2020-02-29	2020-03-31	2020-09-30 00:00:00	-30.6600	1	6.0000	7
BL_KIO	2021-01-31	2021-01-31	2021-02-28 00:00:00	-3.2400	0	1.0000	1
BL_KIO	2021-06-30	2022-10-31	2024-01-31 00:00:00	-25.4300	16	15.0000	31
BL_KIO	2024-06-30	2024-06-30	2024-07-31 00:00:00	-3.9000	0	1.0000	1
BL_KIO	2024-11-30	2024-11-30	2024-12-31 00:00:00	-0.8500	0	1.0000	1
BL_KIO	2025-03-31	2025-04-30	2025-06-30 00:00:00	-5.9300	1	2.0000	3
BL_KIO	2025-09-30	2025-09-30	2025-10-31 00:00:00	-2.0600	0	1.0000	1
BL_KIO	2025-11-30	2025-11-30	2025-12-31 00:00:00	-1.4700	0	1.0000	1
BL_KIO	2026-01-31	2026-03-31	Not recovered	-6.5600	2	nan	4
EW	2018-04-30	2018-12-31	2020-09-30 00:00:00	-30.3600	8	21.0000	29
EW	2021-01-31	2021-01-31	2021-02-28 00:00:00	-6.2900	0	1.0000	1
EW	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.5800	2	5.0000	7

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
EW	2022-02-28	2022-10-31	2024-09-30 00:00:00	-36.5500	8	23.0000	31
EW	2024-10-31	2024-11-30	2025-02-28 00:00:00	-4.6700	1	3.0000	4
EW	2025-04-30	2025-04-30	2025-05-31 00:00:00	-4.8700	0	1.0000	1
EW	2025-09-30	2026-03-31	Not recovered	-11.1000	6	nan	8
RP	2018-04-30	2020-03-31	2020-09-30 00:00:00	-32.1400	23	6.0000	29
RP	2021-01-31	2021-01-31	2021-02-28 00:00:00	-6.0000	0	1.0000	1
RP	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.7700	2	5.0000	7
RP	2022-02-28	2022-10-31	2024-09-30 00:00:00	-35.7200	8	23.0000	31
RP	2024-10-31	2024-11-30	2025-02-28 00:00:00	-4.5500	1	3.0000	4
RP	2025-04-30	2025-04-30	2025-05-31 00:00:00	-5.3300	0	1.0000	1
RP	2025-09-30	2026-03-31	Not recovered	-11.7400	6	nan	8

7. Kiểm định Phân phối Lợi nhuận – 7 Danh mục

Nguồn dữ liệu: output/distribution_jb_tests.csv | output/distribution_lb_tests.csv

7a. Jarque-Bera (Chuẩn hóa)

H₀: lợi nhuận có phân phối chuẩn. p < 0.05 = bác bỏ H₀ (fat tail / skewness đáng kể). Hầu hết danh mục KHÔNG chuẩn → cần Sortino/MDD thay vì chỉ dùng Sharpe.

Portfolio	JB_stat	JB_pval	Skewness	Excess_Kurtosis	is_normal_5pct
MKT	10.142	0.0063	-0.1755	1.6061	No ✗
TAN	16.282	0.0003	-0.2122	2.0242	No ✗
MV	83.773	0.0000	0.1440	4.6169	No ✗
BL	4.496	0.1056	-0.1157	1.0897	Yes
BL_KIO	16.267	0.0003	-0.0136	2.0714	No ✗
EW	7.280	0.0263	0.1984	1.3413	No ✗
RP	10.779	0.0046	0.1505	1.6675	No ✗

7b. Ljung-Box (Tự tương quan, lag 5/10/20)

H₀: không có tự tương quan (white noise). p < 0.05 = tự tương quan có ý nghĩa (có thể khai thác được). no_autocorr = True: không bác bỏ H₀.

Portfolio	LB5_stat	LB5_pval	LB5_no_autocorr	LB10_stat	LB10_pval	LB10_no_autocorr	LB20_stat	LB20_pval	LB20_no_autocorr
MKT	8.868	0.1145	True	19.718	0.0320	False	34.265	0.0244	False
TAN	4.586	0.4685	True	11.580	0.3141	True	25.547	0.1813	True
MV	2.071	0.8392	True	16.285	0.0918	True	25.796	0.1726	True
BL	6.850	0.2320	True	14.510	0.1510	True	24.659	0.2148	True
BL_KIO	5.981	0.3081	True	12.418	0.2580	True	22.777	0.2999	True
EW	3.048	0.6927	True	12.875	0.2308	True	23.944	0.2448	True
RP	2.675	0.7500	True	13.328	0.2059	True	24.201	0.2337	True

8. Monte Carlo Robustness – Nhiều trong View BL_KIO (N=2,000 sims/bước)

Nguồn dữ liệu: output/monte_carlo_v2.csv

Thêm nhiễu Gaussian vào vector view q ($\delta = -10\% \dots +10\%$) để kiểm tra độ nhạy của Sharpe kỳ vọng BL_KIO. Tính chất tuyến tính $\mu_{BL_KIO}(q') = \text{base_vec} + A_mat @ q'$ cho phép 2,000 mô phỏng mà không cần tối ưu lại. BL_KIO và TAN đều ổn định qua dải nhiễu.

Delta	N_sims	BL_KIO Mean	BL_KIO Std	BL_KIO Min	BL_KIO 5%	BL_KIO 95%	BL_KIO Max	TAN Mean
-0.1000	2000	0.8282	0.0509	0.6457	0.7431	0.9113	0.9925	1.0271
-0.0500	2000	0.8314	0.0253	0.7532	0.7897	0.8728	0.9095	1.0271
0.000000	2000	0.8311	0.005100	0.8087	0.8226	0.8395	0.8486	1.0271
0.0500	2000	0.8307	0.0254	0.7532	0.7903	0.8736	0.9147	1.0271
0.1000	2000	0.8304	0.0519	0.6518	0.7449	0.9143	1.0355	1.0271

9. Chuỗi Lợi nhuận OOS theo Tháng – 7 Danh mục (106 tháng)

Nguồn dữ liệu: output/oos_returns_v2.csv

Lợi nhuận tháng (simple return) của 7 danh mục và lãi suất phi rủi ro VN10Y/12. Chuỗi này là dữ liệu gốc để tính toàn bộ các chỉ số hiệu suất ở các bảng trên.

Date	RF_monthly	MKT	TAN	MV	BL	BL_KIO	EW	RP
2017-08-31	+0.483%	-2.659%	+2.180%	+0.053%	-3.993%	-5.077%	-2.470%	-2.021%
2017-09-30	+0.483%	+0.860%	+1.761%	+1.229%	+0.405%	+0.729%	+1.408%	+1.238%
2017-10-31	+0.483%	+5.992%	+5.025%	+2.923%	+6.017%	-2.052%	+1.733%	+1.888%
2017-11-30	+0.483%	+18.441%	+15.273%	+16.398%	+21.010%	+17.578%	+18.707%	+18.217%
2017-12-31	+0.483%	+4.809%	+5.063%	+3.436%	+3.446%	-1.136%	+1.888%	+2.005%
2018-01-31	+0.382%	+24.375%	+23.053%	+29.404%	+23.073%	+26.798%	+26.710%	+26.684%
2018-02-28	+0.382%	+12.034%	+9.928%	+7.759%	+14.681%	+13.449%	+9.847%	+9.636%
2018-03-31	+0.382%	+1.671%	+1.784%	+1.593%	+2.548%	+4.164%	+2.054%	+2.176%
2018-04-30	+0.382%	-14.673%	-12.626%	-12.380%	-15.163%	-13.806%	-13.185%	-13.183%
2018-05-31	+0.382%	-11.489%	-7.325%	-7.619%	-12.469%	-11.464%	-11.176%	-10.378%
2018-06-30	+0.382%	-3.259%	-5.307%	-6.419%	-4.485%	-6.371%	-6.996%	-6.960%
2018-07-31	+0.382%	+0.745%	+2.262%	+2.295%	-0.302%	-0.547%	+0.637%	+0.934%
2018-08-31	+0.382%	+12.304%	+6.097%	+5.542%	+15.732%	+17.230%	+9.577%	+9.094%
2018-09-30	+0.382%	+1.855%	+0.270%	+0.425%	+3.271%	+1.297%	+4.066%	+3.391%
2018-10-31	+0.382%	-12.180%	-9.598%	-9.537%	-13.296%	-12.646%	-11.574%	-11.164%
2018-11-30	+0.382%	+0.754%	-1.314%	-1.319%	+1.040%	-0.462%	-0.572%	-0.726%
2018-12-31	+0.382%	-1.858%	-4.582%	-4.485%	-3.189%	-5.144%	-3.755%	-4.236%
2019-01-31	+0.382%	+1.298%	+4.809%	+5.657%	+0.128%	-0.094%	+2.109%	+2.844%
2019-02-28	+0.382%	+4.145%	+1.971%	+1.666%	+4.692%	+3.464%	+2.401%	+2.325%
2019-03-31	+0.382%	+8.747%	+6.303%	+5.207%	+8.830%	+7.480%	+5.215%	+5.291%
2019-04-30	+0.382%	-0.991%	-0.234%	-0.157%	-2.247%	-0.796%	-1.423%	-1.266%
2019-05-31	+0.382%	-4.029%	-3.664%	-2.884%	-4.915%	-3.387%	-4.528%	-4.234%
2019-06-30	+0.382%	+1.882%	+1.549%	+1.144%	+0.162%	+0.437%	-0.635%	-0.354%
2019-07-31	+0.382%	+10.357%	+7.936%	+6.167%	+9.225%	+9.916%	+4.577%	+4.904%
2019-08-31	+0.382%	+0.318%	+0.413%	-0.787%	+0.654%	+1.402%	-0.321%	-0.454%
2019-09-30	+0.382%	+4.438%	+5.440%	+5.418%	+4.007%	+5.384%	+4.261%	+4.472%

Date	RF_monthly	MKT	TAN	MV	BL	BL_KIO	EW	RP
2019-10-31	+0.382%	+4.767%	+3.552%	+3.482%	+4.277%	+4.481%	+2.898%	+3.036%
2019-11-30	+0.382%	-1.992%	-2.871%	-3.877%	-2.622%	-2.400%	-4.268%	-4.250%
2019-12-31	+0.382%	+6.745%	+1.179%	-0.157%	+7.144%	+4.588%	+3.297%	+2.689%
2020-01-31	+0.250%	+4.005%	+0.932%	-0.000%	+9.149%	+8.121%	+6.401%	+5.373%
2020-02-29	+0.250%	-4.779%	-4.118%	+2.962%	-0.645%	-2.616%	+8.919%	+7.805%
2020-03-31	+0.250%	-27.611%	-30.087%	-30.149%	-27.870%	-28.797%	-24.025%	-25.622%
2020-04-30	+0.250%	+13.756%	+15.059%	+15.597%	+14.607%	+13.451%	+17.849%	+17.158%
2020-05-31	+0.250%	+16.876%	+8.950%	+8.916%	+16.264%	+6.818%	+10.404%	+11.281%
2020-06-30	+0.250%	-4.693%	-5.545%	-3.566%	-4.131%	-1.521%	-1.896%	-1.753%
2020-07-31	+0.250%	-4.033%	-5.332%	-4.802%	-3.392%	-5.146%	-3.520%	-3.318%
2020-08-31	+0.250%	+12.180%	+11.411%	+12.560%	+13.679%	+14.649%	+14.658%	+14.543%
2020-09-30	+0.250%	+3.401%	+5.437%	+6.103%	+3.436%	+13.094%	+9.758%	+9.666%
2020-10-31	+0.250%	-0.066%	+1.375%	+3.432%	-0.496%	+1.736%	+3.421%	+3.794%
2020-11-30	+0.250%	+11.447%	+9.143%	+11.126%	+11.743%	+10.958%	+10.725%	+10.780%
2020-12-31	+0.250%	+10.064%	+15.732%	+16.505%	+12.057%	+12.892%	+13.994%	+14.560%
2021-01-31	+0.211%	-7.875%	-8.081%	-8.084%	-10.119%	-3.235%	-6.293%	-5.996%
2021-02-28	+0.211%	+12.842%	+10.445%	+5.662%	+15.287%	+13.255%	+14.491%	+12.726%
2021-03-31	+0.211%	+4.393%	+19.556%	+22.408%	+5.906%	+23.825%	+14.756%	+16.488%
2021-04-30	+0.211%	+6.969%	+8.767%	+3.903%	+8.980%	+15.522%	+8.930%	+6.233%
2021-05-31	+0.211%	+15.472%	+22.096%	+14.404%	+23.773%	+20.119%	+23.371%	+22.860%
2021-06-30	+0.211%	+4.495%	+0.467%	-1.814%	-1.144%	-6.826%	-0.465%	-0.648%
2021-07-31	+0.211%	-11.312%	-12.333%	-11.108%	-11.747%	-9.448%	-9.112%	-9.276%
2021-08-31	+0.211%	-3.462%	-3.753%	-0.353%	-3.421%	-3.871%	-4.476%	-4.334%
2021-09-30	+0.211%	+0.100%	+1.140%	+2.552%	+0.366%	-2.856%	+0.418%	+0.747%
2021-10-31	+0.211%	+3.057%	+3.196%	+2.879%	+3.122%	+3.240%	+4.012%	+3.811%
2021-11-30	+0.211%	+3.291%	+5.750%	+3.024%	+4.867%	+5.914%	+5.247%	+5.167%
2021-12-31	+0.211%	+2.077%	+8.329%	+3.033%	+0.969%	+3.157%	+3.287%	+3.459%
2022-01-31	+0.333%	+10.426%	+3.134%	+4.337%	+10.356%	+3.504%	+7.623%	+7.484%
2022-02-28	+0.333%	-4.341%	-1.939%	-2.726%	-5.103%	-2.025%	-3.816%	-3.748%
2022-03-31	+0.333%	-2.061%	-2.219%	-1.697%	-2.109%	+6.368%	+0.465%	+0.222%

Date	RF_monthly	MKT	TAN	MV	BL	BL_KIO	EW	RP
2022-04-30	+0.333%	-8.279%	-14.738%	-9.927%	-9.942%	-10.749%	-10.918%	-10.596%
2022-05-31	+0.333%	-7.245%	-10.355%	-4.418%	-7.494%	-4.157%	-8.141%	-7.535%
2022-06-30	+0.333%	-6.843%	-13.601%	-7.078%	-6.487%	-8.278%	-8.776%	-8.749%
2022-07-31	+0.333%	+4.429%	+9.032%	+1.811%	+5.420%	+6.572%	+5.926%	+5.549%
2022-08-31	+0.333%	+7.404%	+1.302%	+4.695%	+6.416%	+8.815%	+5.077%	+5.052%
2022-09-30	+0.333%	-14.347%	-12.778%	-10.973%	-16.022%	-13.872%	-13.673%	-13.330%
2022-10-31	+0.333%	-5.387%	-5.715%	-3.329%	-7.205%	-0.742%	-8.450%	-8.075%
2022-11-30	+0.333%	+8.320%	-8.462%	+0.206%	+9.858%	+2.979%	+2.860%	+2.738%
2022-12-31	+0.333%	-0.674%	-7.964%	-2.836%	+3.758%	+0.918%	-1.040%	-1.188%
2023-01-31	+0.292%	+14.812%	+15.756%	+14.366%	+13.971%	+14.969%	+15.249%	+15.303%
2023-02-28	+0.292%	-4.766%	-8.025%	+0.523%	-8.970%	-5.260%	-6.174%	-5.459%
2023-03-31	+0.292%	+4.718%	+7.017%	-0.702%	+9.178%	+5.082%	+5.252%	+4.730%
2023-04-30	+0.292%	-2.011%	-3.752%	-0.649%	-3.371%	-2.190%	-0.678%	-0.750%
2023-05-31	+0.292%	+2.165%	+3.166%	+3.714%	+1.112%	+1.952%	+2.372%	+2.412%
2023-06-30	+0.292%	+4.657%	+4.878%	+4.535%	+5.020%	+4.528%	+5.637%	+5.244%
2023-07-31	+0.292%	+7.293%	+5.816%	+4.218%	+6.791%	+4.436%	+6.479%	+6.365%
2023-08-31	+0.292%	-0.262%	+1.185%	-1.744%	+0.362%	+2.990%	+1.644%	+1.258%
2023-09-30	+0.292%	-3.168%	-3.095%	-1.904%	-2.911%	-3.904%	-4.587%	-4.149%
2023-10-31	+0.292%	-5.983%	-0.353%	-3.397%	-8.178%	-5.435%	-7.582%	-7.052%
2023-11-30	+0.292%	+2.447%	+0.989%	+0.735%	+4.824%	+1.888%	+4.446%	+4.095%
2023-12-31	+0.292%	+2.353%	+0.820%	+1.722%	+5.886%	+2.975%	+3.621%	+3.755%
2024-01-31	+0.255%	+9.194%	+7.580%	+7.830%	+10.565%	+9.987%	+7.271%	+7.664%
2024-02-29	+0.255%	+9.584%	+4.304%	+6.342%	+8.423%	+5.151%	+6.580%	+7.064%
2024-03-31	+0.255%	+0.385%	+0.388%	-0.098%	+2.368%	+0.689%	+0.425%	+0.479%
2024-04-30	+0.255%	-4.866%	-1.304%	-4.212%	-8.591%	+1.727%	-2.959%	-3.392%
2024-05-31	+0.255%	-0.366%	+0.264%	+1.082%	-1.540%	+1.015%	+2.250%	+2.086%
2024-06-30	+0.255%	-1.878%	-2.256%	-2.024%	-1.398%	-3.901%	-1.274%	-1.287%
2024-07-31	+0.255%	+4.863%	+5.279%	+1.935%	+4.507%	+4.774%	+3.720%	+3.825%
2024-08-31	+0.255%	+2.529%	+5.198%	+0.116%	+2.846%	+3.314%	+1.848%	+1.633%
2024-09-30	+0.255%	+2.692%	+1.214%	-1.535%	+4.621%	+3.665%	+3.906%	+3.423%

Date	RF_monthly	MKT	TAN	MV	BL	BL_KIO	EW	RP
2024-10-31	+0.255%	-0.823%	-1.522%	-1.122%	-0.534%	+1.031%	-2.480%	-2.227%
2024-11-30	+0.255%	-2.001%	-2.191%	-1.321%	-3.584%	-0.848%	-2.248%	-2.372%
2024-12-31	+0.255%	+1.831%	+9.313%	+1.160%	+3.737%	+4.058%	+3.025%	+2.988%
2025-01-31	+0.258%	+1.884%	+4.951%	+3.789%	+2.755%	+6.369%	+1.155%	+1.050%
2025-02-28	+0.258%	+2.817%	+1.463%	+3.205%	+3.328%	+3.937%	+4.535%	+4.475%
2025-03-31	+0.258%	+0.076%	-3.929%	+0.603%	+2.805%	-2.500%	+0.025%	+0.134%
2025-04-30	+0.258%	-7.686%	-5.466%	-6.310%	-0.637%	-3.521%	-4.870%	-5.327%
2025-05-31	+0.258%	+4.214%	+2.615%	+1.578%	+7.293%	+4.098%	+7.746%	+6.819%
2025-06-30	+0.258%	+3.861%	+1.889%	+1.241%	+8.650%	+6.379%	+2.949%	+2.929%
2025-07-31	+0.258%	+10.102%	+8.619%	+9.323%	+5.009%	+11.597%	+15.647%	+15.350%
2025-08-31	+0.258%	+19.146%	+17.411%	+9.314%	+20.399%	+9.256%	+15.577%	+15.287%
2025-09-30	+0.258%	-6.185%	-2.685%	-8.440%	-5.905%	-2.058%	-5.900%	-6.524%
2025-10-31	+0.258%	-4.223%	-0.663%	-2.756%	-6.305%	+2.428%	-1.992%	-2.268%
2025-11-30	+0.258%	-1.988%	-1.871%	-3.721%	-0.030%	-1.466%	-1.647%	-1.781%
2025-12-31	+0.258%	+2.524%	-0.666%	-0.180%	+1.631%	+5.338%	+1.635%	+1.447%
2026-01-31	+0.362%	+9.818%	+0.062%	+4.737%	+8.402%	-1.515%	+3.410%	+4.235%
2026-02-28	+0.362%	-2.301%	+0.706%	-1.392%	-0.249%	+1.309%	-0.432%	-0.750%
2026-03-31	+0.362%	-9.397%	-6.475%	-4.237%	-10.492%	-6.353%	-6.343%	-6.273%
2026-04-30	+0.362%	+2.959%	+9.154%	+2.915%	+0.801%	+3.329%	+3.621%	+3.370%
2026-05-31	+0.362%	+2.971%	+7.932%	+1.104%	+1.927%	+1.029%	+1.183%	+0.997%

10. Tín hiệu Phân cụm K-means (k=4) – Nhãn & Vị thế từng Cổ phiếu

Nguồn dữ liệu: output/cluster_signals_v2.csv

Nhãn cụm và vị thế view (Long / Short / Neutral) tại mỗi kỳ tái cân bằng. Features K-means (chuẩn hoá): (idiosyncratic momentum 6-1 tháng, -ann_vol). Long = cụm tốt nhất → nhận view dương trong BL_KIO. Short = cụm tệ nhất → nhận view âm trong BL_KIO.

Tổng quan sát: 1,561 dòng (106 bước × 25 cổ phiếu tối đa)

Date	Ticker	Cluster_Label	Assigned_View_Position
2017-08-31	VCB	0	Short
2017-08-31	BID	2	Neutral
2017-08-31	CTG	2	Neutral
2017-08-31	MBB	3	Long
2017-08-31	ACB	1	Neutral
2017-08-31	STB	1	Neutral
2017-08-31	SHB	1	Neutral
2017-09-30	VCB	2	Short
2017-09-30	BID	0	Neutral
2017-09-30	CTG	0	Neutral
2017-09-30	MBB	1	Long
2017-09-30	ACB	3	Neutral
2017-09-30	STB	0	Neutral
2017-09-30	SHB	3	Neutral
2017-10-31	VCB	2	Short
2017-10-31	BID	0	Neutral
2017-10-31	CTG	0	Neutral
2017-10-31	MBB	3	Long
2017-10-31	ACB	2	Short
2017-10-31	STB	0	Neutral
2017-10-31	SHB	1	Neutral
2017-11-30	VCB	0	Short
2017-11-30	BID	2	Neutral
2017-11-30	CTG	2	Neutral
2017-11-30	MBB	3	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2017-11-30	ACB	1	Neutral
2017-11-30	STB	2	Neutral
2017-11-30	SHB	1	Neutral
2017-12-31	VCB	2	Short
2017-12-31	BID	0	Neutral
2017-12-31	CTG	0	Neutral
2017-12-31	MBB	3	Long
2017-12-31	ACB	1	Neutral
2017-12-31	STB	0	Neutral
2017-12-31	SHB	1	Neutral
2018-01-31	VCB	0	Neutral
2018-01-31	BID	3	Neutral
2018-01-31	CTG	1	Short
2018-01-31	MBB	2	Long
2018-01-31	ACB	0	Neutral
2018-01-31	STB	1	Short
2018-01-31	SHB	1	Short
2018-02-28	VCB	1	Neutral
2018-02-28	BID	0	Neutral
2018-02-28	CTG	0	Neutral
2018-02-28	MBB	2	Long
2018-02-28	ACB	1	Neutral
2018-02-28	STB	3	Short
2018-02-28	SHB	1	Neutral
2018-03-31	VCB	2	Long
2018-03-31	BID	0	Neutral
2018-03-31	CTG	0	Neutral
2018-03-31	MBB	3	Neutral
2018-03-31	ACB	2	Long
2018-03-31	STB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-03-31	SHB	0	Neutral
2018-04-30	VCB	0	Long
2018-04-30	BID	2	Neutral
2018-04-30	CTG	1	Short
2018-04-30	MBB	3	Neutral
2018-04-30	ACB	0	Long
2018-04-30	STB	1	Short
2018-04-30	SHB	2	Neutral
2018-05-31	VCB	2	Long
2018-05-31	BID	0	Neutral
2018-05-31	CTG	0	Neutral
2018-05-31	MBB	3	Neutral
2018-05-31	ACB	3	Neutral
2018-05-31	STB	1	Short
2018-05-31	SHB	0	Neutral
2018-06-30	VCB	0	Neutral
2018-06-30	BID	3	Long
2018-06-30	CTG	1	Neutral
2018-06-30	MBB	0	Neutral
2018-06-30	ACB	0	Neutral
2018-06-30	STB	2	Short
2018-06-30	SHB	1	Neutral
2018-07-31	VCB	0	Neutral
2018-07-31	BID	1	Long
2018-07-31	CTG	1	Long
2018-07-31	MBB	2	Neutral
2018-07-31	ACB	0	Neutral
2018-07-31	STB	3	Short
2018-07-31	SHB	1	Long
2018-08-31	VCB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-08-31	BID	0	Neutral
2018-08-31	CTG	0	Neutral
2018-08-31	MBB	2	Neutral
2018-08-31	ACB	3	Long
2018-08-31	STB	1	Short
2018-08-31	SHB	0	Neutral
2018-09-30	VCB	2	Long
2018-09-30	BID	0	Neutral
2018-09-30	CTG	0	Neutral
2018-09-30	MBB	2	Long
2018-09-30	ACB	2	Long
2018-09-30	STB	3	Short
2018-09-30	SHB	1	Neutral
2018-10-31	VCB	2	Long
2018-10-31	BID	3	Neutral
2018-10-31	CTG	0	Neutral
2018-10-31	MBB	2	Long
2018-10-31	ACB	2	Long
2018-10-31	STB	3	Neutral
2018-10-31	SHB	1	Short
2018-11-30	VCB	2	Long
2018-11-30	BID	3	Neutral
2018-11-30	CTG	0	Neutral
2018-11-30	MBB	2	Long
2018-11-30	ACB	2	Long
2018-11-30	STB	1	Short
2018-11-30	SHB	1	Short
2018-12-31	VCB	2	Long
2018-12-31	BID	3	Neutral
2018-12-31	CTG	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-12-31	MBB	0	Neutral
2018-12-31	ACB	2	Long
2018-12-31	STB	3	Neutral
2018-12-31	SHB	1	Short
2019-01-31	VCB	0	Long
2019-01-31	BID	3	Neutral
2019-01-31	CTG	2	Neutral
2019-01-31	MBB	0	Long
2019-01-31	ACB	2	Neutral
2019-01-31	STB	3	Neutral
2019-01-31	SHB	1	Short
2019-02-28	VCB	2	Long
2019-02-28	BID	3	Neutral
2019-02-28	CTG	0	Short
2019-02-28	MBB	1	Neutral
2019-02-28	ACB	1	Neutral
2019-02-28	STB	3	Neutral
2019-02-28	SHB	0	Short
2019-03-31	VCB	0	Neutral
2019-03-31	BID	2	Long
2019-03-31	CTG	1	Short
2019-03-31	MBB	0	Neutral
2019-03-31	ACB	0	Neutral
2019-03-31	STB	3	Neutral
2019-03-31	SHB	1	Short
2019-04-30	VCB	0	Neutral
2019-04-30	BID	2	Long
2019-04-30	CTG	1	Short
2019-04-30	MBB	0	Neutral
2019-04-30	ACB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-04-30	STB	3	Neutral
2019-04-30	SHB	1	Short
2019-05-31	VCB	2	Neutral
2019-05-31	BID	3	Neutral
2019-05-31	CTG	0	Short
2019-05-31	MBB	2	Neutral
2019-05-31	ACB	2	Neutral
2019-05-31	STB	1	Long
2019-05-31	SHB	3	Neutral
2019-06-30	VCB	0	Long
2019-06-30	BID	2	Neutral
2019-06-30	CTG	1	Short
2019-06-30	MBB	3	Neutral
2019-06-30	ACB	3	Neutral
2019-06-30	STB	1	Short
2019-06-30	SHB	1	Short
2019-07-31	VCB	2	Long
2019-07-31	BID	3	Neutral
2019-07-31	CTG	0	Short
2019-07-31	MBB	1	Neutral
2019-07-31	ACB	1	Neutral
2019-07-31	STB	0	Short
2019-07-31	SHB	0	Short
2019-08-31	VCB	2	Long
2019-08-31	BID	3	Neutral
2019-08-31	CTG	0	Short
2019-08-31	MBB	1	Neutral
2019-08-31	ACB	1	Neutral
2019-08-31	STB	0	Short
2019-08-31	SHB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-09-30	VCB	2	Long
2019-09-30	BID	3	Short
2019-09-30	CTG	0	Neutral
2019-09-30	MBB	1	Neutral
2019-09-30	ACB	1	Neutral
2019-09-30	STB	0	Neutral
2019-09-30	SHB	3	Short
2019-10-31	VCB	2	Long
2019-10-31	BID	0	Neutral
2019-10-31	CTG	0	Neutral
2019-10-31	MBB	1	Neutral
2019-10-31	ACB	1	Neutral
2019-10-31	STB	3	Short
2019-10-31	SHB	3	Short
2019-11-30	VCB	0	Long
2019-11-30	BID	2	Neutral
2019-11-30	CTG	1	Short
2019-11-30	MBB	3	Neutral
2019-11-30	ACB	3	Neutral
2019-11-30	STB	1	Short
2019-11-30	SHB	1	Short
2019-12-31	VCB	0	Long
2019-12-31	BID	2	Neutral
2019-12-31	CTG	1	Short
2019-12-31	MBB	0	Long
2019-12-31	ACB	3	Neutral
2019-12-31	STB	1	Short
2019-12-31	SHB	1	Short
2020-01-31	VCB	0	Long
2020-01-31	BID	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-01-31	CTG	1	Neutral
2020-01-31	MBB	2	Neutral
2020-01-31	ACB	2	Neutral
2020-01-31	STB	3	Short
2020-01-31	SHB	3	Short
2020-02-29	VCB	1	Neutral
2020-02-29	BID	2	Long
2020-02-29	CTG	0	Neutral
2020-02-29	MBB	1	Neutral
2020-02-29	ACB	1	Neutral
2020-02-29	STB	3	Short
2020-02-29	SHB	0	Neutral
2020-03-31	VCB	0	Neutral
2020-03-31	BID	2	Long
2020-03-31	CTG	3	Short
2020-03-31	MBB	1	Neutral
2020-03-31	ACB	1	Neutral
2020-03-31	VIB	0	Neutral
2020-03-31	STB	1	Neutral
2020-03-31	SHB	3	Short
2020-04-30	VCB	0	Neutral
2020-04-30	BID	2	Long
2020-04-30	CTG	0	Neutral
2020-04-30	MBB	1	Short
2020-04-30	ACB	1	Short
2020-04-30	VIB	0	Neutral
2020-04-30	STB	1	Short
2020-04-30	SHB	3	Neutral
2020-05-31	VCB	2	Neutral
2020-05-31	BID	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-05-31	CTG	0	Neutral
2020-05-31	MBB	1	Short
2020-05-31	ACB	2	Neutral
2020-05-31	VIB	1	Short
2020-05-31	STB	1	Short
2020-05-31	SHB	3	Long
2020-06-30	VCB	2	Neutral
2020-06-30	BID	0	Neutral
2020-06-30	CTG	0	Neutral
2020-06-30	MBB	3	Short
2020-06-30	ACB	2	Neutral
2020-06-30	VIB	3	Short
2020-06-30	STB	0	Neutral
2020-06-30	SHB	1	Long
2020-07-31	VCB	2	Neutral
2020-07-31	BID	0	Neutral
2020-07-31	CTG	0	Neutral
2020-07-31	MBB	3	Short
2020-07-31	ACB	2	Neutral
2020-07-31	VIB	3	Short
2020-07-31	STB	0	Neutral
2020-07-31	SHB	1	Long
2020-08-31	VCB	3	Neutral
2020-08-31	BID	2	Neutral
2020-08-31	CTG	2	Neutral
2020-08-31	MBB	0	Short
2020-08-31	ACB	3	Neutral
2020-08-31	VIB	0	Short
2020-08-31	STB	2	Neutral
2020-08-31	SHB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-08-31	KLB	3	Neutral
2020-09-30	VCB	0	Neutral
2020-09-30	BID	2	Short
2020-09-30	CTG	3	Neutral
2020-09-30	MBB	0	Neutral
2020-09-30	ACB	0	Neutral
2020-09-30	VIB	3	Neutral
2020-09-30	STB	3	Neutral
2020-09-30	SHB	1	Long
2020-09-30	KLB	0	Neutral
2020-10-31	VCB	0	Neutral
2020-10-31	BID	1	Short
2020-10-31	CTG	1	Short
2020-10-31	MBB	0	Neutral
2020-10-31	ACB	0	Neutral
2020-10-31	VPB	1	Short
2020-10-31	VIB	3	Neutral
2020-10-31	STB	3	Neutral
2020-10-31	SHB	2	Long
2020-10-31	KLB	0	Neutral
2020-11-30	VCB	0	Neutral
2020-11-30	BID	1	Short
2020-11-30	CTG	1	Short
2020-11-30	MBB	0	Neutral
2020-11-30	ACB	0	Neutral
2020-11-30	VPB	1	Short
2020-11-30	VIB	2	Neutral
2020-11-30	STB	1	Short
2020-11-30	SHB	3	Long
2020-11-30	KLB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-12-31	VCB	1	Neutral
2020-12-31	BID	2	Neutral
2020-12-31	CTG	2	Neutral
2020-12-31	MBB	1	Neutral
2020-12-31	ACB	1	Neutral
2020-12-31	VPB	2	Neutral
2020-12-31	VIB	0	Long
2020-12-31	STB	0	Long
2020-12-31	SHB	2	Neutral
2020-12-31	LPB	0	Long
2020-12-31	KLB	3	Short
2021-01-31	VCB	0	Neutral
2021-01-31	BID	1	Short
2021-01-31	CTG	2	Neutral
2021-01-31	MBB	0	Neutral
2021-01-31	ACB	0	Neutral
2021-01-31	VPB	1	Short
2021-01-31	VIB	3	Long
2021-01-31	STB	2	Neutral
2021-01-31	SHB	1	Short
2021-01-31	LPB	2	Neutral
2021-01-31	KLB	0	Neutral
2021-02-28	VCB	1	Short
2021-02-28	BID	1	Short
2021-02-28	CTG	0	Neutral
2021-02-28	MBB	0	Neutral
2021-02-28	ACB	0	Neutral
2021-02-28	VPB	1	Short
2021-02-28	VIB	3	Long
2021-02-28	STB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-02-28	SHB	1	Short
2021-02-28	LPB	0	Neutral
2021-02-28	BAB	2	Neutral
2021-02-28	KLB	0	Neutral
2021-03-31	VCB	2	Short
2021-03-31	BID	2	Short
2021-03-31	CTG	3	Neutral
2021-03-31	MBB	3	Neutral
2021-03-31	ACB	3	Neutral
2021-03-31	VPB	3	Neutral
2021-03-31	HDB	3	Neutral
2021-03-31	VIB	0	Long
2021-03-31	STB	3	Neutral
2021-03-31	SHB	2	Short
2021-03-31	LPB	3	Neutral
2021-03-31	BAB	1	Neutral
2021-03-31	KLB	0	Long
2021-04-30	VCB	1	Short
2021-04-30	BID	1	Short
2021-04-30	CTG	2	Neutral
2021-04-30	MBB	2	Neutral
2021-04-30	ACB	2	Neutral
2021-04-30	VPB	2	Neutral
2021-04-30	HDB	2	Neutral
2021-04-30	VIB	0	Long
2021-04-30	STB	2	Neutral
2021-04-30	SHB	1	Short
2021-04-30	LPB	2	Neutral
2021-04-30	BAB	3	Neutral
2021-04-30	KLB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-05-31	VCB	2	Neutral
2021-05-31	BID	1	Short
2021-05-31	CTG	0	Neutral
2021-05-31	MBB	0	Neutral
2021-05-31	ACB	0	Neutral
2021-05-31	VPB	3	Long
2021-05-31	HDB	0	Neutral
2021-05-31	VIB	3	Long
2021-05-31	STB	0	Neutral
2021-05-31	SHB	1	Short
2021-05-31	LPB	0	Neutral
2021-05-31	BAB	2	Neutral
2021-05-31	KLB	0	Neutral
2021-06-30	VCB	0	Short
2021-06-30	BID	0	Short
2021-06-30	CTG	1	Neutral
2021-06-30	MBB	1	Neutral
2021-06-30	ACB	3	Neutral
2021-06-30	VPB	2	Long
2021-06-30	HDB	1	Neutral
2021-06-30	VIB	2	Long
2021-06-30	TPB	3	Neutral
2021-06-30	STB	1	Neutral
2021-06-30	SHB	2	Long
2021-06-30	LPB	1	Neutral
2021-06-30	BAB	3	Neutral
2021-06-30	KLB	1	Neutral
2021-07-31	VCB	2	Neutral
2021-07-31	BID	1	Short
2021-07-31	CTG	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-07-31	MBB	3	Neutral
2021-07-31	ACB	2	Neutral
2021-07-31	VPB	0	Long
2021-07-31	HDB	1	Short
2021-07-31	VIB	3	Neutral
2021-07-31	TPB	2	Neutral
2021-07-31	STB	3	Neutral
2021-07-31	SHB	0	Long
2021-07-31	LPB	3	Neutral
2021-07-31	BAB	2	Neutral
2021-07-31	KLB	3	Neutral
2021-08-31	VCB	0	Neutral
2021-08-31	BID	2	Short
2021-08-31	CTG	2	Short
2021-08-31	MBB	1	Long
2021-08-31	TCB	1	Long
2021-08-31	ACB	0	Neutral
2021-08-31	VPB	3	Neutral
2021-08-31	HDB	2	Short
2021-08-31	VIB	1	Long
2021-08-31	TPB	0	Neutral
2021-08-31	STB	1	Long
2021-08-31	SHB	3	Neutral
2021-08-31	LPB	1	Long
2021-08-31	BAB	0	Neutral
2021-08-31	KLB	1	Long
2021-09-30	VCB	0	Neutral
2021-09-30	BID	1	Short
2021-09-30	CTG	1	Short
2021-09-30	MBB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-09-30	TCB	1	Short
2021-09-30	ACB	0	Neutral
2021-09-30	VPB	3	Neutral
2021-09-30	HDB	1	Short
2021-09-30	VIB	2	Long
2021-09-30	TPB	0	Neutral
2021-09-30	STB	3	Neutral
2021-09-30	SHB	3	Neutral
2021-09-30	LPB	2	Long
2021-09-30	BAB	0	Neutral
2021-09-30	KLB	1	Short
2021-10-31	VCB	0	Neutral
2021-10-31	BID	2	Short
2021-10-31	CTG	2	Short
2021-10-31	MBB	1	Neutral
2021-10-31	TCB	1	Neutral
2021-10-31	ACB	1	Neutral
2021-10-31	VPB	3	Long
2021-10-31	HDB	1	Neutral
2021-10-31	VIB	1	Neutral
2021-10-31	TPB	0	Neutral
2021-10-31	STB	3	Long
2021-10-31	SHB	3	Long
2021-10-31	LPB	1	Neutral
2021-10-31	BAB	0	Neutral
2021-10-31	KLB	2	Short
2021-11-30	VCB	3	Neutral
2021-11-30	BID	0	Short
2021-11-30	CTG	2	Neutral
2021-11-30	MBB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-11-30	TCB	2	Neutral
2021-11-30	ACB	3	Neutral
2021-11-30	VPB	1	Long
2021-11-30	HDB	2	Neutral
2021-11-30	VIB	2	Neutral
2021-11-30	TPB	3	Neutral
2021-11-30	STB	1	Long
2021-11-30	SHB	1	Long
2021-11-30	LPB	1	Long
2021-11-30	BAB	3	Neutral
2021-11-30	KLB	2	Neutral
2021-12-31	VCB	1	Short
2021-12-31	BID	3	Neutral
2021-12-31	CTG	3	Neutral
2021-12-31	MBB	0	Long
2021-12-31	TCB	0	Long
2021-12-31	ACB	0	Long
2021-12-31	VPB	2	Neutral
2021-12-31	HDB	0	Long
2021-12-31	VIB	3	Neutral
2021-12-31	TPB	0	Long
2021-12-31	STB	2	Neutral
2021-12-31	SHB	2	Neutral
2021-12-31	LPB	0	Long
2021-12-31	BAB	1	Short
2021-12-31	KLB	3	Neutral
2022-01-31	VCB	1	Short
2022-01-31	BID	0	Neutral
2022-01-31	CTG	0	Neutral
2022-01-31	MBB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-01-31	TCB	3	Neutral
2022-01-31	ACB	3	Neutral
2022-01-31	VPB	0	Neutral
2022-01-31	HDB	3	Neutral
2022-01-31	VIB	0	Neutral
2022-01-31	TPB	2	Long
2022-01-31	STB	0	Neutral
2022-01-31	SHB	0	Neutral
2022-01-31	LPB	3	Neutral
2022-01-31	BAB	1	Short
2022-01-31	KLB	0	Neutral
2022-02-28	VCB	0	Long
2022-02-28	BID	1	Neutral
2022-02-28	CTG	1	Neutral
2022-02-28	MBB	0	Long
2022-02-28	TCB	1	Neutral
2022-02-28	ACB	0	Long
2022-02-28	VPB	2	Neutral
2022-02-28	HDB	0	Long
2022-02-28	VIB	1	Neutral
2022-02-28	TPB	0	Long
2022-02-28	STB	1	Neutral
2022-02-28	SHB	2	Neutral
2022-02-28	LPB	1	Neutral
2022-02-28	BAB	3	Short
2022-02-28	KLB	1	Neutral
2022-03-31	VCB	0	Neutral
2022-03-31	BID	1	Neutral
2022-03-31	CTG	2	Short
2022-03-31	MBB	2	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-03-31	TCB	1	Neutral
2022-03-31	ACB	0	Neutral
2022-03-31	VPB	1	Neutral
2022-03-31	HDB	1	Neutral
2022-03-31	VIB	1	Neutral
2022-03-31	TPB	3	Long
2022-03-31	STB	1	Neutral
2022-03-31	SHB	1	Neutral
2022-03-31	LPB	2	Short
2022-03-31	BAB	0	Neutral
2022-03-31	KLB	1	Neutral
2022-04-30	VCB	0	Neutral
2022-04-30	BID	1	Long
2022-04-30	CTG	2	Neutral
2022-04-30	MBB	2	Neutral
2022-04-30	TCB	2	Neutral
2022-04-30	ACB	0	Neutral
2022-04-30	VPB	3	Short
2022-04-30	HDB	2	Neutral
2022-04-30	VIB	2	Neutral
2022-04-30	TPB	1	Long
2022-04-30	STB	3	Short
2022-04-30	SHB	3	Short
2022-04-30	LPB	2	Neutral
2022-04-30	BAB	0	Neutral
2022-04-30	KLB	1	Long
2022-05-31	VCB	0	Neutral
2022-05-31	BID	2	Neutral
2022-05-31	CTG	1	Short
2022-05-31	MBB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-05-31	TCB	1	Short
2022-05-31	ACB	0	Neutral
2022-05-31	VPB	1	Short
2022-05-31	HDB	1	Short
2022-05-31	VIB	2	Neutral
2022-05-31	TPB	3	Long
2022-05-31	STB	2	Neutral
2022-05-31	SHB	1	Short
2022-05-31	LPB	1	Short
2022-05-31	BAB	0	Neutral
2022-05-31	KLB	2	Neutral
2022-06-30	VCB	0	Neutral
2022-06-30	BID	1	Neutral
2022-06-30	CTG	2	Short
2022-06-30	MBB	1	Neutral
2022-06-30	TCB	2	Short
2022-06-30	ACB	0	Neutral
2022-06-30	VPB	2	Short
2022-06-30	HDB	1	Neutral
2022-06-30	VIB	1	Neutral
2022-06-30	TPB	1	Neutral
2022-06-30	STB	1	Neutral
2022-06-30	SHB	2	Short
2022-06-30	LPB	2	Short
2022-06-30	BAB	0	Neutral
2022-06-30	KLB	3	Long
2022-07-31	VCB	2	Neutral
2022-07-31	BID	0	Long
2022-07-31	CTG	3	Neutral
2022-07-31	MBB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-07-31	TCB	3	Neutral
2022-07-31	ACB	2	Neutral
2022-07-31	VPB	3	Neutral
2022-07-31	HDB	3	Neutral
2022-07-31	VIB	0	Long
2022-07-31	TPB	2	Neutral
2022-07-31	STB	0	Long
2022-07-31	SHB	1	Short
2022-07-31	LPB	3	Neutral
2022-07-31	BAB	2	Neutral
2022-07-31	KLB	0	Long
2022-08-31	VCB	0	Neutral
2022-08-31	BID	2	Long
2022-08-31	CTG	3	Neutral
2022-08-31	MBB	2	Long
2022-08-31	TCB	3	Neutral
2022-08-31	ACB	0	Neutral
2022-08-31	VPB	3	Neutral
2022-08-31	HDB	3	Neutral
2022-08-31	VIB	3	Neutral
2022-08-31	TPB	2	Long
2022-08-31	STB	3	Neutral
2022-08-31	SHB	1	Short
2022-08-31	LPB	3	Neutral
2022-08-31	BAB	0	Neutral
2022-08-31	KLB	2	Long
2022-09-30	VCB	2	Long
2022-09-30	BID	0	Neutral
2022-09-30	CTG	3	Neutral
2022-09-30	MBB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-09-30	TCB	3	Neutral
2022-09-30	ACB	2	Long
2022-09-30	VPB	0	Neutral
2022-09-30	HDB	3	Neutral
2022-09-30	VIB	1	Short
2022-09-30	TPB	3	Neutral
2022-09-30	STB	1	Short
2022-09-30	SHB	1	Short
2022-09-30	LPB	3	Neutral
2022-09-30	BAB	2	Long
2022-09-30	KLB	0	Neutral
2022-09-30	VBB	1	Short
2022-10-31	VCB	0	Long
2022-10-31	BID	2	Neutral
2022-10-31	CTG	2	Neutral
2022-10-31	MBB	2	Neutral
2022-10-31	TCB	2	Neutral
2022-10-31	ACB	0	Long
2022-10-31	VPB	2	Neutral
2022-10-31	HDB	2	Neutral
2022-10-31	VIB	2	Neutral
2022-10-31	TPB	1	Short
2022-10-31	STB	3	Neutral
2022-10-31	SHB	3	Neutral
2022-10-31	LPB	1	Short
2022-10-31	BAB	0	Long
2022-10-31	KLB	2	Neutral
2022-10-31	VBB	1	Short
2022-11-30	VCB	0	Long
2022-11-30	BID	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-11-30	CTG	3	Neutral
2022-11-30	MBB	3	Neutral
2022-11-30	TCB	1	Neutral
2022-11-30	ACB	0	Long
2022-11-30	VPB	1	Neutral
2022-11-30	HDB	3	Neutral
2022-11-30	VIB	1	Neutral
2022-11-30	TPB	2	Short
2022-11-30	STB	1	Neutral
2022-11-30	SHB	1	Neutral
2022-11-30	LPB	1	Neutral
2022-11-30	BAB	0	Long
2022-11-30	KLB	3	Neutral
2022-11-30	VBB	2	Short
2022-12-31	VCB	0	Long
2022-12-31	BID	2	Neutral
2022-12-31	CTG	2	Neutral
2022-12-31	MBB	2	Neutral
2022-12-31	TCB	1	Neutral
2022-12-31	ACB	0	Long
2022-12-31	VPB	1	Neutral
2022-12-31	HDB	2	Neutral
2022-12-31	VIB	1	Neutral
2022-12-31	TPB	3	Short
2022-12-31	STB	1	Neutral
2022-12-31	SHB	1	Neutral
2022-12-31	LPB	1	Neutral
2022-12-31	BAB	0	Long
2022-12-31	KLB	1	Neutral
2022-12-31	VBB	3	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-01-31	VCB	0	Long
2023-01-31	BID	2	Neutral
2023-01-31	CTG	2	Neutral
2023-01-31	MBB	3	Neutral
2023-01-31	TCB	1	Short
2023-01-31	ACB	0	Long
2023-01-31	VPB	1	Short
2023-01-31	HDB	2	Neutral
2023-01-31	VIB	1	Short
2023-01-31	TPB	3	Neutral
2023-01-31	STB	1	Short
2023-01-31	SHB	1	Short
2023-01-31	LPB	3	Neutral
2023-01-31	BAB	0	Long
2023-01-31	KLB	1	Short
2023-01-31	VBB	3	Neutral
2023-02-28	VCB	3	Neutral
2023-02-28	BID	0	Long
2023-02-28	CTG	0	Long
2023-02-28	MBB	1	Short
2023-02-28	TCB	1	Short
2023-02-28	ACB	3	Neutral
2023-02-28	VPB	2	Neutral
2023-02-28	HDB	1	Short
2023-02-28	VIB	2	Neutral
2023-02-28	TPB	1	Short
2023-02-28	STB	2	Neutral
2023-02-28	SHB	2	Neutral
2023-02-28	LPB	0	Long
2023-02-28	BAB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-02-28	KLB	2	Neutral
2023-02-28	VBB	1	Short
2023-03-31	VCB	3	Neutral
2023-03-31	BID	1	Long
2023-03-31	CTG	1	Long
2023-03-31	MBB	0	Neutral
2023-03-31	TCB	0	Neutral
2023-03-31	ACB	3	Neutral
2023-03-31	VPB	1	Long
2023-03-31	HDB	0	Neutral
2023-03-31	VIB	1	Long
2023-03-31	TPB	0	Neutral
2023-03-31	STB	1	Long
2023-03-31	SHB	1	Long
2023-03-31	LPB	1	Long
2023-03-31	BAB	3	Neutral
2023-03-31	KLB	2	Short
2023-03-31	VBB	0	Neutral
2023-04-30	VCB	0	Long
2023-04-30	BID	2	Neutral
2023-04-30	CTG	2	Neutral
2023-04-30	MBB	2	Neutral
2023-04-30	TCB	1	Neutral
2023-04-30	ACB	0	Long
2023-04-30	VPB	1	Neutral
2023-04-30	HDB	2	Neutral
2023-04-30	VIB	1	Neutral
2023-04-30	TPB	2	Neutral
2023-04-30	STB	1	Neutral
2023-04-30	SHB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-04-30	LPB	2	Neutral
2023-04-30	BAB	0	Long
2023-04-30	KLB	3	Short
2023-04-30	VBB	2	Neutral
2023-05-31	VCB	0	Long
2023-05-31	BID	0	Long
2023-05-31	CTG	0	Long
2023-05-31	MBB	3	Neutral
2023-05-31	TCB	3	Neutral
2023-05-31	ACB	0	Long
2023-05-31	VPB	3	Neutral
2023-05-31	HDB	0	Long
2023-05-31	VIB	1	Neutral
2023-05-31	TPB	3	Neutral
2023-05-31	STB	1	Neutral
2023-05-31	SHB	3	Neutral
2023-05-31	LPB	1	Neutral
2023-05-31	BAB	0	Long
2023-05-31	KLB	2	Short
2023-05-31	VBB	1	Neutral
2023-06-30	VCB	3	Neutral
2023-06-30	BID	2	Long
2023-06-30	CTG	2	Long
2023-06-30	MBB	3	Neutral
2023-06-30	TCB	0	Neutral
2023-06-30	ACB	3	Neutral
2023-06-30	VPB	2	Long
2023-06-30	HDB	3	Neutral
2023-06-30	VIB	0	Neutral
2023-06-30	TPB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-06-30	STB	0	Neutral
2023-06-30	SHB	0	Neutral
2023-06-30	LPB	2	Long
2023-06-30	BAB	3	Neutral
2023-06-30	KLB	1	Short
2023-06-30	VBB	0	Neutral
2023-07-31	VCB	0	Neutral
2023-07-31	BID	2	Neutral
2023-07-31	CTG	2	Neutral
2023-07-31	MBB	0	Neutral
2023-07-31	TCB	2	Neutral
2023-07-31	ACB	0	Neutral
2023-07-31	VPB	2	Neutral
2023-07-31	HDB	0	Neutral
2023-07-31	VIB	1	Long
2023-07-31	TPB	2	Neutral
2023-07-31	STB	1	Long
2023-07-31	SHB	1	Long
2023-07-31	LPB	1	Long
2023-07-31	BAB	0	Neutral
2023-07-31	KLB	3	Short
2023-07-31	VBB	1	Long
2023-08-31	VCB	0	Long
2023-08-31	BID	2	Neutral
2023-08-31	CTG	2	Neutral
2023-08-31	MBB	2	Neutral
2023-08-31	TCB	1	Neutral
2023-08-31	ACB	0	Long
2023-08-31	VPB	1	Neutral
2023-08-31	HDB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-08-31	VIB	1	Neutral
2023-08-31	TPB	1	Neutral
2023-08-31	STB	1	Neutral
2023-08-31	SHB	1	Neutral
2023-08-31	LPB	1	Neutral
2023-08-31	BAB	2	Neutral
2023-08-31	KLB	3	Short
2023-08-31	VBB	1	Neutral
2023-09-30	VCB	2	Long
2023-09-30	BID	0	Neutral
2023-09-30	CTG	0	Neutral
2023-09-30	MBB	2	Long
2023-09-30	TCB	1	Neutral
2023-09-30	ACB	2	Long
2023-09-30	VPB	0	Neutral
2023-09-30	HDB	2	Long
2023-09-30	VIB	1	Neutral
2023-09-30	TPB	1	Neutral
2023-09-30	STB	1	Neutral
2023-09-30	SHB	1	Neutral
2023-09-30	LPB	0	Neutral
2023-09-30	BAB	0	Neutral
2023-09-30	KLB	3	Short
2023-09-30	VBB	1	Neutral
2023-09-30	BVB	1	Neutral
2023-10-31	VCB	1	Neutral
2023-10-31	BID	3	Short
2023-10-31	CTG	3	Short
2023-10-31	MBB	1	Neutral
2023-10-31	TCB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-10-31	ACB	3	Short
2023-10-31	VPB	1	Neutral
2023-10-31	HDB	1	Neutral
2023-10-31	VIB	0	Neutral
2023-10-31	TPB	1	Neutral
2023-10-31	STB	0	Neutral
2023-10-31	SHB	2	Long
2023-10-31	LPB	0	Neutral
2023-10-31	BAB	3	Short
2023-10-31	KLB	0	Neutral
2023-10-31	VBB	0	Neutral
2023-10-31	BVB	0	Neutral
2023-11-30	VCB	1	Neutral
2023-11-30	BID	1	Neutral
2023-11-30	CTG	0	Neutral
2023-11-30	MBB	0	Neutral
2023-11-30	TCB	0	Neutral
2023-11-30	ACB	1	Neutral
2023-11-30	VPB	0	Neutral
2023-11-30	HDB	1	Neutral
2023-11-30	VIB	0	Neutral
2023-11-30	TPB	0	Neutral
2023-11-30	STB	2	Long
2023-11-30	SHB	2	Long
2023-11-30	LPB	2	Long
2023-11-30	BAB	1	Neutral
2023-11-30	KLB	3	Short
2023-11-30	VBB	3	Short
2023-11-30	BVB	3	Short
2023-12-31	VCB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-12-31	BID	0	Neutral
2023-12-31	CTG	0	Neutral
2023-12-31	MBB	0	Neutral
2023-12-31	TCB	2	Neutral
2023-12-31	ACB	0	Neutral
2023-12-31	VPB	2	Neutral
2023-12-31	HDB	0	Neutral
2023-12-31	VIB	2	Neutral
2023-12-31	TPB	2	Neutral
2023-12-31	STB	2	Neutral
2023-12-31	SHB	2	Neutral
2023-12-31	LPB	2	Neutral
2023-12-31	NAB	3	Long
2023-12-31	BAB	0	Neutral
2023-12-31	KLB	1	Short
2023-12-31	VBB	1	Short
2023-12-31	SGB	2	Neutral
2023-12-31	BVB	1	Short
2024-01-31	VCB	0	Neutral
2024-01-31	BID	3	Neutral
2024-01-31	CTG	3	Neutral
2024-01-31	MBB	0	Neutral
2024-01-31	TCB	3	Neutral
2024-01-31	ACB	0	Neutral
2024-01-31	VPB	3	Neutral
2024-01-31	HDB	0	Neutral
2024-01-31	VIB	1	Short
2024-01-31	TPB	3	Neutral
2024-01-31	STB	1	Short
2024-01-31	SHB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-01-31	LPB	2	Long
2024-01-31	NAB	2	Long
2024-01-31	BAB	3	Neutral
2024-01-31	KLB	1	Short
2024-01-31	VBB	1	Short
2024-01-31	SGB	3	Neutral
2024-01-31	BVB	1	Short
2024-02-29	VCB	0	Neutral
2024-02-29	BID	3	Neutral
2024-02-29	CTG	0	Neutral
2024-02-29	MBB	0	Neutral
2024-02-29	TCB	3	Neutral
2024-02-29	ACB	0	Neutral
2024-02-29	VPB	3	Neutral
2024-02-29	HDB	0	Neutral
2024-02-29	VIB	3	Neutral
2024-02-29	TPB	3	Neutral
2024-02-29	STB	1	Short
2024-02-29	SHB	1	Short
2024-02-29	LPB	2	Long
2024-02-29	MSB	0	Neutral
2024-02-29	NAB	2	Long
2024-02-29	BAB	3	Neutral
2024-02-29	KLB	1	Short
2024-02-29	PGB	1	Short
2024-02-29	VBB	1	Short
2024-02-29	SGB	3	Neutral
2024-02-29	ABB	3	Neutral
2024-02-29	BVB	1	Short
2024-03-31	VCB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-03-31	BID	2	Neutral
2024-03-31	CTG	2	Neutral
2024-03-31	MBB	2	Neutral
2024-03-31	TCB	0	Neutral
2024-03-31	ACB	2	Neutral
2024-03-31	VPB	0	Neutral
2024-03-31	HDB	1	Long
2024-03-31	VIB	0	Neutral
2024-03-31	TPB	0	Neutral
2024-03-31	STB	3	Short
2024-03-31	SHB	0	Neutral
2024-03-31	LPB	1	Long
2024-03-31	MSB	2	Neutral
2024-03-31	OCB	2	Neutral
2024-03-31	NAB	1	Long
2024-03-31	BAB	2	Neutral
2024-03-31	KLB	3	Short
2024-03-31	PGB	3	Short
2024-03-31	VBB	3	Short
2024-03-31	SGB	0	Neutral
2024-03-31	ABB	0	Neutral
2024-03-31	BVB	3	Short
2024-04-30	VCB	0	Neutral
2024-04-30	BID	2	Neutral
2024-04-30	CTG	2	Neutral
2024-04-30	MBB	2	Neutral
2024-04-30	TCB	0	Neutral
2024-04-30	ACB	2	Neutral
2024-04-30	VPB	0	Neutral
2024-04-30	HDB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-04-30	VIB	0	Neutral
2024-04-30	TPB	0	Neutral
2024-04-30	STB	1	Short
2024-04-30	SHB	1	Short
2024-04-30	LPB	3	Long
2024-04-30	MSB	0	Neutral
2024-04-30	OCB	2	Neutral
2024-04-30	NAB	3	Long
2024-04-30	BAB	0	Neutral
2024-04-30	KLB	1	Short
2024-04-30	PGB	1	Short
2024-04-30	VBB	1	Short
2024-04-30	SGB	0	Neutral
2024-04-30	ABB	0	Neutral
2024-04-30	BVB	1	Short
2024-05-31	VCB	0	Neutral
2024-05-31	BID	2	Long
2024-05-31	CTG	2	Long
2024-05-31	MBB	2	Long
2024-05-31	TCB	2	Long
2024-05-31	ACB	0	Neutral
2024-05-31	VPB	3	Neutral
2024-05-31	HDB	2	Long
2024-05-31	VIB	2	Long
2024-05-31	TPB	3	Neutral
2024-05-31	STB	1	Short
2024-05-31	SHB	3	Neutral
2024-05-31	LPB	1	Short
2024-05-31	SSB	3	Neutral
2024-05-31	MSB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-05-31	OCB	2	Long
2024-05-31	NAB	2	Long
2024-05-31	BAB	0	Neutral
2024-05-31	KLB	1	Short
2024-05-31	PGB	1	Short
2024-05-31	VBB	1	Short
2024-05-31	SGB	3	Neutral
2024-05-31	ABB	3	Neutral
2024-05-31	BVB	1	Short
2024-06-30	VCB	0	Neutral
2024-06-30	BID	2	Long
2024-06-30	CTG	2	Long
2024-06-30	MBB	2	Long
2024-06-30	TCB	2	Long
2024-06-30	ACB	0	Neutral
2024-06-30	VPB	3	Neutral
2024-06-30	HDB	2	Long
2024-06-30	VIB	2	Long
2024-06-30	TPB	3	Neutral
2024-06-30	STB	1	Short
2024-06-30	SHB	3	Neutral
2024-06-30	LPB	2	Long
2024-06-30	SSB	3	Neutral
2024-06-30	MSB	3	Neutral
2024-06-30	OCB	3	Neutral
2024-06-30	NAB	2	Long
2024-06-30	BAB	0	Neutral
2024-06-30	KLB	1	Short
2024-06-30	PGB	1	Short
2024-06-30	VBB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-06-30	SGB	3	Neutral
2024-06-30	ABB	3	Neutral
2024-06-30	BVB	1	Short
2024-07-31	VCB	0	Neutral
2024-07-31	BID	1	Long
2024-07-31	CTG	1	Long
2024-07-31	MBB	1	Long
2024-07-31	TCB	1	Long
2024-07-31	ACB	3	Neutral
2024-07-31	VPB	0	Neutral
2024-07-31	HDB	1	Long
2024-07-31	VIB	1	Long
2024-07-31	TPB	0	Neutral
2024-07-31	STB	2	Short
2024-07-31	SHB	0	Neutral
2024-07-31	LPB	1	Long
2024-07-31	SSB	0	Neutral
2024-07-31	MSB	0	Neutral
2024-07-31	OCB	0	Neutral
2024-07-31	NAB	1	Long
2024-07-31	BAB	3	Neutral
2024-07-31	KLB	2	Short
2024-07-31	PGB	2	Short
2024-07-31	VBB	1	Long
2024-07-31	SGB	3	Neutral
2024-07-31	ABB	0	Neutral
2024-07-31	BVB	0	Neutral
2024-08-31	VCB	3	Neutral
2024-08-31	BID	0	Neutral
2024-08-31	CTG	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-08-31	MBB	0	Neutral
2024-08-31	TCB	2	Long
2024-08-31	ACB	0	Neutral
2024-08-31	VPB	3	Neutral
2024-08-31	HDB	0	Neutral
2024-08-31	VIB	0	Neutral
2024-08-31	TPB	0	Neutral
2024-08-31	STB	1	Short
2024-08-31	SHB	0	Neutral
2024-08-31	LPB	2	Long
2024-08-31	SSB	3	Neutral
2024-08-31	MSB	0	Neutral
2024-08-31	OCB	0	Neutral
2024-08-31	NAB	0	Neutral
2024-08-31	BAB	3	Neutral
2024-08-31	KLB	1	Short
2024-08-31	PGB	1	Short
2024-08-31	VBB	0	Neutral
2024-08-31	SGB	3	Neutral
2024-08-31	ABB	0	Neutral
2024-08-31	BVB	0	Neutral
2024-09-30	VCB	3	Neutral
2024-09-30	BID	0	Neutral
2024-09-30	CTG	0	Neutral
2024-09-30	MBB	0	Neutral
2024-09-30	TCB	1	Long
2024-09-30	ACB	3	Neutral
2024-09-30	VPB	0	Neutral
2024-09-30	HDB	0	Neutral
2024-09-30	VIB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-09-30	TPB	0	Neutral
2024-09-30	STB	2	Short
2024-09-30	SHB	0	Neutral
2024-09-30	LPB	1	Long
2024-09-30	SSB	3	Neutral
2024-09-30	MSB	0	Neutral
2024-09-30	OCB	0	Neutral
2024-09-30	NAB	0	Neutral
2024-09-30	BAB	3	Neutral
2024-09-30	KLB	2	Short
2024-09-30	PGB	2	Short
2024-09-30	VAB	1	Long
2024-09-30	VBB	0	Neutral
2024-09-30	SGB	3	Neutral
2024-09-30	ABB	0	Neutral
2024-09-30	BVB	0	Neutral
2024-10-31	VCB	1	Neutral
2024-10-31	BID	3	Neutral
2024-10-31	CTG	1	Neutral
2024-10-31	MBB	2	Long
2024-10-31	TCB	2	Long
2024-10-31	ACB	1	Neutral
2024-10-31	VPB	1	Neutral
2024-10-31	HDB	2	Long
2024-10-31	VIB	3	Neutral
2024-10-31	TPB	3	Neutral
2024-10-31	STB	3	Neutral
2024-10-31	SHB	3	Neutral
2024-10-31	LPB	2	Long
2024-10-31	SSB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-10-31	MSB	1	Neutral
2024-10-31	OCB	3	Neutral
2024-10-31	NAB	2	Long
2024-10-31	BAB	1	Neutral
2024-10-31	KLB	3	Neutral
2024-10-31	PGB	0	Short
2024-10-31	VAB	2	Long
2024-10-31	VBB	3	Neutral
2024-10-31	SGB	1	Neutral
2024-10-31	ABB	1	Neutral
2024-10-31	BVB	2	Long
2024-11-30	VCB	3	Neutral
2024-11-30	BID	0	Neutral
2024-11-30	CTG	0	Neutral
2024-11-30	MBB	0	Neutral
2024-11-30	TCB	0	Neutral
2024-11-30	ACB	3	Neutral
2024-11-30	VPB	0	Neutral
2024-11-30	HDB	0	Neutral
2024-11-30	VIB	0	Neutral
2024-11-30	TPB	0	Neutral
2024-11-30	STB	0	Neutral
2024-11-30	SHB	0	Neutral
2024-11-30	LPB	2	Long
2024-11-30	SSB	3	Neutral
2024-11-30	MSB	0	Neutral
2024-11-30	OCB	0	Neutral
2024-11-30	NAB	0	Neutral
2024-11-30	BAB	3	Neutral
2024-11-30	KLB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-11-30	PGB	1	Short
2024-11-30	VAB	0	Neutral
2024-11-30	VBB	0	Neutral
2024-11-30	SGB	3	Neutral
2024-11-30	ABB	0	Neutral
2024-11-30	BVB	0	Neutral
2024-12-31	VCB	2	Neutral
2024-12-31	BID	0	Neutral
2024-12-31	CTG	0	Neutral
2024-12-31	MBB	0	Neutral
2024-12-31	TCB	0	Neutral
2024-12-31	ACB	2	Neutral
2024-12-31	VPB	0	Neutral
2024-12-31	HDB	0	Neutral
2024-12-31	VIB	0	Neutral
2024-12-31	TPB	0	Neutral
2024-12-31	STB	0	Neutral
2024-12-31	SHB	0	Neutral
2024-12-31	LPB	1	Long
2024-12-31	SSB	2	Neutral
2024-12-31	MSB	0	Neutral
2024-12-31	OCB	0	Neutral
2024-12-31	NAB	0	Neutral
2024-12-31	BAB	2	Neutral
2024-12-31	KLB	0	Neutral
2024-12-31	PGB	3	Short
2024-12-31	VAB	0	Neutral
2024-12-31	VBB	0	Neutral
2024-12-31	SGB	2	Neutral
2024-12-31	ABB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-12-31	BVB	0	Neutral
2025-01-31	VCB	3	Neutral
2025-01-31	BID	0	Short
2025-01-31	CTG	2	Neutral
2025-01-31	MBB	2	Neutral
2025-01-31	TCB	0	Short
2025-01-31	ACB	3	Neutral
2025-01-31	VPB	2	Neutral
2025-01-31	HDB	2	Neutral
2025-01-31	VIB	0	Short
2025-01-31	TPB	2	Neutral
2025-01-31	STB	1	Long
2025-01-31	SHB	0	Short
2025-01-31	LPB	1	Long
2025-01-31	SSB	3	Neutral
2025-01-31	MSB	2	Neutral
2025-01-31	OCB	0	Short
2025-01-31	NAB	2	Neutral
2025-01-31	BAB	3	Neutral
2025-01-31	KLB	0	Short
2025-01-31	PGB	0	Short
2025-01-31	VAB	0	Short
2025-01-31	VBB	0	Short
2025-01-31	SGB	3	Neutral
2025-01-31	ABB	3	Neutral
2025-01-31	BVB	2	Neutral
2025-02-28	VCB	1	Neutral
2025-02-28	BID	0	Neutral
2025-02-28	CTG	3	Long
2025-02-28	MBB	3	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-02-28	TCB	0	Neutral
2025-02-28	ACB	1	Neutral
2025-02-28	VPB	1	Neutral
2025-02-28	HDB	3	Long
2025-02-28	VIB	0	Neutral
2025-02-28	TPB	3	Long
2025-02-28	STB	3	Long
2025-02-28	SHB	2	Short
2025-02-28	LPB	3	Long
2025-02-28	SSB	2	Short
2025-02-28	MSB	1	Neutral
2025-02-28	OCB	2	Short
2025-02-28	NAB	3	Long
2025-02-28	BAB	1	Neutral
2025-02-28	KLB	0	Neutral
2025-02-28	PGB	0	Neutral
2025-02-28	VAB	2	Short
2025-02-28	VBB	0	Neutral
2025-02-28	SGB	2	Short
2025-02-28	ABB	2	Short
2025-02-28	BVB	2	Short
2025-03-31	VCB	2	Neutral
2025-03-31	BID	1	Neutral
2025-03-31	CTG	3	Long
2025-03-31	MBB	1	Neutral
2025-03-31	TCB	1	Neutral
2025-03-31	ACB	2	Neutral
2025-03-31	VPB	1	Neutral
2025-03-31	HDB	3	Long
2025-03-31	VIB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-03-31	TPB	3	Long
2025-03-31	STB	3	Long
2025-03-31	SHB	1	Neutral
2025-03-31	LPB	3	Long
2025-03-31	SSB	2	Neutral
2025-03-31	MSB	1	Neutral
2025-03-31	OCB	1	Neutral
2025-03-31	NAB	3	Long
2025-03-31	BAB	2	Neutral
2025-03-31	KLB	0	Short
2025-03-31	PGB	0	Short
2025-03-31	VAB	1	Neutral
2025-03-31	VBB	3	Long
2025-03-31	SGB	2	Neutral
2025-03-31	ABB	1	Neutral
2025-03-31	BVB	1	Neutral
2025-04-30	VCB	2	Neutral
2025-04-30	BID	0	Neutral
2025-04-30	CTG	0	Neutral
2025-04-30	MBB	0	Neutral
2025-04-30	TCB	0	Neutral
2025-04-30	ACB	2	Neutral
2025-04-30	VPB	0	Neutral
2025-04-30	HDB	0	Neutral
2025-04-30	VIB	0	Neutral
2025-04-30	TPB	0	Neutral
2025-04-30	STB	3	Long
2025-04-30	SHB	1	Short
2025-04-30	LPB	3	Long
2025-04-30	SSB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-04-30	MSB	0	Neutral
2025-04-30	OCB	1	Short
2025-04-30	NAB	0	Neutral
2025-04-30	BAB	2	Neutral
2025-04-30	KLB	1	Short
2025-04-30	PGB	1	Short
2025-04-30	VAB	1	Short
2025-04-30	VBB	0	Neutral
2025-04-30	SGB	2	Neutral
2025-04-30	ABB	1	Short
2025-04-30	BVB	0	Neutral
2025-05-31	VCB	2	Neutral
2025-05-31	BID	0	Neutral
2025-05-31	CTG	1	Long
2025-05-31	MBB	0	Neutral
2025-05-31	TCB	1	Long
2025-05-31	ACB	2	Neutral
2025-05-31	VPB	0	Neutral
2025-05-31	HDB	0	Neutral
2025-05-31	VIB	1	Long
2025-05-31	TPB	1	Long
2025-05-31	STB	1	Long
2025-05-31	SHB	0	Neutral
2025-05-31	LPB	1	Long
2025-05-31	SSB	2	Neutral
2025-05-31	MSB	0	Neutral
2025-05-31	OCB	0	Neutral
2025-05-31	NAB	0	Neutral
2025-05-31	BAB	2	Neutral
2025-05-31	KLB	3	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-05-31	PGB	3	Short
2025-05-31	VAB	1	Long
2025-05-31	VBB	1	Long
2025-05-31	SGB	2	Neutral
2025-05-31	ABB	0	Neutral
2025-05-31	BVB	1	Long
2025-06-30	VCB	2	Neutral
2025-06-30	BID	0	Neutral
2025-06-30	CTG	1	Long
2025-06-30	MBB	1	Long
2025-06-30	TCB	1	Long
2025-06-30	ACB	2	Neutral
2025-06-30	VPB	0	Neutral
2025-06-30	HDB	0	Neutral
2025-06-30	VIB	1	Long
2025-06-30	TPB	0	Neutral
2025-06-30	STB	1	Long
2025-06-30	SHB	1	Long
2025-06-30	LPB	1	Long
2025-06-30	SSB	2	Neutral
2025-06-30	MSB	0	Neutral
2025-06-30	OCB	0	Neutral
2025-06-30	NAB	0	Neutral
2025-06-30	BAB	2	Neutral
2025-06-30	KLB	3	Short
2025-06-30	PGB	3	Short
2025-06-30	VAB	1	Long
2025-06-30	VBB	1	Long
2025-06-30	SGB	2	Neutral
2025-06-30	ABB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-06-30	BVB	1	Long
2025-07-31	VCB	1	Neutral
2025-07-31	BID	0	Short
2025-07-31	CTG	1	Neutral
2025-07-31	MBB	2	Long
2025-07-31	TCB	2	Long
2025-07-31	ACB	1	Neutral
2025-07-31	VPB	0	Short
2025-07-31	HDB	0	Short
2025-07-31	VIB	0	Short
2025-07-31	TPB	0	Short
2025-07-31	STB	2	Long
2025-07-31	SHB	2	Long
2025-07-31	LPB	2	Long
2025-07-31	SSB	1	Neutral
2025-07-31	MSB	0	Short
2025-07-31	OCB	0	Short
2025-07-31	NAB	2	Long
2025-07-31	BAB	1	Neutral
2025-07-31	KLB	3	Neutral
2025-07-31	PGB	3	Neutral
2025-07-31	VAB	2	Long
2025-07-31	VBB	2	Long
2025-07-31	SGB	1	Neutral
2025-07-31	ABB	1	Neutral
2025-07-31	BVB	2	Long
2025-08-31	VCB	2	Neutral
2025-08-31	BID	1	Short
2025-08-31	CTG	1	Short
2025-08-31	MBB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-08-31	TCB	0	Neutral
2025-08-31	ACB	2	Neutral
2025-08-31	VPB	1	Short
2025-08-31	HDB	1	Short
2025-08-31	VIB	1	Short
2025-08-31	TPB	1	Short
2025-08-31	STB	0	Neutral
2025-08-31	SHB	3	Long
2025-08-31	LPB	0	Neutral
2025-08-31	SSB	2	Neutral
2025-08-31	MSB	1	Short
2025-08-31	OCB	1	Short
2025-08-31	NAB	1	Short
2025-08-31	BAB	2	Neutral
2025-08-31	KLB	3	Long
2025-08-31	PGB	0	Neutral
2025-08-31	VAB	3	Long
2025-08-31	VBB	0	Neutral
2025-08-31	SGB	2	Neutral
2025-08-31	ABB	1	Short
2025-08-31	BVB	1	Short
2025-09-30	VCB	3	Neutral
2025-09-30	BID	2	Short
2025-09-30	CTG	0	Neutral
2025-09-30	MBB	0	Neutral
2025-09-30	TCB	1	Long
2025-09-30	ACB	3	Neutral
2025-09-30	VPB	2	Short
2025-09-30	HDB	2	Short
2025-09-30	VIB	2	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-09-30	TPB	2	Short
2025-09-30	STB	1	Long
2025-09-30	SHB	1	Long
2025-09-30	LPB	0	Neutral
2025-09-30	SSB	3	Neutral
2025-09-30	MSB	0	Neutral
2025-09-30	OCB	0	Neutral
2025-09-30	NAB	0	Neutral
2025-09-30	BAB	3	Neutral
2025-09-30	KLB	1	Long
2025-09-30	PGB	2	Short
2025-09-30	VAB	1	Long
2025-09-30	VBB	1	Long
2025-09-30	SGB	3	Neutral
2025-09-30	ABB	0	Neutral
2025-09-30	BVB	0	Neutral
2025-10-31	VCB	1	Short
2025-10-31	BID	0	Neutral
2025-10-31	CTG	0	Neutral
2025-10-31	MBB	0	Neutral
2025-10-31	TCB	3	Neutral
2025-10-31	ACB	1	Short
2025-10-31	VPB	3	Neutral
2025-10-31	HDB	0	Neutral
2025-10-31	VIB	0	Neutral
2025-10-31	TPB	0	Neutral
2025-10-31	STB	3	Neutral
2025-10-31	SHB	2	Long
2025-10-31	LPB	0	Neutral
2025-10-31	SSB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-10-31	MSB	0	Neutral
2025-10-31	OCB	0	Neutral
2025-10-31	NAB	0	Neutral
2025-10-31	BAB	1	Short
2025-10-31	KLB	2	Long
2025-10-31	PGB	3	Neutral
2025-10-31	VAB	2	Long
2025-10-31	VBB	3	Neutral
2025-10-31	SGB	1	Short
2025-10-31	ABB	2	Long
2025-10-31	BVB	0	Neutral
2025-11-30	VCB	3	Neutral
2025-11-30	BID	2	Short
2025-11-30	CTG	3	Neutral
2025-11-30	MBB	0	Neutral
2025-11-30	TCB	0	Neutral
2025-11-30	ACB	3	Neutral
2025-11-30	VPB	1	Long
2025-11-30	HDB	0	Neutral
2025-11-30	VIB	0	Neutral
2025-11-30	TPB	0	Neutral
2025-11-30	STB	0	Neutral
2025-11-30	SHB	1	Long
2025-11-30	LPB	0	Neutral
2025-11-30	SSB	3	Neutral
2025-11-30	MSB	0	Neutral
2025-11-30	OCB	0	Neutral
2025-11-30	NAB	2	Short
2025-11-30	BAB	3	Neutral
2025-11-30	KLB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-11-30	PGB	2	Short
2025-11-30	VAB	1	Long
2025-11-30	VBB	0	Neutral
2025-11-30	SGB	3	Neutral
2025-11-30	ABB	1	Long
2025-11-30	BVB	2	Short
2025-12-31	VCB	1	Short
2025-12-31	BID	1	Short
2025-12-31	CTG	1	Short
2025-12-31	MBB	0	Neutral
2025-12-31	TCB	0	Neutral
2025-12-31	ACB	1	Short
2025-12-31	VPB	2	Neutral
2025-12-31	HDB	0	Neutral
2025-12-31	VIB	0	Neutral
2025-12-31	TPB	0	Neutral
2025-12-31	STB	0	Neutral
2025-12-31	SHB	2	Neutral
2025-12-31	LPB	2	Neutral
2025-12-31	SSB	1	Short
2025-12-31	MSB	0	Neutral
2025-12-31	OCB	0	Neutral
2025-12-31	NAB	1	Short
2025-12-31	BAB	1	Short
2025-12-31	KLB	3	Long
2025-12-31	PGB	0	Neutral
2025-12-31	VAB	2	Neutral
2025-12-31	VBB	0	Neutral
2025-12-31	SGB	1	Short
2025-12-31	ABB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-12-31	BVB	1	Short
2026-01-31	VCB	0	Short
2026-01-31	BID	0	Short
2026-01-31	CTG	2	Neutral
2026-01-31	MBB	2	Neutral
2026-01-31	TCB	2	Neutral
2026-01-31	ACB	0	Short
2026-01-31	VPB	1	Neutral
2026-01-31	HDB	1	Neutral
2026-01-31	VIB	2	Neutral
2026-01-31	TPB	1	Neutral
2026-01-31	STB	2	Neutral
2026-01-31	SHB	1	Neutral
2026-01-31	LPB	1	Neutral
2026-01-31	SSB	0	Short
2026-01-31	MSB	2	Neutral
2026-01-31	OCB	2	Neutral
2026-01-31	NAB	2	Neutral
2026-01-31	BAB	0	Short
2026-01-31	KLB	3	Long
2026-01-31	PGB	1	Neutral
2026-01-31	VAB	1	Neutral
2026-01-31	VBB	1	Neutral
2026-01-31	SGB	0	Short
2026-01-31	ABB	1	Neutral
2026-01-31	BVB	0	Short
2026-02-28	VCB	0	Short
2026-02-28	BID	0	Short
2026-02-28	CTG	2	Neutral
2026-02-28	MBB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-02-28	TCB	2	Neutral
2026-02-28	ACB	0	Short
2026-02-28	VPB	1	Long
2026-02-28	HDB	1	Long
2026-02-28	VIB	2	Neutral
2026-02-28	TPB	1	Long
2026-02-28	STB	2	Neutral
2026-02-28	SHB	1	Long
2026-02-28	LPB	1	Long
2026-02-28	SSB	0	Short
2026-02-28	MSB	1	Long
2026-02-28	OCB	2	Neutral
2026-02-28	NAB	2	Neutral
2026-02-28	BAB	0	Short
2026-02-28	KLB	3	Neutral
2026-02-28	PGB	3	Neutral
2026-02-28	VAB	2	Neutral
2026-02-28	VBB	3	Neutral
2026-02-28	SGB	0	Short
2026-02-28	ABB	1	Long
2026-02-28	BVB	0	Short
2026-03-31	VCB	2	Short
2026-03-31	BID	0	Neutral
2026-03-31	CTG	0	Neutral
2026-03-31	MBB	3	Neutral
2026-03-31	TCB	0	Neutral
2026-03-31	ACB	2	Short
2026-03-31	VPB	1	Long
2026-03-31	HDB	1	Long
2026-03-31	VIB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-03-31	TPB	3	Neutral
2026-03-31	STB	3	Neutral
2026-03-31	SHB	3	Neutral
2026-03-31	LPB	3	Neutral
2026-03-31	SSB	2	Short
2026-03-31	MSB	3	Neutral
2026-03-31	OCB	0	Neutral
2026-03-31	NAB	0	Neutral
2026-03-31	BAB	2	Short
2026-03-31	KLB	1	Long
2026-03-31	PGB	3	Neutral
2026-03-31	VAB	0	Neutral
2026-03-31	VBB	0	Neutral
2026-03-31	SGB	2	Short
2026-03-31	ABB	1	Long
2026-03-31	BVB	0	Neutral
2026-04-30	VCB	1	Long
2026-04-30	BID	1	Long
2026-04-30	CTG	1	Long
2026-04-30	MBB	1	Long
2026-04-30	TCB	0	Neutral
2026-04-30	ACB	3	Short
2026-04-30	VPB	2	Neutral
2026-04-30	HDB	1	Long
2026-04-30	VIB	3	Short
2026-04-30	TPB	0	Neutral
2026-04-30	STB	1	Long
2026-04-30	SHB	0	Neutral
2026-04-30	LPB	2	Neutral
2026-04-30	SSB	3	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-04-30	MSB	0	Neutral
2026-04-30	OCB	3	Short
2026-04-30	NAB	3	Short
2026-04-30	BAB	3	Short
2026-04-30	KLB	2	Neutral
2026-04-30	PGB	2	Neutral
2026-04-30	VAB	0	Neutral
2026-04-30	VBB	0	Neutral
2026-04-30	SGB	3	Short
2026-04-30	ABB	2	Neutral
2026-04-30	BVB	3	Short
2026-05-31	VCB	0	Neutral
2026-05-31	BID	3	Neutral
2026-05-31	CTG	2	Long
2026-05-31	MBB	2	Long
2026-05-31	TCB	1	Short
2026-05-31	ACB	0	Neutral
2026-05-31	VPB	1	Short
2026-05-31	HDB	3	Neutral
2026-05-31	VIB	0	Neutral
2026-05-31	TPB	1	Short
2026-05-31	STB	2	Long
2026-05-31	SHB	1	Short
2026-05-31	LPB	1	Short
2026-05-31	SSB	0	Neutral
2026-05-31	MSB	1	Short
2026-05-31	OCB	0	Neutral
2026-05-31	NAB	0	Neutral
2026-05-31	BAB	0	Neutral
2026-05-31	KLB	3	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-05-31	PGB	1	Short
2026-05-31	VAB	1	Short
2026-05-31	VBB	0	Neutral
2026-05-31	SGB	0	Neutral
2026-05-31	ABB	3	Neutral
2026-05-31	BVB	0	Neutral

11. Lịch sử Tỷ trọng Danh mục (7 Danh mục × 106 bước)

Nguồn dữ liệu: output/weights_v2.csv

Tỷ trọng tại mỗi kỳ tái cân bằng cho từng cổ phiếu trong 7 danh mục. Ràng buộc: $0 \leq w \leq 30\%$, $\sum w = 1$ (áp dụng cho TAN, MV, BL, BL_KIO, RP). $EW = 1/N$ (không ràng buộc per-asset, tự nhiên $\leq 1/N_{min}$).

Tổng quan sát: 1,561 dòng

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2017-08-31	VCB	0.3733	0.3000	0.2284	0.3000	0.000000	0.1429	0.1434
2017-08-31	BID	0.2359	0.0781	0.000000	0.3000	0.1000	0.1429	0.1165
2017-08-31	CTG	0.1383	0.000000	0.000000	0.3000	0.000000	0.1429	0.1261
2017-08-31	MBB	0.0854	0.3000	0.3000	0.000000	0.3000	0.1429	0.2006
2017-08-31	ACB	0.0670	0.3000	0.1806	0.0339	0.000000	0.1429	0.1482
2017-08-31	STB	0.0707	0.000000	0.000000	0.0661	0.3000	0.1429	0.1265
2017-08-31	SHB	0.0292	0.0219	0.2910	0.000000	0.3000	0.1429	0.1386
2017-09-30	VCB	0.3769	0.3000	0.2371	0.3000	0.000000	0.1429	0.1441
2017-09-30	BID	0.2333	0.0360	0.000000	0.3000	0.1000	0.1429	0.1161
2017-09-30	CTG	0.1326	0.000000	0.000000	0.3000	0.000000	0.1429	0.1266
2017-09-30	MBB	0.0876	0.3000	0.3000	0.000000	0.3000	0.1429	0.2006
2017-09-30	ACB	0.0761	0.3000	0.1762	0.0401	0.000000	0.1429	0.1488
2017-09-30	STB	0.0649	0.000000	0.005906	0.0599	0.3000	0.1429	0.1268
2017-09-30	SHB	0.0286	0.0640	0.2808	0.000000	0.3000	0.1429	0.1370
2017-10-31	VCB	0.3871	0.3000	0.2461	0.3000	0.000000	0.1429	0.1461
2017-10-31	BID	0.2190	0.0793	0.000000	0.3000	0.1000	0.1429	0.1162
2017-10-31	CTG	0.1326	0.000000	0.000000	0.3000	0.000000	0.1429	0.1257
2017-10-31	MBB	0.0834	0.3000	0.3000	0.000000	0.3000	0.1429	0.2011
2017-10-31	ACB	0.0804	0.3000	0.1687	0.0322	0.000000	0.1429	0.1472
2017-10-31	STB	0.0687	0.000000	0.000000	0.0678	0.3000	0.1429	0.1249
2017-10-31	SHB	0.0287	0.0207	0.2852	0.000000	0.3000	0.1429	0.1388
2017-11-30	VCB	0.4020	0.3000	0.2581	0.3000	0.000000	0.1429	0.1489
2017-11-30	BID	0.2302	0.0349	0.000000	0.3000	0.1000	0.1429	0.1169
2017-11-30	CTG	0.1267	0.000000	0.000000	0.3000	0.000000	0.1429	0.1261
2017-11-30	MBB	0.0817	0.3000	0.3000	0.000000	0.3000	0.1429	0.1981

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2017-11-30	ACB	0.0759	0.3000	0.1603	0.0313	0.000000	0.1429	0.1473
2017-11-30	STB	0.0570	0.000000	0.000000	0.0687	0.3000	0.1429	0.1238
2017-11-30	SHB	0.0264	0.0651	0.2816	0.000000	0.3000	0.1429	0.1390
2017-12-31	VCB	0.3976	0.3000	0.2480	0.3000	0.000000	0.1429	0.1482
2017-12-31	BID	0.2283	0.0709	0.000000	0.3000	0.1206	0.1429	0.1163
2017-12-31	CTG	0.1387	0.000000	0.000000	0.3000	0.000000	0.1429	0.1267
2017-12-31	MBB	0.0782	0.3000	0.3000	0.000000	0.3000	0.1429	0.1989
2017-12-31	ACB	0.0730	0.3000	0.1679	0.0423	0.000000	0.1429	0.1479
2017-12-31	STB	0.0574	0.000000	0.000000	0.0577	0.2794	0.1429	0.1215
2017-12-31	SHB	0.0268	0.0291	0.2841	0.000000	0.3000	0.1429	0.1405
2018-01-31	VCB	0.4247	0.3000	0.2348	0.3000	0.3000	0.1429	0.1487
2018-01-31	BID	0.2166	0.0706	0.000000	0.3000	0.3000	0.1429	0.1194
2018-01-31	CTG	0.1318	0.000000	0.000000	0.3000	0.000000	0.1429	0.1210
2018-01-31	MBB	0.0741	0.3000	0.3000	0.000000	0.3000	0.1429	0.2017
2018-01-31	ACB	0.0738	0.3000	0.2081	0.0221	0.1000	0.1429	0.1511
2018-01-31	STB	0.0537	0.000000	0.000000	0.0779	0.000000	0.1429	0.1207
2018-01-31	SHB	0.0253	0.0294	0.2570	0.000000	0.000000	0.1429	0.1374
2018-02-28	VCB	0.4189	0.3000	0.2323	0.3000	0.1983	0.1429	0.1503
2018-02-28	BID	0.2334	0.0331	0.000000	0.3000	0.3000	0.1429	0.1190
2018-02-28	CTG	0.1191	0.000000	0.000000	0.3000	0.2017	0.1429	0.1208
2018-02-28	MBB	0.0764	0.3000	0.3000	0.000000	0.3000	0.1429	0.2003
2018-02-28	ACB	0.0670	0.3000	0.2091	0.0191	0.000000	0.1429	0.1504
2018-02-28	STB	0.0546	0.000000	0.000000	0.0809	0.000000	0.1429	0.1205
2018-02-28	SHB	0.0306	0.0669	0.2586	0.000000	0.000000	0.1429	0.1386
2018-03-31	VCB	0.4143	0.3000	0.2667	0.3000	0.3000	0.1429	0.1519
2018-03-31	BID	0.2376	0.0220	0.000000	0.3000	0.1877	0.1429	0.1300
2018-03-31	CTG	0.1313	0.000000	0.000000	0.2996	0.2123	0.1429	0.1400
2018-03-31	MBB	0.0736	0.3000	0.3000	0.000000	0.000000	0.1429	0.1814
2018-03-31	ACB	0.0674	0.3000	0.3000	0.000000	0.3000	0.1429	0.1598
2018-03-31	STB	0.0483	0.000000	0.000000	0.1004	0.000000	0.1429	0.1194

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2018-03-31	SHB	0.0275	0.0780	0.1333	0.000000	0.000000	0.1429	0.1176
2018-04-30	VCB	0.3893	0.3000	0.2541	0.3000	0.3000	0.1429	0.1528
2018-04-30	BID	0.2601	0.0194	0.000000	0.3000	0.3000	0.1429	0.1298
2018-04-30	CTG	0.1330	0.000000	0.000000	0.3000	0.1000	0.1429	0.1340
2018-04-30	MBB	0.0729	0.3000	0.3000	0.000000	0.000000	0.1429	0.1835
2018-04-30	ACB	0.0724	0.3000	0.3000	0.001565	0.3000	0.1429	0.1597
2018-04-30	STB	0.0457	0.000000	0.000000	0.0971	0.000000	0.1429	0.1211
2018-04-30	SHB	0.0266	0.0806	0.1459	0.001361	0.000000	0.1429	0.1192
2018-05-31	VCB	0.3883	0.3000	0.2617	0.3000	0.3000	0.1429	0.1542
2018-05-31	BID	0.2536	0.0568	0.000000	0.3000	0.3000	0.1429	0.1288
2018-05-31	CTG	0.1320	0.000000	0.000000	0.2852	0.3000	0.1429	0.1367
2018-05-31	MBB	0.0737	0.3000	0.3000	0.000000	0.000000	0.1429	0.1836
2018-05-31	ACB	0.0770	0.3000	0.3000	0.008514	0.1000	0.1429	0.1582
2018-05-31	STB	0.0479	0.000000	0.000000	0.1005	0.000000	0.1429	0.1200
2018-05-31	SHB	0.0276	0.0432	0.1383	0.005787	0.000000	0.1429	0.1186
2018-06-30	VCB	0.3956	0.3000	0.2429	0.3000	0.2471	0.1429	0.1513
2018-06-30	BID	0.2321	0.0219	0.000000	0.3000	0.3000	0.1429	0.1276
2018-06-30	CTG	0.1379	0.000000	0.000000	0.3000	0.3000	0.1429	0.1363
2018-06-30	MBB	0.0810	0.3000	0.3000	0.000000	0.000000	0.1429	0.1773
2018-06-30	ACB	0.0821	0.3000	0.3000	0.000884	0.1529	0.1429	0.1607
2018-06-30	STB	0.0459	0.000000	0.000000	0.0991	0.000000	0.1429	0.1246
2018-06-30	SHB	0.0254	0.0781	0.1571	0.000000	0.000000	0.1429	0.1221
2018-07-31	VCB	0.4384	0.3000	0.3000	0.3000	0.1820	0.1429	0.1557
2018-07-31	BID	0.2133	0.000000	0.000000	0.3000	0.3000	0.1429	0.1225
2018-07-31	CTG	0.1276	0.0206	0.000000	0.3000	0.3000	0.1429	0.1367
2018-07-31	MBB	0.0754	0.3000	0.3000	0.000000	0.000000	0.1429	0.1779
2018-07-31	ACB	0.0755	0.3000	0.3000	0.000000	0.009195	0.1429	0.1678
2018-07-31	STB	0.0466	0.000000	0.000000	0.1000	0.000000	0.1429	0.1220
2018-07-31	SHB	0.0232	0.0794	0.1000	0.000000	0.2088	0.1429	0.1174
2018-08-31	VCB	0.4389	0.3000	0.3000	0.3000	0.1000	0.1429	0.1619

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2018-08-31	BID	0.2141	0.000000	0.000000	0.3000	0.3000	0.1429	0.1227
2018-08-31	CTG	0.1232	0.0473	0.000000	0.3000	0.3000	0.1429	0.1358
2018-08-31	MBB	0.0797	0.3000	0.3000	0.000000	0.000000	0.1429	0.1751
2018-08-31	ACB	0.0754	0.3000	0.3000	0.000000	0.3000	0.1429	0.1637
2018-08-31	STB	0.0459	0.000000	0.0129	0.1000	0.000000	0.1429	0.1239
2018-08-31	SHB	0.0227	0.0527	0.0871	0.000000	0.000000	0.1429	0.1168
2018-09-30	VCB	0.4175	0.3000	0.3000	0.3000	0.1000	0.1429	0.1608
2018-09-30	BID	0.2494	0.000000	0.000000	0.3000	0.3000	0.1429	0.1246
2018-09-30	CTG	0.1263	0.000000	0.000000	0.3000	0.3000	0.1429	0.1393
2018-09-30	MBB	0.0717	0.3000	0.3000	0.000000	0.000000	0.1429	0.1734
2018-09-30	ACB	0.0734	0.3000	0.3000	0.000000	0.3000	0.1429	0.1629
2018-09-30	STB	0.0405	0.000000	0.0132	0.1000	0.000000	0.1429	0.1231
2018-09-30	SHB	0.0212	0.1000	0.0868	0.000000	0.000000	0.1429	0.1160
2018-10-31	VCB	0.4132	0.3000	0.3000	0.3000	0.3000	0.1429	0.1672
2018-10-31	BID	0.2506	0.000000	0.000000	0.3000	0.1000	0.1429	0.1131
2018-10-31	CTG	0.1258	0.000000	0.000000	0.3000	0.3000	0.1429	0.1376
2018-10-31	MBB	0.0686	0.3000	0.3000	0.000000	0.000000	0.1429	0.1731
2018-10-31	ACB	0.0722	0.3000	0.3000	0.000000	0.3000	0.1429	0.1705
2018-10-31	STB	0.0473	0.000000	0.007476	0.1000	0.000000	0.1429	0.1226
2018-10-31	SHB	0.0224	0.1000	0.0925	0.000000	0.000000	0.1429	0.1159
2018-11-30	VCB	0.4207	0.3000	0.3000	0.3000	0.3000	0.1429	0.1678
2018-11-30	BID	0.2391	0.000000	0.000000	0.3000	0.1000	0.1429	0.1136
2018-11-30	CTG	0.1217	0.000000	0.000000	0.3000	0.3000	0.1429	0.1382
2018-11-30	MBB	0.0741	0.3000	0.3000	0.000000	0.000000	0.1429	0.1744
2018-11-30	ACB	0.0731	0.3000	0.3000	0.000000	0.3000	0.1429	0.1715
2018-11-30	STB	0.0498	0.000000	0.0133	0.1000	0.000000	0.1429	0.1186
2018-11-30	SHB	0.0215	0.1000	0.0867	0.000000	0.000000	0.1429	0.1160
2018-12-31	VCB	0.4159	0.3000	0.3000	0.3000	0.3000	0.1429	0.1687
2018-12-31	BID	0.2527	0.000000	0.000000	0.3000	0.1000	0.1429	0.1124
2018-12-31	CTG	0.1187	0.000000	0.000000	0.3000	0.3000	0.1429	0.1361

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2018-12-31	MBB	0.0724	0.3000	0.3000	0.000000	0.000000	0.1429	0.1781
2018-12-31	ACB	0.0714	0.3000	0.3000	0.000000	0.3000	0.1429	0.1689
2018-12-31	STB	0.0481	0.000000	0.0395	0.0964	0.000000	0.1429	0.1208
2018-12-31	SHB	0.0208	0.1000	0.0605	0.003627	0.000000	0.1429	0.1150
2019-01-31	VCB	0.4093	0.3000	0.3000	0.3000	0.3000	0.1429	0.1707
2019-01-31	BID	0.2819	0.0737	0.000000	0.3000	0.3000	0.1429	0.1133
2019-01-31	CTG	0.1014	0.000000	0.000000	0.3000	0.3000	0.1429	0.1364
2019-01-31	MBB	0.0666	0.3000	0.3000	0.000000	0.000000	0.1429	0.1746
2019-01-31	ACB	0.0722	0.3000	0.3000	0.000000	0.1000	0.1429	0.1678
2019-01-31	STB	0.0482	0.000000	0.0779	0.0547	0.000000	0.1429	0.1247
2019-01-31	SHB	0.0204	0.0263	0.0221	0.0453	0.000000	0.1429	0.1125
2019-02-28	VCB	0.4290	0.3000	0.3000	0.3000	0.3000	0.1429	0.1713
2019-02-28	BID	0.2549	0.1000	0.000000	0.3000	0.3000	0.1429	0.1137
2019-02-28	CTG	0.1023	0.000000	0.000000	0.3000	0.1000	0.1429	0.1313
2019-02-28	MBB	0.0740	0.3000	0.3000	0.000000	0.000000	0.1429	0.1702
2019-02-28	ACB	0.0708	0.3000	0.3000	0.000000	0.000000	0.1429	0.1691
2019-02-28	STB	0.0490	0.000000	0.0939	0.0536	0.3000	0.1429	0.1309
2019-02-28	SHB	0.0201	0.000000	0.006109	0.0464	0.000000	0.1429	0.1134
2019-03-31	VCB	0.4387	0.3000	0.3000	0.3000	0.3000	0.1429	0.1712
2019-03-31	BID	0.2526	0.1000	0.000000	0.3000	0.3000	0.1429	0.1173
2019-03-31	CTG	0.1037	0.000000	0.000000	0.3000	0.000000	0.1429	0.1313
2019-03-31	MBB	0.0697	0.3000	0.3000	0.000000	0.1000	0.1429	0.1654
2019-03-31	ACB	0.0686	0.3000	0.3000	0.000000	0.3000	0.1429	0.1688
2019-03-31	STB	0.0470	0.000000	0.1000	0.0468	0.000000	0.1429	0.1336
2019-03-31	SHB	0.0198	0.000000	0.000000	0.0532	0.000000	0.1429	0.1124
2019-04-30	VCB	0.4487	0.3000	0.3000	0.3000	0.3000	0.1429	0.1710
2019-04-30	BID	0.2558	0.1000	0.000000	0.3000	0.3000	0.1429	0.1175
2019-04-30	CTG	0.1028	0.000000	0.000000	0.3000	0.000000	0.1429	0.1311
2019-04-30	MBB	0.0668	0.3000	0.3000	0.000000	0.1000	0.1429	0.1650
2019-04-30	ACB	0.0645	0.3000	0.3000	0.000000	0.3000	0.1429	0.1690

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2019-04-30	STB	0.0429	0.000000	0.1000	0.0469	0.000000	0.1429	0.1339
2019-04-30	SHB	0.0185	0.000000	0.000000	0.0531	0.000000	0.1429	0.1125
2019-05-31	VCB	0.4573	0.3000	0.3000	0.3000	0.2838	0.1429	0.1678
2019-05-31	BID	0.2511	0.1000	0.000000	0.3000	0.000000	0.1429	0.1172
2019-05-31	CTG	0.0984	0.000000	0.000000	0.3000	0.000000	0.1429	0.1314
2019-05-31	MBB	0.0675	0.3000	0.3000	0.000000	0.1215	0.1429	0.1650
2019-05-31	ACB	0.0646	0.3000	0.3000	0.000000	0.000000	0.1429	0.1720
2019-05-31	STB	0.0424	0.000000	0.1000	0.0551	0.3000	0.1429	0.1344
2019-05-31	SHB	0.0187	0.000000	0.000000	0.0449	0.2947	0.1429	0.1123
2019-06-30	VCB	0.4716	0.3000	0.3000	0.3000	0.3000	0.1429	0.1707
2019-06-30	BID	0.2369	0.1000	0.000000	0.3000	0.3000	0.1429	0.1166
2019-06-30	CTG	0.0979	0.000000	0.000000	0.3000	0.3000	0.1429	0.1302
2019-06-30	MBB	0.0672	0.3000	0.3000	0.000000	0.000000	0.1429	0.1647
2019-06-30	ACB	0.0649	0.3000	0.3000	0.000000	0.1000	0.1429	0.1719
2019-06-30	STB	0.0435	0.000000	0.1000	0.0550	0.000000	0.1429	0.1339
2019-06-30	SHB	0.0181	0.000000	0.000000	0.0450	0.000000	0.1429	0.1119
2019-07-31	VCB	0.4856	0.3000	0.3000	0.3000	0.3000	0.1429	0.1711
2019-07-31	BID	0.2340	0.1000	0.000000	0.3000	0.3000	0.1429	0.1154
2019-07-31	CTG	0.0923	0.000000	0.000000	0.3000	0.3000	0.1429	0.1310
2019-07-31	MBB	0.0662	0.3000	0.3000	0.000000	0.000000	0.1429	0.1641
2019-07-31	ACB	0.0635	0.3000	0.3000	0.000000	0.1000	0.1429	0.1713
2019-07-31	STB	0.0412	0.000000	0.1000	0.0489	0.000000	0.1429	0.1352
2019-07-31	SHB	0.0173	0.000000	0.000000	0.0511	0.000000	0.1429	0.1119
2019-08-31	VCB	0.5005	0.3000	0.3000	0.3000	0.3000	0.1429	0.1705
2019-08-31	BID	0.2374	0.1000	0.000000	0.3000	0.3000	0.1429	0.1156
2019-08-31	CTG	0.0890	0.000000	0.000000	0.3000	0.3000	0.1429	0.1311
2019-08-31	MBB	0.0637	0.3000	0.3000	0.000000	0.1000	0.1429	0.1654
2019-08-31	ACB	0.0590	0.3000	0.3000	0.000000	0.000000	0.1429	0.1708
2019-08-31	STB	0.0352	0.000000	0.1000	0.0525	0.000000	0.1429	0.1348
2019-08-31	SHB	0.0152	0.000000	0.000000	0.0475	0.000000	0.1429	0.1117

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2019-09-30	VCB	0.4834	0.3000	0.3000	0.3000	0.3000	0.1429	0.1709
2019-09-30	BID	0.2584	0.1000	0.000000	0.3000	0.000000	0.1429	0.1146
2019-09-30	CTG	0.0863	0.000000	0.000000	0.3000	0.3000	0.1429	0.1310
2019-09-30	MBB	0.0660	0.3000	0.3000	0.0171	0.1000	0.1429	0.1649
2019-09-30	ACB	0.0573	0.3000	0.3000	0.000000	0.3000	0.1429	0.1728
2019-09-30	STB	0.0341	0.000000	0.1000	0.0339	0.000000	0.1429	0.1342
2019-09-30	SHB	0.0145	0.000000	0.000000	0.0490	0.000000	0.1429	0.1117
2019-10-31	VCB	0.4891	0.3000	0.3000	0.3000	0.3000	0.1429	0.1696
2019-10-31	BID	0.2515	0.1000	0.000000	0.3000	0.3000	0.1429	0.1149
2019-10-31	CTG	0.0866	0.000000	0.000000	0.3000	0.3000	0.1429	0.1307
2019-10-31	MBB	0.0674	0.3000	0.3000	0.0210	0.000000	0.1429	0.1648
2019-10-31	ACB	0.0578	0.3000	0.3000	0.000000	0.1000	0.1429	0.1718
2019-10-31	STB	0.0331	0.000000	0.1000	0.0370	0.000000	0.1429	0.1351
2019-10-31	SHB	0.0145	0.000000	0.000000	0.0420	0.000000	0.1429	0.1132
2019-11-30	VCB	0.4986	0.3000	0.3000	0.3000	0.3000	0.1429	0.1691
2019-11-30	BID	0.2474	0.1000	0.000000	0.3000	0.3000	0.1429	0.1143
2019-11-30	CTG	0.0860	0.000000	0.000000	0.3000	0.3000	0.1429	0.1300
2019-11-30	MBB	0.0649	0.3000	0.2847	0.0305	0.000000	0.1429	0.1635
2019-11-30	ACB	0.0568	0.3000	0.3000	0.000000	0.1000	0.1429	0.1706
2019-11-30	STB	0.0324	0.000000	0.1153	0.0382	0.000000	0.1429	0.1370
2019-11-30	SHB	0.0138	0.000000	0.000000	0.0313	0.000000	0.1429	0.1155
2019-12-31	VCB	0.4949	0.3000	0.3000	0.3000	0.3000	0.1429	0.1704
2019-12-31	BID	0.2604	0.1000	0.000000	0.3000	0.3000	0.1429	0.1139
2019-12-31	CTG	0.0815	0.000000	0.000000	0.3000	0.1000	0.1429	0.1290
2019-12-31	MBB	0.0638	0.3000	0.2803	0.0230	0.3000	0.1429	0.1625
2019-12-31	ACB	0.0556	0.3000	0.3000	0.000000	0.000000	0.1429	0.1696
2019-12-31	STB	0.0309	0.000000	0.1197	0.0208	0.000000	0.1429	0.1368
2019-12-31	SHB	0.0129	0.000000	0.000000	0.0562	0.000000	0.1429	0.1178
2020-01-31	VCB	0.4947	0.3000	0.2900	0.3000	0.3000	0.1429	0.1695
2020-01-31	BID	0.2780	0.1000	0.000000	0.3000	0.3000	0.1429	0.1139

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2020-01-31	CTG	0.0780	0.000000	0.000000	0.3000	0.3000	0.1429	0.1278
2020-01-31	MBB	0.0561	0.3000	0.2693	0.0219	0.000000	0.1429	0.1637
2020-01-31	ACB	0.0514	0.3000	0.3000	0.000000	0.1000	0.1429	0.1674
2020-01-31	STB	0.0288	0.000000	0.1407	0.007210	0.000000	0.1429	0.1414
2020-01-31	SHB	0.0130	0.000000	0.000000	0.0709	0.000000	0.1429	0.1162
2020-02-29	VCB	0.4677	0.3000	0.2768	0.3000	0.3000	0.1429	0.1671
2020-02-29	BID	0.2983	0.2559	0.000000	0.3000	0.3000	0.1429	0.1128
2020-02-29	CTG	0.0877	0.000000	0.000000	0.2880	0.1505	0.1429	0.1274
2020-02-29	MBB	0.0540	0.2804	0.2372	0.0388	0.0154	0.1429	0.1594
2020-02-29	ACB	0.0498	0.1637	0.3000	0.000000	0.2341	0.1429	0.1650
2020-02-29	STB	0.0281	0.000000	0.1859	0.0123	0.000000	0.1429	0.1536
2020-02-29	SHB	0.0145	0.000000	0.000000	0.0609	0.000000	0.1429	0.1147
2020-03-31	VCB	0.4480	0.3000	0.2935	0.3000	0.3000	0.1250	0.1414
2020-03-31	BID	0.2717	0.3000	0.000000	0.3000	0.3000	0.1250	0.0960
2020-03-31	CTG	0.0953	0.000000	0.000000	0.2410	0.000000	0.1250	0.1126
2020-03-31	MBB	0.0545	0.3000	0.1392	0.0503	0.000000	0.1250	0.1334
2020-03-31	ACB	0.0563	0.1000	0.3000	0.000000	0.2979	0.1250	0.1721
2020-03-31	VIB	0.0180	0.000000	0.0618	0.0491	0.1021	0.1250	0.1201
2020-03-31	STB	0.0330	0.000000	0.2055	0.0198	0.000000	0.1250	0.1301
2020-03-31	SHB	0.0231	0.000000	0.000000	0.0397	0.000000	0.1250	0.0944
2020-04-30	VCB	0.4657	0.3000	0.3000	0.3000	0.3000	0.1250	0.1435
2020-04-30	BID	0.2545	0.1912	0.000000	0.3000	0.3000	0.1250	0.0962
2020-04-30	CTG	0.0880	0.000000	0.000000	0.2271	0.1808	0.1250	0.1152
2020-04-30	MBB	0.0501	0.1922	0.1473	0.0697	0.000000	0.1250	0.1363
2020-04-30	ACB	0.0550	0.2517	0.3000	0.000000	0.0319	0.1250	0.1740
2020-04-30	VIB	0.0176	0.000000	0.0626	0.0501	0.1872	0.1250	0.1228
2020-04-30	STB	0.0287	0.000000	0.1901	0.0230	0.000000	0.1250	0.1292
2020-04-30	SHB	0.0405	0.0649	0.000000	0.0301	0.000000	0.1250	0.0828
2020-05-31	VCB	0.4490	0.3000	0.3000	0.3000	0.1437	0.1250	0.1486
2020-05-31	BID	0.2582	0.3000	0.000000	0.3000	0.000000	0.1250	0.1023

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2020-05-31	CTG	0.0899	0.000000	0.000000	0.2975	0.2333	0.1250	0.1162
2020-05-31	MBB	0.0518	0.000000	0.1292	0.0322	0.000000	0.1250	0.1282
2020-05-31	ACB	0.0557	0.1000	0.3000	0.000000	0.1370	0.1250	0.1576
2020-05-31	VIB	0.0176	0.000000	0.003073	0.0344	0.1860	0.1250	0.1270
2020-05-31	STB	0.0314	0.000000	0.000000	0.0360	0.000000	0.1250	0.1206
2020-05-31	SHB	0.0465	0.3000	0.2677	0.000000	0.3000	0.1250	0.0995
2020-06-30	VCB	0.4814	0.3000	0.3000	0.3000	0.0947	0.1250	0.1509
2020-06-30	BID	0.2488	0.3000	0.000000	0.3000	0.000000	0.1250	0.1033
2020-06-30	CTG	0.0866	0.000000	0.000000	0.2470	0.005330	0.1250	0.1159
2020-06-30	MBB	0.0478	0.000000	0.0923	0.0593	0.000000	0.1250	0.1267
2020-06-30	ACB	0.0532	0.1000	0.3000	0.000000	0.3000	0.1250	0.1564
2020-06-30	VIB	0.0168	0.000000	0.0567	0.0346	0.000000	0.1250	0.1289
2020-06-30	STB	0.0304	0.000000	0.000000	0.0592	0.3000	0.1250	0.1162
2020-06-30	SHB	0.0350	0.3000	0.2510	0.000000	0.3000	0.1250	0.1016
2020-07-31	VCB	0.4760	0.3000	0.3000	0.3000	0.000000	0.1250	0.1431
2020-07-31	BID	0.2468	0.3000	0.000000	0.3000	0.000000	0.1250	0.1043
2020-07-31	CTG	0.0870	0.000000	0.000000	0.2327	0.1000	0.1250	0.1164
2020-07-31	MBB	0.0484	0.000000	0.1103	0.0536	0.000000	0.1250	0.1320
2020-07-31	ACB	0.0555	0.1000	0.3000	0.000000	0.3000	0.1250	0.1565
2020-07-31	VIB	0.0189	0.000000	0.0236	0.0410	0.000000	0.1250	0.1301
2020-07-31	STB	0.0333	0.000000	0.000000	0.0727	0.3000	0.1250	0.1163
2020-07-31	SHB	0.0341	0.3000	0.2660	0.000000	0.3000	0.1250	0.1013
2020-08-31	VCB	0.4703	0.3000	0.2393	0.3000	0.000000	0.1111	0.1233
2020-08-31	BID	0.2478	0.2560	0.000000	0.3000	0.000000	0.1111	0.0906
2020-08-31	CTG	0.0889	0.000000	0.000000	0.2643	0.0621	0.1111	0.1013
2020-08-31	MBB	0.0477	0.000000	0.000000	0.0447	0.000000	0.1111	0.1184
2020-08-31	ACB	0.0569	0.004559	0.3000	0.000000	0.2968	0.1111	0.1360
2020-08-31	VIB	0.0203	0.000000	0.000000	0.0307	0.000000	0.1111	0.1125
2020-08-31	STB	0.0325	0.000000	0.000000	0.0603	0.0411	0.1111	0.1019
2020-08-31	SHB	0.0322	0.3000	0.1607	0.000000	0.3000	0.1111	0.0874

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2020-08-31	KLB	0.003515	0.1394	0.3000	0.000000	0.3000	0.1111	0.1285
2020-09-30	VCB	0.4554	0.3000	0.2465	0.3000	0.000000	0.1111	0.1224
2020-09-30	BID	0.2468	0.1441	0.000000	0.3000	0.000000	0.1111	0.0905
2020-09-30	CTG	0.0960	0.000000	0.000000	0.2819	0.000000	0.1111	0.1011
2020-09-30	MBB	0.0484	0.000000	0.000000	0.0425	0.000000	0.1111	0.1184
2020-09-30	ACB	0.0621	0.0609	0.3000	0.000000	0.1000	0.1111	0.1356
2020-09-30	VIB	0.0221	0.000000	0.000000	0.0127	0.000000	0.1111	0.1122
2020-09-30	STB	0.0321	0.000000	0.000000	0.0629	0.3000	0.1111	0.1016
2020-09-30	SHB	0.0338	0.3000	0.1535	0.000000	0.3000	0.1111	0.0867
2020-09-30	KLB	0.003228	0.1950	0.3000	0.000000	0.3000	0.1111	0.1316
2020-10-31	VCB	0.4098	0.3000	0.3000	0.3000	0.000000	0.1000	0.1134
2020-10-31	BID	0.2184	0.1217	0.000000	0.3000	0.000000	0.1000	0.0837
2020-10-31	CTG	0.0881	0.000000	0.000000	0.1947	0.000000	0.1000	0.0914
2020-10-31	MBB	0.0473	0.000000	0.000000	0.000000	0.000000	0.1000	0.1076
2020-10-31	ACB	0.0586	0.000000	0.2403	0.000000	0.1000	0.1000	0.1201
2020-10-31	VPB	0.0817	0.000000	0.000000	0.1801	0.000000	0.1000	0.0860
2020-10-31	VIB	0.0247	0.0432	0.000000	0.000000	0.000000	0.1000	0.1010
2020-10-31	STB	0.0351	0.000000	0.000000	0.0251	0.3000	0.1000	0.0944
2020-10-31	SHB	0.0332	0.3000	0.1597	0.000000	0.3000	0.1000	0.0797
2020-10-31	KLB	0.003084	0.2351	0.3000	0.000000	0.3000	0.1000	0.1227
2020-11-30	VCB	0.4048	0.3000	0.3000	0.3000	0.1058	0.1000	0.1144
2020-11-30	BID	0.2062	0.0137	0.000000	0.3000	0.000000	0.1000	0.0848
2020-11-30	CTG	0.0963	0.000000	0.000000	0.1894	0.000000	0.1000	0.0921
2020-11-30	MBB	0.0491	0.000000	0.000000	0.000000	0.000000	0.1000	0.1076
2020-11-30	ACB	0.0628	0.000000	0.2408	0.000000	0.2646	0.1000	0.1211
2020-11-30	VPB	0.0812	0.000000	0.000000	0.1654	0.0296	0.1000	0.0868
2020-11-30	VIB	0.0293	0.1465	0.000000	0.000000	0.000000	0.1000	0.0971
2020-11-30	STB	0.0339	0.000000	0.000000	0.0452	0.000000	0.1000	0.0917
2020-11-30	SHB	0.0332	0.2700	0.1592	0.000000	0.3000	0.1000	0.0800
2020-11-30	KLB	0.003303	0.2698	0.3000	0.000000	0.3000	0.1000	0.1246

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2020-12-31	VCB	0.4002	0.3000	0.3000	0.2839	0.0384	0.0909	0.1043
2020-12-31	BID	0.1971	0.000000	0.000000	0.3000	0.3000	0.0909	0.0769
2020-12-31	CTG	0.0982	0.000000	0.000000	0.1934	0.1561	0.0909	0.0829
2020-12-31	MBB	0.0495	0.000000	0.000000	0.000000	0.1013	0.0909	0.0975
2020-12-31	ACB	0.0625	0.000000	0.2404	0.000000	0.000000	0.0909	0.1092
2020-12-31	VPB	0.0813	0.000000	0.000000	0.1628	0.1408	0.0909	0.0790
2020-12-31	VIB	0.0261	0.2370	0.000000	0.000000	0.001454	0.0909	0.0861
2020-12-31	STB	0.0329	0.000000	0.000000	0.0599	0.002530	0.0909	0.0843
2020-12-31	SHB	0.0324	0.2522	0.1596	0.000000	0.000000	0.0909	0.0725
2020-12-31	LPB	0.0166	0.000000	0.000000	0.000000	0.2594	0.0909	0.0944
2020-12-31	KLB	0.003180	0.2108	0.3000	0.000000	0.000000	0.0909	0.1128
2021-01-31	VCB	0.3859	0.2693	0.3000	0.2686	0.000000	0.0909	0.1049
2021-01-31	BID	0.2092	0.000000	0.000000	0.3000	0.000000	0.0909	0.0773
2021-01-31	CTG	0.0933	0.000000	0.000000	0.1541	0.000000	0.0909	0.0857
2021-01-31	MBB	0.0508	0.000000	0.000000	0.000000	0.2361	0.0909	0.0963
2021-01-31	ACB	0.0587	0.000000	0.2214	0.000000	0.000000	0.0909	0.1086
2021-01-31	VPB	0.0896	0.000000	0.000000	0.2546	0.000000	0.0909	0.0773
2021-01-31	VIB	0.0278	0.2250	0.000000	0.000000	0.3000	0.0909	0.0854
2021-01-31	STB	0.0345	0.000000	0.000000	0.0226	0.000000	0.0909	0.0852
2021-01-31	SHB	0.0294	0.2057	0.1642	0.000000	0.000000	0.0909	0.0729
2021-01-31	LPB	0.0165	0.000000	0.0144	0.000000	0.1639	0.0909	0.0945
2021-01-31	KLB	0.004227	0.3000	0.3000	0.000000	0.3000	0.0909	0.1118
2021-02-28	VCB	0.3914	0.1895	0.2553	0.2770	0.000000	0.0833	0.0847
2021-02-28	BID	0.1860	0.000000	0.000000	0.2950	0.000000	0.0833	0.0608
2021-02-28	CTG	0.0880	0.000000	0.000000	0.1371	0.000000	0.0833	0.0683
2021-02-28	MBB	0.0554	0.000000	0.000000	0.0263	0.003297	0.0833	0.0757
2021-02-28	ACB	0.0598	0.000000	0.0923	0.000000	0.0869	0.0833	0.0869
2021-02-28	VPB	0.0883	0.000000	0.000000	0.2105	0.000000	0.0833	0.0600
2021-02-28	VIB	0.0290	0.3000	0.000000	0.000000	0.3000	0.0833	0.0670
2021-02-28	STB	0.0369	0.000000	0.000000	0.0540	0.000000	0.0833	0.0668

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2021-02-28	SHB	0.0270	0.2105	0.1219	0.000000	0.000000	0.0833	0.0582
2021-02-28	LPB	0.0180	0.000000	0.000000	0.000000	0.1527	0.0833	0.0749
2021-02-28	BAB	0.0163	0.000000	0.3000	0.000000	0.3000	0.0833	0.2203
2021-02-28	KLB	0.004019	0.3000	0.2305	0.000000	0.1571	0.0833	0.0762
2021-03-31	VCB	0.3558	0.2616	0.3000	0.0225	0.000000	0.0769	0.0799
2021-03-31	BID	0.1750	0.000000	0.000000	0.3000	0.000000	0.0769	0.0563
2021-03-31	CTG	0.0925	0.000000	0.000000	0.3000	0.000000	0.0769	0.0610
2021-03-31	MBB	0.0562	0.000000	0.000000	0.000000	0.000000	0.0769	0.0733
2021-03-31	ACB	0.0634	0.000000	0.0482	0.000000	0.000000	0.0769	0.0791
2021-03-31	VPB	0.1014	0.000000	0.000000	0.3000	0.000000	0.0769	0.0565
2021-03-31	HDB	0.0287	0.000000	0.000000	0.0580	0.000000	0.0769	0.0686
2021-03-31	VIB	0.0313	0.3000	0.000000	0.000000	0.3000	0.0769	0.0637
2021-03-31	STB	0.0345	0.000000	0.000000	0.0195	0.000000	0.0769	0.0632
2021-03-31	SHB	0.0253	0.1357	0.1619	0.000000	0.000000	0.0769	0.0576
2021-03-31	LPB	0.0180	0.000000	0.0150	0.000000	0.1000	0.0769	0.0762
2021-03-31	BAB	0.0146	0.002657	0.3000	0.000000	0.3000	0.0769	0.1974
2021-03-31	KLB	0.003442	0.3000	0.1749	0.000000	0.3000	0.0769	0.0673
2021-04-30	VCB	0.3290	0.0353	0.3000	0.000000	0.000000	0.0769	0.0815
2021-04-30	BID	0.1651	0.000000	0.000000	0.3000	0.000000	0.0769	0.0572
2021-04-30	CTG	0.0955	0.000000	0.000000	0.3000	0.000000	0.0769	0.0618
2021-04-30	MBB	0.0548	0.000000	0.000000	0.000000	0.2566	0.0769	0.0718
2021-04-30	ACB	0.0612	0.000000	0.0401	0.000000	0.000000	0.0769	0.0763
2021-04-30	VPB	0.1084	0.000000	0.000000	0.3000	0.000000	0.0769	0.0539
2021-04-30	HDB	0.0288	0.000000	0.000000	0.0715	0.000000	0.0769	0.0680
2021-04-30	VIB	0.0366	0.3000	0.000000	0.000000	0.3000	0.0769	0.0617
2021-04-30	STB	0.0386	0.000000	0.000000	0.0285	0.000000	0.0769	0.0633
2021-04-30	SHB	0.0392	0.0981	0.1653	0.000000	0.000000	0.0769	0.0578
2021-04-30	LPB	0.0203	0.000000	0.006097	0.000000	0.3000	0.0769	0.0749
2021-04-30	BAB	0.0186	0.2666	0.3000	0.000000	0.0356	0.0769	0.2039
2021-04-30	KLB	0.003893	0.3000	0.1886	0.000000	0.1078	0.0769	0.0679

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2021-05-31	VCB	0.3244	0.004001	0.3000	0.000000	0.0951	0.0769	0.0898
2021-05-31	BID	0.1474	0.000000	0.000000	0.3000	0.000000	0.0769	0.0635
2021-05-31	CTG	0.0907	0.000000	0.000000	0.3000	0.000000	0.0769	0.0676
2021-05-31	MBB	0.0552	0.000000	0.000000	0.000000	0.3000	0.0769	0.0790
2021-05-31	ACB	0.0596	0.000000	0.1324	0.000000	0.000000	0.0769	0.0848
2021-05-31	VPB	0.1328	0.000000	0.000000	0.3000	0.2718	0.0769	0.0590
2021-05-31	HDB	0.0278	0.000000	0.000000	0.0307	0.0330	0.0769	0.0747
2021-05-31	VIB	0.0393	0.3000	0.000000	0.000000	0.3000	0.0769	0.0684
2021-05-31	STB	0.0402	0.000000	0.000000	0.0693	0.000000	0.0769	0.0682
2021-05-31	SHB	0.0391	0.1313	0.0740	0.000000	0.000000	0.0769	0.0544
2021-05-31	LPB	0.0236	0.000000	0.000000	0.000000	0.000000	0.0769	0.0798
2021-05-31	BAB	0.0157	0.3000	0.3000	0.000000	0.000000	0.0769	0.1380
2021-05-31	KLB	0.004379	0.2647	0.1936	0.000000	0.000000	0.0769	0.0728
2021-06-30	VCB	0.2699	0.2497	0.2927	0.0227	0.000000	0.0714	0.0851
2021-06-30	BID	0.1477	0.000000	0.000000	0.2434	0.000000	0.0714	0.0588
2021-06-30	CTG	0.0996	0.000000	0.000000	0.2423	0.000000	0.0714	0.0627
2021-06-30	MBB	0.0583	0.000000	0.000000	0.000000	0.000000	0.0714	0.0731
2021-06-30	ACB	0.0600	0.000000	0.000000	0.000000	0.000000	0.0714	0.0776
2021-06-30	VPB	0.1325	0.000000	0.000000	0.3000	0.1703	0.0714	0.0520
2021-06-30	HDB	0.0291	0.000000	0.000000	0.1129	0.000000	0.0714	0.0676
2021-06-30	VIB	0.0409	0.3000	0.000000	0.0168	0.000000	0.0714	0.0628
2021-06-30	TPB	0.0257	0.000000	0.2302	0.000000	0.000000	0.0714	0.0965
2021-06-30	STB	0.0480	0.000000	0.000000	0.0619	0.000000	0.0714	0.0618
2021-06-30	SHB	0.0402	0.2369	0.0965	0.000000	0.3000	0.0714	0.0496
2021-06-30	LPB	0.0277	0.000000	0.000000	0.000000	0.2172	0.0714	0.0682
2021-06-30	BAB	0.0159	0.0718	0.3000	0.000000	0.1852	0.0714	0.1181
2021-06-30	KLB	0.004547	0.1416	0.0805	0.000000	0.1273	0.0714	0.0661
2021-07-31	VCB	0.3049	0.2884	0.3000	0.1568	0.000000	0.0714	0.0909
2021-07-31	BID	0.1371	0.000000	0.000000	0.1721	0.000000	0.0714	0.0625
2021-07-31	CTG	0.0946	0.000000	0.000000	0.2357	0.000000	0.0714	0.0619

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2021-07-31	MBB	0.0637	0.000000	0.000000	0.0282	0.000000	0.0714	0.0720
2021-07-31	ACB	0.0619	0.000000	0.0117	0.000000	0.000000	0.0714	0.0782
2021-07-31	VPB	0.1240	0.000000	0.000000	0.2206	0.3000	0.0714	0.0548
2021-07-31	HDB	0.0292	0.000000	0.000000	0.1342	0.000000	0.0714	0.0677
2021-07-31	VIB	0.0395	0.3000	0.000000	0.003513	0.000000	0.0714	0.0681
2021-07-31	TPB	0.0241	0.004652	0.1674	0.000000	0.000000	0.0714	0.0851
2021-07-31	STB	0.0415	0.000000	0.000000	0.0488	0.000000	0.0714	0.0602
2021-07-31	SHB	0.0363	0.2805	0.1224	0.000000	0.3000	0.0714	0.0527
2021-07-31	LPB	0.0265	0.000000	0.000000	0.000000	0.000000	0.0714	0.0676
2021-07-31	BAB	0.0126	0.000000	0.3000	0.000000	0.2350	0.0714	0.1115
2021-07-31	KLB	0.003799	0.1264	0.0986	0.000000	0.1650	0.0714	0.0670
2021-08-31	VCB	0.2549	0.3000	0.3000	0.1209	0.000000	0.0667	0.0832
2021-08-31	BID	0.1184	0.000000	0.000000	0.1225	0.000000	0.0667	0.0603
2021-08-31	CTG	0.0791	0.000000	0.000000	0.1977	0.000000	0.0667	0.0598
2021-08-31	MBB	0.0569	0.000000	0.000000	0.0322	0.000000	0.0667	0.0687
2021-08-31	TCB	0.1194	0.000000	0.000000	0.1809	0.0349	0.0667	0.0601
2021-08-31	ACB	0.0623	0.000000	0.0543	0.000000	0.000000	0.0667	0.0762
2021-08-31	VPB	0.1110	0.000000	0.000000	0.1886	0.000000	0.0667	0.0519
2021-08-31	HDB	0.0276	0.000000	0.000000	0.1523	0.000000	0.0667	0.0644
2021-08-31	VIB	0.0330	0.3000	0.000000	0.000000	0.0784	0.0667	0.0651
2021-08-31	TPB	0.0228	0.0224	0.1373	0.000000	0.000000	0.0667	0.0808
2021-08-31	STB	0.0403	0.000000	0.000000	0.004875	0.0322	0.0667	0.0564
2021-08-31	SHB	0.0349	0.3000	0.1202	0.000000	0.000000	0.0667	0.0507
2021-08-31	LPB	0.0251	0.000000	0.000000	0.000000	0.3000	0.0667	0.0652
2021-08-31	BAB	0.0111	0.000000	0.3000	0.000000	0.2545	0.0667	0.0952
2021-08-31	KLB	0.003134	0.0776	0.0882	0.000000	0.3000	0.0667	0.0622
2021-09-30	VCB	0.2678	0.0892	0.3000	0.1499	0.000000	0.0667	0.0794
2021-09-30	BID	0.1163	0.000000	0.000000	0.1197	0.000000	0.0667	0.0599
2021-09-30	CTG	0.0758	0.000000	0.000000	0.2125	0.000000	0.0667	0.0588
2021-09-30	MBB	0.0569	0.000000	0.000000	0.0279	0.3000	0.0667	0.0686

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2021-09-30	TCB	0.1174	0.000000	0.000000	0.1635	0.000000	0.0667	0.0617
2021-09-30	ACB	0.0571	0.000000	0.0655	0.000000	0.0334	0.0667	0.0774
2021-09-30	VPB	0.1161	0.000000	0.000000	0.1994	0.000000	0.0667	0.0524
2021-09-30	HDB	0.0275	0.000000	0.000000	0.1149	0.000000	0.0667	0.0652
2021-09-30	VIB	0.0298	0.3000	0.000000	0.0105	0.3000	0.0667	0.0629
2021-09-30	TPB	0.0230	0.1758	0.1673	0.000000	0.000000	0.0667	0.0828
2021-09-30	STB	0.0382	0.000000	0.000000	0.001780	0.000000	0.0667	0.0573
2021-09-30	SHB	0.0358	0.3000	0.1327	0.000000	0.0666	0.0667	0.0514
2021-09-30	LPB	0.0235	0.1350	0.000000	0.000000	0.3000	0.0667	0.0676
2021-09-30	BAB	0.0116	0.000000	0.3000	0.000000	0.000000	0.0667	0.0940
2021-09-30	KLB	0.003277	0.000000	0.0344	0.000000	0.000000	0.0667	0.0605
2021-10-31	VCB	0.2616	0.1077	0.3000	0.1224	0.000000	0.0667	0.0797
2021-10-31	BID	0.1176	0.000000	0.000000	0.1205	0.000000	0.0667	0.0637
2021-10-31	CTG	0.0724	0.000000	0.000000	0.2223	0.000000	0.0667	0.0592
2021-10-31	MBB	0.0568	0.000000	0.000000	0.0258	0.000000	0.0667	0.0683
2021-10-31	TCB	0.1200	0.000000	0.000000	0.1760	0.000000	0.0667	0.0617
2021-10-31	ACB	0.0563	0.000000	0.0536	0.000000	0.0295	0.0667	0.0755
2021-10-31	VPB	0.1241	0.000000	0.000000	0.2118	0.3000	0.0667	0.0525
2021-10-31	HDB	0.0269	0.000000	0.000000	0.1095	0.000000	0.0667	0.0652
2021-10-31	VIB	0.0290	0.2204	0.000000	0.0117	0.000000	0.0667	0.0610
2021-10-31	TPB	0.0281	0.1184	0.1832	0.000000	0.000000	0.0667	0.0826
2021-10-31	STB	0.0359	0.000000	0.000000	0.000000	0.2512	0.0667	0.0567
2021-10-31	SHB	0.0346	0.3000	0.1361	0.000000	0.3000	0.0667	0.0513
2021-10-31	LPB	0.0220	0.2268	0.000000	0.000000	0.1193	0.0667	0.0669
2021-10-31	BAB	0.0116	0.000000	0.3000	0.000000	0.000000	0.0667	0.0940
2021-10-31	KLB	0.003333	0.0267	0.0271	0.000000	0.000000	0.0667	0.0618
2021-11-30	VCB	0.2559	0.0897	0.3000	0.1076	0.000000	0.0667	0.0798
2021-11-30	BID	0.1201	0.000000	0.000000	0.1223	0.000000	0.0667	0.0639
2021-11-30	CTG	0.0728	0.000000	0.000000	0.2327	0.000000	0.0667	0.0592
2021-11-30	MBB	0.0563	0.000000	0.000000	0.0256	0.000000	0.0667	0.0686

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2021-11-30	TCB	0.1211	0.000000	0.000000	0.1751	0.000000	0.0667	0.0624
2021-11-30	ACB	0.0571	0.000000	0.0583	0.000000	0.000000	0.0667	0.0756
2021-11-30	VPB	0.1251	0.000000	0.000000	0.2142	0.000000	0.0667	0.0526
2021-11-30	HDB	0.0259	0.000000	0.000000	0.1128	0.000000	0.0667	0.0653
2021-11-30	VIB	0.0295	0.3000	0.000000	0.009712	0.000000	0.0667	0.0610
2021-11-30	TPB	0.0294	0.3000	0.1785	0.000000	0.000000	0.0667	0.0789
2021-11-30	STB	0.0360	0.000000	0.000000	0.000000	0.2870	0.0667	0.0577
2021-11-30	SHB	0.0353	0.3000	0.1364	0.000000	0.3000	0.0667	0.0513
2021-11-30	LPB	0.0205	0.0103	0.000000	0.000000	0.3000	0.0667	0.0668
2021-11-30	BAB	0.0111	0.000000	0.3000	0.000000	0.000000	0.0667	0.0943
2021-11-30	KLB	0.003884	0.000000	0.0268	0.000000	0.1130	0.0667	0.0626
2021-12-31	VCB	0.2501	0.1985	0.3000	0.1432	0.000000	0.0667	0.0806
2021-12-31	BID	0.1244	0.000000	0.000000	0.1221	0.000000	0.0667	0.0651
2021-12-31	CTG	0.0750	0.000000	0.000000	0.2083	0.000000	0.0667	0.0599
2021-12-31	MBB	0.0555	0.000000	0.000000	0.0434	0.0892	0.0667	0.0679
2021-12-31	TCB	0.1177	0.000000	0.000000	0.1769	0.1132	0.0667	0.0619
2021-12-31	ACB	0.0558	0.000000	0.0863	0.000000	0.3000	0.0667	0.0765
2021-12-31	VPB	0.1200	0.000000	0.000000	0.1753	0.0788	0.0667	0.0543
2021-12-31	HDB	0.0303	0.000000	0.000000	0.0773	0.1105	0.0667	0.0666
2021-12-31	VIB	0.0326	0.1986	0.000000	0.0249	0.000000	0.0667	0.0609
2021-12-31	TPB	0.0286	0.3000	0.1517	0.000000	0.0962	0.0667	0.0775
2021-12-31	STB	0.0379	0.000000	0.000000	0.0287	0.0695	0.0667	0.0572
2021-12-31	SHB	0.0355	0.3000	0.1523	0.000000	0.000000	0.0667	0.0520
2021-12-31	LPB	0.0214	0.000000	0.000000	0.000000	0.1425	0.0667	0.0667
2021-12-31	BAB	0.0115	0.000000	0.3000	0.000000	0.000000	0.0667	0.0927
2021-12-31	KLB	0.003745	0.002853	0.009623	0.000000	0.000000	0.0667	0.0601
2022-01-31	VCB	0.2521	0.1626	0.3000	0.1576	0.000000	0.0667	0.0808
2022-01-31	BID	0.1287	0.000000	0.000000	0.1298	0.000000	0.0667	0.0652
2022-01-31	CTG	0.0761	0.000000	0.000000	0.2055	0.000000	0.0667	0.0599
2022-01-31	MBB	0.0542	0.000000	0.000000	0.0477	0.000000	0.0667	0.0682

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2022-01-31	TCB	0.1113	0.000000	0.000000	0.1763	0.3000	0.0667	0.0621
2022-01-31	ACB	0.0566	0.000000	0.0833	0.000000	0.3000	0.0667	0.0768
2022-01-31	VPB	0.1116	0.000000	0.000000	0.1624	0.004191	0.0667	0.0545
2022-01-31	HDB	0.0300	0.000000	0.000000	0.0555	0.000000	0.0667	0.0654
2022-01-31	VIB	0.0357	0.2374	0.000000	0.0250	0.000000	0.0667	0.0606
2022-01-31	TPB	0.0344	0.3000	0.1534	0.000000	0.3000	0.0667	0.0779
2022-01-31	STB	0.0404	0.000000	0.000000	0.0402	0.0464	0.0667	0.0572
2022-01-31	SHB	0.0330	0.3000	0.1530	0.000000	0.000000	0.0667	0.0523
2022-01-31	LPB	0.0209	0.000000	0.000000	0.000000	0.000000	0.0667	0.0667
2022-01-31	BAB	0.0109	0.000000	0.3000	0.000000	0.000000	0.0667	0.0922
2022-01-31	KLB	0.003886	0.000000	0.0103	0.000000	0.0494	0.0667	0.0604
2022-02-28	VCB	0.2579	0.1320	0.3000	0.1035	0.3000	0.0667	0.0811
2022-02-28	BID	0.1506	0.000000	0.000000	0.1603	0.0113	0.0667	0.0655
2022-02-28	CTG	0.0750	0.000000	0.000000	0.2499	0.000000	0.0667	0.0622
2022-02-28	MBB	0.0572	0.000000	0.000000	0.0635	0.2426	0.0667	0.0695
2022-02-28	TCB	0.1064	0.000000	0.000000	0.1753	0.000000	0.0667	0.0619
2022-02-28	ACB	0.0513	0.000000	0.0859	0.000000	0.3000	0.0667	0.0769
2022-02-28	VPB	0.1035	0.000000	0.000000	0.1624	0.0604	0.0667	0.0544
2022-02-28	HDB	0.0273	0.000000	0.000000	0.0564	0.0857	0.0667	0.0653
2022-02-28	VIB	0.0325	0.2680	0.000000	0.007846	0.000000	0.0667	0.0609
2022-02-28	TPB	0.0319	0.3000	0.1457	0.000000	0.000000	0.0667	0.0743
2022-02-28	STB	0.0413	0.000000	0.000000	0.0207	0.000000	0.0667	0.0570
2022-02-28	SHB	0.0301	0.3000	0.1547	0.000000	0.000000	0.0667	0.0522
2022-02-28	LPB	0.0212	0.000000	0.000000	0.000000	0.000000	0.0667	0.0665
2022-02-28	BAB	0.009849	0.000000	0.3000	0.000000	0.000000	0.0667	0.0918
2022-02-28	KLB	0.003886	0.000000	0.0137	0.000000	0.000000	0.0667	0.0606
2022-03-31	VCB	0.2559	0.1572	0.3000	0.1186	0.0377	0.0667	0.0803
2022-03-31	BID	0.1434	0.000000	0.000000	0.1705	0.000000	0.0667	0.0625
2022-03-31	CTG	0.0705	0.000000	0.000000	0.2248	0.000000	0.0667	0.0623
2022-03-31	MBB	0.0611	0.000000	0.000000	0.0968	0.000000	0.0667	0.0691

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2022-03-31	TCB	0.1062	0.000000	0.000000	0.1589	0.1735	0.0667	0.0622
2022-03-31	ACB	0.0527	0.000000	0.0856	0.000000	0.000000	0.0667	0.0773
2022-03-31	VPB	0.1122	0.000000	0.000000	0.1609	0.000000	0.0667	0.0547
2022-03-31	HDB	0.0265	0.000000	0.000000	0.0219	0.000000	0.0667	0.0656
2022-03-31	VIB	0.0336	0.2428	0.000000	0.000000	0.000000	0.0667	0.0614
2022-03-31	TPB	0.0335	0.3000	0.1476	0.000000	0.3000	0.0667	0.0747
2022-03-31	STB	0.0399	0.000000	0.000000	0.0475	0.000000	0.0667	0.0569
2022-03-31	SHB	0.0303	0.3000	0.1559	0.000000	0.000000	0.0667	0.0524
2022-03-31	LPB	0.0201	0.000000	0.000000	0.000000	0.000000	0.0667	0.0673
2022-03-31	BAB	0.0101	0.000000	0.3000	0.000000	0.2720	0.0667	0.0922
2022-03-31	KLB	0.003959	0.000000	0.0109	0.000000	0.2168	0.0667	0.0613
2022-04-30	VCB	0.2539	0.1030	0.3000	0.1012	0.3000	0.0667	0.0803
2022-04-30	BID	0.1458	0.000000	0.000000	0.1807	0.3000	0.0667	0.0620
2022-04-30	CTG	0.0704	0.000000	0.000000	0.2425	0.000000	0.0667	0.0616
2022-04-30	MBB	0.0597	0.000000	0.000000	0.0811	0.000000	0.0667	0.0697
2022-04-30	TCB	0.1066	0.000000	0.000000	0.1659	0.000000	0.0667	0.0622
2022-04-30	ACB	0.0524	0.000000	0.0840	0.000000	0.000000	0.0667	0.0775
2022-04-30	VPB	0.1121	0.000000	0.000000	0.1586	0.000000	0.0667	0.0550
2022-04-30	HDB	0.0267	0.000000	0.000000	0.0227	0.000000	0.0667	0.0655
2022-04-30	VIB	0.0344	0.2970	0.000000	0.000000	0.000000	0.0667	0.0615
2022-04-30	TPB	0.0325	0.3000	0.1522	0.000000	0.2920	0.0667	0.0751
2022-04-30	STB	0.0393	0.000000	0.000000	0.0473	0.000000	0.0667	0.0567
2022-04-30	SHB	0.0306	0.3000	0.1539	0.000000	0.000000	0.0667	0.0525
2022-04-30	LPB	0.0196	0.000000	0.000000	0.000000	0.000000	0.0667	0.0662
2022-04-30	BAB	0.0103	0.000000	0.3000	0.000000	0.000000	0.0667	0.0925
2022-04-30	KLB	0.005605	0.000000	0.009866	0.000000	0.1080	0.0667	0.0617
2022-05-31	VCB	0.2728	0.000000	0.3000	0.1001	0.3000	0.0667	0.0816
2022-05-31	BID	0.1362	0.000000	0.000000	0.1717	0.2586	0.0667	0.0628
2022-05-31	CTG	0.0656	0.000000	0.000000	0.2264	0.000000	0.0667	0.0620
2022-05-31	MBB	0.0589	0.000000	0.000000	0.0974	0.000000	0.0667	0.0696

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2022-05-31	TCB	0.1032	0.000000	0.000000	0.1623	0.0411	0.0667	0.0627
2022-05-31	ACB	0.0565	0.000000	0.0914	0.000000	0.000000	0.0667	0.0776
2022-05-31	VPB	0.1206	0.000000	0.000000	0.1763	0.000000	0.0667	0.0553
2022-05-31	HDB	0.0260	0.000000	0.000000	0.0331	0.000000	0.0667	0.0658
2022-05-31	VIB	0.0343	0.2846	0.000000	0.000000	0.000000	0.0667	0.0618
2022-05-31	TPB	0.0297	0.3000	0.1268	0.000000	0.3000	0.0667	0.0748
2022-05-31	STB	0.0375	0.000000	0.000000	0.0328	0.000000	0.0667	0.0569
2022-05-31	SHB	0.0254	0.2978	0.1424	0.000000	0.000000	0.0667	0.0527
2022-05-31	LPB	0.0179	0.000000	0.000000	0.000000	0.000000	0.0667	0.0661
2022-05-31	BAB	0.0100	0.000000	0.3000	0.000000	0.1003	0.0667	0.0930
2022-05-31	KLB	0.005406	0.1176	0.0394	0.000000	0.000000	0.0667	0.0573
2022-06-30	VCB	0.2868	0.0129	0.3000	0.0680	0.2977	0.0667	0.0833
2022-06-30	BID	0.1391	0.000000	0.000000	0.1939	0.000000	0.0667	0.0625
2022-06-30	CTG	0.0711	0.000000	0.000000	0.2565	0.000000	0.0667	0.0619
2022-06-30	MBB	0.0592	0.000000	0.000000	0.1149	0.000000	0.0667	0.0699
2022-06-30	TCB	0.0940	0.000000	0.000000	0.1560	0.000000	0.0667	0.0633
2022-06-30	ACB	0.0588	0.000000	0.1876	0.000000	0.000000	0.0667	0.0793
2022-06-30	VPB	0.1098	0.0394	0.000000	0.1565	0.000000	0.0667	0.0568
2022-06-30	HDB	0.0289	0.000000	0.000000	0.0446	0.000000	0.0667	0.0664
2022-06-30	VIB	0.0311	0.3000	0.000000	0.000000	0.000000	0.0667	0.0621
2022-06-30	TPB	0.0308	0.2140	0.0480	0.000000	0.1023	0.0667	0.0721
2022-06-30	STB	0.0326	0.000000	0.000000	0.009582	0.000000	0.0667	0.0570
2022-06-30	SHB	0.0249	0.2354	0.1054	0.000000	0.000000	0.0667	0.0505
2022-06-30	LPB	0.0177	0.000000	0.000000	0.000000	0.000000	0.0667	0.0649
2022-06-30	BAB	0.0105	0.000000	0.3000	0.000000	0.3000	0.0667	0.0924
2022-06-30	KLB	0.004839	0.1984	0.0590	0.000000	0.3000	0.0667	0.0575
2022-07-31	VCB	0.2919	0.0142	0.3000	0.0758	0.3000	0.0667	0.0840
2022-07-31	BID	0.1417	0.000000	0.000000	0.1993	0.3000	0.0667	0.0636
2022-07-31	CTG	0.0716	0.000000	0.000000	0.2355	0.000000	0.0667	0.0627
2022-07-31	MBB	0.0553	0.000000	0.000000	0.0959	0.1724	0.0667	0.0702

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2022-07-31	TCB	0.0965	0.000000	0.000000	0.1723	0.1880	0.0667	0.0622
2022-07-31	ACB	0.0600	0.1811	0.2055	0.000000	0.000000	0.0667	0.0801
2022-07-31	VPB	0.1103	0.000000	0.000000	0.1733	0.000000	0.0667	0.0558
2022-07-31	HDB	0.0285	0.000000	0.000000	0.000000	0.000000	0.0667	0.0672
2022-07-31	VIB	0.0267	0.3000	0.000000	0.000000	0.0397	0.0667	0.0607
2022-07-31	TPB	0.0278	0.0830	0.0400	0.000000	0.000000	0.0667	0.0732
2022-07-31	STB	0.0337	0.000000	0.000000	0.0478	0.000000	0.0667	0.0552
2022-07-31	SHB	0.0243	0.2760	0.1079	0.000000	0.000000	0.0667	0.0508
2022-07-31	LPB	0.0165	0.000000	0.000000	0.000000	0.000000	0.0667	0.0656
2022-07-31	BAB	0.0104	0.000000	0.3000	0.000000	0.000000	0.0667	0.0929
2022-07-31	KLB	0.004721	0.1457	0.0466	0.000000	0.000000	0.0667	0.0560
2022-08-31	VCB	0.2791	0.000000	0.3000	0.000000	0.3000	0.0667	0.0848
2022-08-31	BID	0.1511	0.000000	0.000000	0.2189	0.3000	0.0667	0.0642
2022-08-31	CTG	0.0713	0.000000	0.000000	0.2406	0.000000	0.0667	0.0634
2022-08-31	MBB	0.0562	0.000000	0.000000	0.1533	0.1920	0.0667	0.0692
2022-08-31	TCB	0.0985	0.000000	0.000000	0.1677	0.1321	0.0667	0.0637
2022-08-31	ACB	0.0589	0.2561	0.2423	0.000000	0.000000	0.0667	0.0806
2022-08-31	VPB	0.1045	0.000000	0.000000	0.1689	0.000000	0.0667	0.0562
2022-08-31	HDB	0.0275	0.000000	0.000000	0.000000	0.000000	0.0667	0.0674
2022-08-31	VIB	0.0317	0.3000	0.000000	0.0199	0.000000	0.0667	0.0586
2022-08-31	TPB	0.0261	0.000000	0.000000	0.000000	0.0759	0.0667	0.0711
2022-08-31	STB	0.0370	0.000000	0.000000	0.0307	0.000000	0.0667	0.0560
2022-08-31	SHB	0.0253	0.2921	0.1046	0.000000	0.000000	0.0667	0.0511
2022-08-31	LPB	0.0184	0.000000	0.000000	0.000000	0.000000	0.0667	0.0648
2022-08-31	BAB	0.0102	0.000000	0.3000	0.000000	0.000000	0.0667	0.0928
2022-08-31	KLB	0.004285	0.1518	0.0532	0.000000	0.000000	0.0667	0.0562
2022-09-30	VCB	0.2911	0.000000	0.3000	0.000000	0.3000	0.0625	0.0818
2022-09-30	BID	0.1503	0.000000	0.000000	0.2120	0.3000	0.0625	0.0605
2022-09-30	CTG	0.0687	0.000000	0.000000	0.2188	0.000000	0.0625	0.0597
2022-09-30	MBB	0.0579	0.000000	0.000000	0.1399	0.2479	0.0625	0.0651

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2022-09-30	TCB	0.0939	0.000000	0.000000	0.1538	0.000000	0.0625	0.0598
2022-09-30	ACB	0.0548	0.2514	0.2014	0.000000	0.1521	0.0625	0.0758
2022-09-30	VPB	0.1074	0.000000	0.000000	0.1918	0.000000	0.0625	0.0529
2022-09-30	HDB	0.0282	0.000000	0.000000	0.0295	0.000000	0.0625	0.0638
2022-09-30	VIB	0.0276	0.3000	0.000000	0.000000	0.000000	0.0625	0.0541
2022-09-30	TPB	0.0254	0.000000	0.000000	0.000000	0.000000	0.0625	0.0668
2022-09-30	STB	0.0345	0.000000	0.000000	0.0542	0.000000	0.0625	0.0523
2022-09-30	SHB	0.0250	0.3000	0.0878	0.000000	0.000000	0.0625	0.0481
2022-09-30	LPB	0.0177	0.000000	0.000000	0.000000	0.000000	0.0625	0.0601
2022-09-30	BAB	0.009386	0.000000	0.3000	0.000000	0.000000	0.0625	0.0872
2022-09-30	KLB	0.004279	0.1486	0.0261	0.000000	0.000000	0.0625	0.0528
2022-09-30	VBB	0.003771	0.000000	0.0848	0.000000	0.000000	0.0625	0.0593
2022-10-31	VCB	0.2962	0.000000	0.3000	0.000000	0.3000	0.0625	0.0799
2022-10-31	BID	0.1487	0.000000	0.000000	0.2520	0.2281	0.0625	0.0603
2022-10-31	CTG	0.0659	0.000000	0.000000	0.1356	0.000000	0.0625	0.0595
2022-10-31	MBB	0.0569	0.000000	0.000000	0.1741	0.000000	0.0625	0.0642
2022-10-31	TCB	0.0915	0.000000	0.000000	0.1187	0.000000	0.0625	0.0596
2022-10-31	ACB	0.0576	0.3000	0.1850	0.000000	0.3000	0.0625	0.0757
2022-10-31	VPB	0.1065	0.000000	0.000000	0.2121	0.1719	0.0625	0.0526
2022-10-31	HDB	0.0295	0.000000	0.000000	0.0892	0.000000	0.0625	0.0631
2022-10-31	VIB	0.0284	0.004408	0.000000	0.000000	0.000000	0.0625	0.0537
2022-10-31	TPB	0.0265	0.000000	0.000000	0.000000	0.000000	0.0625	0.0664
2022-10-31	STB	0.0334	0.000000	0.000000	0.0183	0.000000	0.0625	0.0522
2022-10-31	SHB	0.0233	0.3000	0.0843	0.000000	0.000000	0.0625	0.0481
2022-10-31	LPB	0.0169	0.1865	0.000000	0.000000	0.000000	0.0625	0.0603
2022-10-31	BAB	0.0103	0.000000	0.3000	0.000000	0.000000	0.0625	0.0868
2022-10-31	KLB	0.004798	0.2091	0.007534	0.000000	0.000000	0.0625	0.0525
2022-10-31	VBB	0.003665	0.000000	0.1232	0.000000	0.000000	0.0625	0.0652
2022-11-30	VCB	0.3144	0.000000	0.3000	0.000000	0.3000	0.0625	0.0795
2022-11-30	BID	0.1602	0.000000	0.000000	0.2364	0.000000	0.0625	0.0602

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2022-11-30	CTG	0.0738	0.000000	0.000000	0.2595	0.000000	0.0625	0.0587
2022-11-30	MBB	0.0533	0.000000	0.000000	0.1076	0.000000	0.0625	0.0636
2022-11-30	TCB	0.0732	0.000000	0.000000	0.0823	0.000000	0.0625	0.0593
2022-11-30	ACB	0.0614	0.3000	0.2016	0.000000	0.3000	0.0625	0.0760
2022-11-30	VPB	0.1022	0.000000	0.000000	0.2451	0.3000	0.0625	0.0527
2022-11-30	HDB	0.0267	0.000000	0.000000	0.000000	0.1000	0.0625	0.0638
2022-11-30	VIB	0.0267	0.0914	0.000000	0.000000	0.000000	0.0625	0.0540
2022-11-30	TPB	0.0232	0.000000	0.000000	0.000000	0.000000	0.0625	0.0670
2022-11-30	STB	0.0277	0.000000	0.000000	0.0690	0.000000	0.0625	0.0520
2022-11-30	SHB	0.0227	0.2866	0.0608	0.000000	0.000000	0.0625	0.0474
2022-11-30	LPB	0.0163	0.0220	0.000000	0.000000	0.000000	0.0625	0.0590
2022-11-30	BAB	0.009837	0.000000	0.3000	0.000000	0.000000	0.0625	0.0884
2022-11-30	KLB	0.004615	0.3000	0.0333	0.000000	0.000000	0.0625	0.0537
2022-11-30	VBB	0.003873	0.000000	0.1044	0.000000	0.000000	0.0625	0.0647
2022-12-31	VCB	0.3198	0.000000	0.3000	0.000000	0.3000	0.0625	0.0813
2022-12-31	BID	0.1759	0.000000	0.000000	0.1814	0.0686	0.0625	0.0612
2022-12-31	CTG	0.0755	0.000000	0.000000	0.2280	0.0314	0.0625	0.0596
2022-12-31	MBB	0.0484	0.000000	0.000000	0.000000	0.000000	0.0625	0.0634
2022-12-31	TCB	0.0711	0.000000	0.000000	0.0594	0.000000	0.0625	0.0563
2022-12-31	ACB	0.0559	0.3000	0.2040	0.000000	0.3000	0.0625	0.0772
2022-12-31	VPB	0.0969	0.000000	0.000000	0.3000	0.3000	0.0625	0.0531
2022-12-31	HDB	0.0234	0.000000	0.000000	0.000000	0.000000	0.0625	0.0631
2022-12-31	VIB	0.0255	0.1000	0.000000	0.000000	0.000000	0.0625	0.0541
2022-12-31	TPB	0.0228	0.000000	0.000000	0.000000	0.000000	0.0625	0.0651
2022-12-31	STB	0.0317	0.000000	0.000000	0.2311	0.000000	0.0625	0.0505
2022-12-31	SHB	0.0212	0.3000	0.0553	0.000000	0.000000	0.0625	0.0478
2022-12-31	LPB	0.0168	0.000000	0.000000	0.000000	0.000000	0.0625	0.0592
2022-12-31	BAB	0.008824	0.000000	0.3000	0.000000	0.000000	0.0625	0.0880
2022-12-31	KLB	0.002975	0.3000	0.0236	0.000000	0.000000	0.0625	0.0541
2022-12-31	VBB	0.003294	0.000000	0.1171	0.000000	0.000000	0.0625	0.0659

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2023-01-31	VCB	0.3180	0.000000	0.3000	0.000000	0.3000	0.0625	0.0809
2023-01-31	BID	0.1663	0.000000	0.000000	0.2937	0.3000	0.0625	0.0601
2023-01-31	CTG	0.0760	0.000000	0.000000	0.1724	0.3000	0.0625	0.0600
2023-01-31	MBB	0.0478	0.000000	0.000000	0.0834	0.000000	0.0625	0.0640
2023-01-31	TCB	0.0715	0.000000	0.000000	0.0915	0.000000	0.0625	0.0567
2023-01-31	ACB	0.0558	0.3000	0.2073	0.000000	0.1000	0.0625	0.0781
2023-01-31	VPB	0.1041	0.000000	0.000000	0.2384	0.000000	0.0625	0.0538
2023-01-31	HDB	0.0241	0.000000	0.000000	0.000000	0.000000	0.0625	0.0642
2023-01-31	VIB	0.0241	0.2024	0.000000	0.000000	0.000000	0.0625	0.0547
2023-01-31	TPB	0.0219	0.000000	0.000000	0.000000	0.000000	0.0625	0.0655
2023-01-31	STB	0.0359	0.000000	0.000000	0.1207	0.000000	0.0625	0.0499
2023-01-31	SHB	0.0210	0.3000	0.0598	0.000000	0.000000	0.0625	0.0486
2023-01-31	LPB	0.0194	0.1570	0.000000	0.000000	0.000000	0.0625	0.0593
2023-01-31	BAB	0.008560	0.000000	0.3000	0.000000	0.000000	0.0625	0.0885
2023-01-31	KLB	0.002337	0.0405	0.0223	0.000000	0.000000	0.0625	0.0496
2023-01-31	VBB	0.003153	0.000000	0.1106	0.000000	0.000000	0.0625	0.0660
2023-02-28	VCB	0.3182	0.000000	0.3000	0.000000	0.1737	0.0625	0.0813
2023-02-28	BID	0.1693	0.000000	0.000000	0.2694	0.3000	0.0625	0.0607
2023-02-28	CTG	0.0741	0.000000	0.000000	0.1793	0.3000	0.0625	0.0599
2023-02-28	MBB	0.0476	0.000000	0.000000	0.1390	0.000000	0.0625	0.0643
2023-02-28	TCB	0.0708	0.000000	0.000000	0.0745	0.000000	0.0625	0.0567
2023-02-28	ACB	0.0578	0.3000	0.1989	0.000000	0.000000	0.0625	0.0781
2023-02-28	VPB	0.0985	0.0728	0.000000	0.2175	0.000000	0.0625	0.0537
2023-02-28	HDB	0.0246	0.000000	0.000000	0.000000	0.000000	0.0625	0.0642
2023-02-28	VIB	0.0260	0.0272	0.000000	0.000000	0.000000	0.0625	0.0544
2023-02-28	TPB	0.0227	0.000000	0.000000	0.000000	0.000000	0.0625	0.0654
2023-02-28	STB	0.0377	0.000000	0.000000	0.1203	0.000000	0.0625	0.0496
2023-02-28	SHB	0.0203	0.3000	0.0595	0.000000	0.1074	0.0625	0.0485
2023-02-28	LPB	0.0191	0.3000	0.000000	0.000000	0.1189	0.0625	0.0587
2023-02-28	BAB	0.007794	0.000000	0.3000	0.000000	0.000000	0.0625	0.0883

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2023-02-28	KLB	0.002051	0.000000	0.0211	0.000000	0.000000	0.0625	0.0495
2023-02-28	VBB	0.003730	0.000000	0.1206	0.000000	0.000000	0.0625	0.0667
2023-03-31	VCB	0.3399	0.000000	0.3000	0.000000	0.3000	0.0625	0.0804
2023-03-31	BID	0.1738	0.000000	0.000000	0.3000	0.000000	0.0625	0.0610
2023-03-31	CTG	0.0699	0.000000	0.000000	0.1781	0.1902	0.0625	0.0614
2023-03-31	MBB	0.0441	0.000000	0.000000	0.2152	0.000000	0.0625	0.0644
2023-03-31	TCB	0.0670	0.000000	0.000000	0.0450	0.000000	0.0625	0.0573
2023-03-31	ACB	0.0568	0.3000	0.2160	0.000000	0.0985	0.0625	0.0767
2023-03-31	VPB	0.0906	0.000000	0.000000	0.1749	0.000000	0.0625	0.0548
2023-03-31	HDB	0.0236	0.000000	0.000000	0.000000	0.000000	0.0625	0.0640
2023-03-31	VIB	0.0249	0.1384	0.000000	0.000000	0.000000	0.0625	0.0540
2023-03-31	TPB	0.0222	0.000000	0.000000	0.000000	0.000000	0.0625	0.0649
2023-03-31	STB	0.0346	0.000000	0.000000	0.0868	0.1767	0.0625	0.0496
2023-03-31	SHB	0.0189	0.2616	0.0722	0.000000	0.000000	0.0625	0.0496
2023-03-31	LPB	0.0189	0.3000	0.000000	0.000000	0.2345	0.0625	0.0595
2023-03-31	BAB	0.008244	0.000000	0.3000	0.000000	0.000000	0.0625	0.0893
2023-03-31	KLB	0.001908	0.000000	0.0587	0.000000	0.000000	0.0625	0.0517
2023-03-31	VBB	0.004547	0.000000	0.0531	0.000000	0.000000	0.0625	0.0614
2023-04-30	VCB	0.3173	0.006709	0.3000	0.000000	0.3000	0.0625	0.0812
2023-04-30	BID	0.1739	0.000000	0.000000	0.2898	0.1000	0.0625	0.0619
2023-04-30	CTG	0.0711	0.000000	0.000000	0.1729	0.3000	0.0625	0.0607
2023-04-30	MBB	0.0446	0.000000	0.000000	0.1419	0.000000	0.0625	0.0632
2023-04-30	TCB	0.0685	0.000000	0.000000	0.0282	0.000000	0.0625	0.0567
2023-04-30	ACB	0.0556	0.3000	0.2081	0.000000	0.3000	0.0625	0.0764
2023-04-30	VPB	0.1068	0.000000	0.000000	0.2351	0.000000	0.0625	0.0550
2023-04-30	HDB	0.0254	0.000000	0.000000	0.000000	0.000000	0.0625	0.0633
2023-04-30	VIB	0.0248	0.3000	0.000000	0.000000	0.000000	0.0625	0.0534
2023-04-30	TPB	0.0224	0.0474	0.000000	0.000000	0.000000	0.0625	0.0644
2023-04-30	STB	0.0365	0.000000	0.000000	0.1320	0.000000	0.0625	0.0492
2023-04-30	SHB	0.0199	0.0459	0.0808	0.000000	0.000000	0.0625	0.0547

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2023-04-30	LPB	0.0199	0.3000	0.000000	0.000000	0.000000	0.0625	0.0591
2023-04-30	BAB	0.007759	0.000000	0.3000	0.000000	0.000000	0.0625	0.0893
2023-04-30	KLB	0.001787	0.000000	0.0528	0.000000	0.000000	0.0625	0.0514
2023-04-30	VBB	0.003741	0.000000	0.0583	0.000000	0.000000	0.0625	0.0601
2023-05-31	VCB	0.3207	0.1496	0.3000	0.2210	0.3000	0.0625	0.0841
2023-05-31	BID	0.1671	0.000000	0.000000	0.1981	0.1563	0.0625	0.0647
2023-05-31	CTG	0.0717	0.000000	0.000000	0.1652	0.2542	0.0625	0.0642
2023-05-31	MBB	0.0460	0.000000	0.000000	0.0967	0.000000	0.0625	0.0689
2023-05-31	TCB	0.0727	0.000000	0.000000	0.1309	0.000000	0.0625	0.0576
2023-05-31	ACB	0.0550	0.3000	0.3000	0.000000	0.2415	0.0625	0.0857
2023-05-31	VPB	0.1031	0.1308	0.000000	0.1101	0.000000	0.0625	0.0581
2023-05-31	HDB	0.0253	0.0686	0.0306	0.000000	0.000000	0.0625	0.0716
2023-05-31	VIB	0.0248	0.0510	0.000000	0.0340	0.000000	0.0625	0.0524
2023-05-31	TPB	0.0245	0.000000	0.000000	0.000000	0.000000	0.0625	0.0602
2023-05-31	STB	0.0360	0.000000	0.000000	0.0439	0.000576	0.0625	0.0508
2023-05-31	SHB	0.0211	0.000000	0.000000	0.000000	0.000000	0.0625	0.0498
2023-05-31	LPB	0.0185	0.3000	0.000000	0.000000	0.0474	0.0625	0.0554
2023-05-31	BAB	0.007803	0.000000	0.3000	0.000000	0.000000	0.0625	0.0788
2023-05-31	KLB	0.001928	0.000000	0.0642	0.000000	0.000000	0.0625	0.0455
2023-05-31	VBB	0.003893	0.000000	0.005159	0.000000	0.000000	0.0625	0.0521
2023-06-30	VCB	0.3260	0.1399	0.3000	0.2047	0.3000	0.0625	0.0837
2023-06-30	BID	0.1654	0.000000	0.000000	0.2143	0.1070	0.0625	0.0646
2023-06-30	CTG	0.0681	0.000000	0.000000	0.1641	0.1992	0.0625	0.0645
2023-06-30	MBB	0.0454	0.000000	0.000000	0.0847	0.000000	0.0625	0.0698
2023-06-30	TCB	0.0723	0.000000	0.000000	0.1441	0.000000	0.0625	0.0575
2023-06-30	ACB	0.0565	0.3000	0.3000	0.000000	0.1209	0.0625	0.0862
2023-06-30	VPB	0.0979	0.0717	0.000000	0.1060	0.000000	0.0625	0.0583
2023-06-30	HDB	0.0256	0.0113	0.0395	0.000000	0.000000	0.0625	0.0719
2023-06-30	VIB	0.0253	0.1561	0.000000	0.0449	0.000000	0.0625	0.0518
2023-06-30	TPB	0.0252	0.1041	0.000000	0.000000	0.000000	0.0625	0.0591

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2023-06-30	STB	0.0385	0.000000	0.000000	0.0371	0.000000	0.0625	0.0513
2023-06-30	SHB	0.0217	0.000000	0.000000	0.000000	0.000000	0.0625	0.0517
2023-06-30	LPB	0.0187	0.2169	0.000000	0.000000	0.2730	0.0625	0.0556
2023-06-30	BAB	0.007864	0.000000	0.3000	0.000000	0.000000	0.0625	0.0778
2023-06-30	KLB	0.001920	0.000000	0.0605	0.000000	0.000000	0.0625	0.0447
2023-06-30	VBB	0.003668	0.000000	0.000000	0.000000	0.000000	0.0625	0.0514
2023-07-31	VCB	0.3314	0.0114	0.3000	0.005463	0.2569	0.0625	0.0916
2023-07-31	BID	0.1557	0.000000	0.000000	0.3000	0.000000	0.0625	0.0643
2023-07-31	CTG	0.0686	0.000000	0.000000	0.1448	0.0484	0.0625	0.0639
2023-07-31	MBB	0.0483	0.000000	0.000000	0.2989	0.000000	0.0625	0.0686
2023-07-31	TCB	0.0746	0.000000	0.000000	0.0265	0.000000	0.0625	0.0582
2023-07-31	ACB	0.0561	0.3000	0.3000	0.000000	0.1462	0.0625	0.0858
2023-07-31	VPB	0.0961	0.0189	0.000000	0.1300	0.000000	0.0625	0.0576
2023-07-31	HDB	0.0247	0.000000	0.0715	0.000000	0.000000	0.0625	0.0726
2023-07-31	VIB	0.0266	0.2433	0.000000	0.0301	0.000000	0.0625	0.0511
2023-07-31	TPB	0.0241	0.1399	0.000000	0.000000	0.000000	0.0625	0.0593
2023-07-31	STB	0.0396	0.000000	0.000000	0.0643	0.2484	0.0625	0.0504
2023-07-31	SHB	0.0220	0.000000	0.000000	0.000000	0.000000	0.0625	0.0513
2023-07-31	LPB	0.0187	0.2864	0.000000	0.000000	0.3000	0.0625	0.0548
2023-07-31	BAB	0.007568	0.000000	0.3000	0.000000	0.000000	0.0625	0.0761
2023-07-31	KLB	0.002190	0.000000	0.0285	0.000000	0.000000	0.0625	0.0441
2023-07-31	VBB	0.003641	0.000000	0.000000	0.000000	0.000000	0.0625	0.0503
2023-08-31	VCB	0.3341	0.1796	0.3000	0.0799	0.3000	0.0625	0.0921
2023-08-31	BID	0.1580	0.000000	0.000000	0.2725	0.1262	0.0625	0.0645
2023-08-31	CTG	0.0650	0.000000	0.000000	0.1306	0.1339	0.0625	0.0640
2023-08-31	MBB	0.0483	0.000000	0.000000	0.2628	0.000000	0.0625	0.0681
2023-08-31	TCB	0.0737	0.000000	0.000000	0.0343	0.000000	0.0625	0.0581
2023-08-31	ACB	0.0544	0.3000	0.3000	0.000000	0.3000	0.0625	0.0858
2023-08-31	VPB	0.1000	0.1842	0.000000	0.1127	0.000000	0.0625	0.0591
2023-08-31	HDB	0.0247	0.000000	0.0799	0.000000	0.0156	0.0625	0.0725

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2023-08-31	VIB	0.0263	0.1802	0.000000	0.0420	0.000000	0.0625	0.0508
2023-08-31	TPB	0.0234	0.0275	0.000000	0.000000	0.000000	0.0625	0.0592
2023-08-31	STB	0.0359	0.000000	0.000000	0.0651	0.000000	0.0625	0.0502
2023-08-31	SHB	0.0244	0.000000	0.000000	0.000000	0.000000	0.0625	0.0514
2023-08-31	LPB	0.0188	0.1285	0.000000	0.000000	0.1243	0.0625	0.0547
2023-08-31	BAB	0.007053	0.000000	0.3000	0.000000	0.000000	0.0625	0.0759
2023-08-31	KLB	0.002055	0.000000	0.0201	0.000000	0.000000	0.0625	0.0433
2023-08-31	VBB	0.003862	0.000000	0.000000	0.000000	0.000000	0.0625	0.0503
2023-09-30	VCB	0.3246	0.3000	0.3000	0.1083	0.2461	0.0588	0.0882
2023-09-30	BID	0.1570	0.000000	0.000000	0.2692	0.1801	0.0588	0.0615
2023-09-30	CTG	0.0702	0.000000	0.000000	0.1241	0.1839	0.0588	0.0613
2023-09-30	MBB	0.0473	0.000000	0.000000	0.2235	0.1466	0.0588	0.0656
2023-09-30	TCB	0.0740	0.000000	0.000000	0.0486	0.0507	0.0588	0.0560
2023-09-30	ACB	0.0535	0.3000	0.3000	0.000000	0.000000	0.0588	0.0824
2023-09-30	VPB	0.0945	0.1973	0.000000	0.1171	0.0636	0.0588	0.0563
2023-09-30	HDB	0.0240	0.000000	0.0792	0.000000	0.000000	0.0588	0.0695
2023-09-30	VIB	0.0257	0.0604	0.000000	0.0567	0.000000	0.0588	0.0486
2023-09-30	TPB	0.0245	0.000000	0.000000	0.000000	0.000000	0.0588	0.0567
2023-09-30	STB	0.0404	0.000000	0.000000	0.0526	0.0744	0.0588	0.0483
2023-09-30	SHB	0.0241	0.000000	0.000000	0.000000	0.000000	0.0588	0.0487
2023-09-30	LPB	0.0237	0.1423	0.000000	0.000000	0.0545	0.0588	0.0523
2023-09-30	BAB	0.006994	0.000000	0.3000	0.000000	0.000000	0.0588	0.0727
2023-09-30	KLB	0.002039	0.000000	0.0208	0.000000	0.000000	0.0588	0.0415
2023-09-30	VBB	0.003609	0.000000	0.000000	0.000000	0.000000	0.0588	0.0476
2023-09-30	BVB	0.003848	0.000000	0.000000	0.000000	0.000000	0.0588	0.0426
2023-10-31	VCB	0.3292	0.2657	0.3000	0.0622	0.1762	0.0588	0.0874
2023-10-31	BID	0.1532	0.000000	0.000000	0.3000	0.000000	0.0588	0.0613
2023-10-31	CTG	0.0671	0.000000	0.000000	0.0771	0.000000	0.0588	0.0627
2023-10-31	MBB	0.0489	0.000000	0.000000	0.2902	0.000000	0.0588	0.0655
2023-10-31	TCB	0.0746	0.000000	0.000000	0.000000	0.000000	0.0588	0.0564

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2023-10-31	ACB	0.0534	0.3000	0.3000	0.000000	0.000000	0.0588	0.0880
2023-10-31	VPB	0.1010	0.0790	0.000000	0.1401	0.1898	0.0588	0.0558
2023-10-31	HDB	0.0255	0.000000	0.0950	0.000000	0.0578	0.0588	0.0708
2023-10-31	VIB	0.0252	0.000000	0.000000	0.0450	0.000000	0.0588	0.0490
2023-10-31	TPB	0.0230	0.0554	0.000000	0.000000	0.000000	0.0588	0.0560
2023-10-31	STB	0.0392	0.000000	0.000000	0.0723	0.0362	0.0588	0.0474
2023-10-31	SHB	0.0222	0.000000	0.000000	0.000000	0.3000	0.0588	0.0488
2023-10-31	LPB	0.0211	0.3000	0.000000	0.000000	0.0981	0.0588	0.0499
2023-10-31	BAB	0.007119	0.000000	0.3000	0.000000	0.1336	0.0588	0.0716
2023-10-31	KLB	0.001934	0.000000	0.004955	0.000000	0.008312	0.0588	0.0408
2023-10-31	VBB	0.003432	0.000000	0.000000	0.0132	0.000000	0.0588	0.0467
2023-10-31	BVB	0.003898	0.000000	0.000000	0.000000	0.000000	0.0588	0.0420
2023-11-30	VCB	0.3473	0.2357	0.3000	0.0535	0.3000	0.0588	0.0870
2023-11-30	BID	0.1477	0.000000	0.000000	0.3000	0.000000	0.0588	0.0608
2023-11-30	CTG	0.0658	0.000000	0.000000	0.0942	0.000000	0.0588	0.0619
2023-11-30	MBB	0.0481	0.000000	0.000000	0.2980	0.000000	0.0588	0.0659
2023-11-30	TCB	0.0653	0.000000	0.000000	0.000000	0.000000	0.0588	0.0564
2023-11-30	ACB	0.0556	0.3000	0.3000	0.000000	0.000000	0.0588	0.0877
2023-11-30	VPB	0.0990	0.3000	0.000000	0.1287	0.000000	0.0588	0.0557
2023-11-30	HDB	0.0269	0.000000	0.0997	0.000000	0.000000	0.0588	0.0716
2023-11-30	VIB	0.0248	0.000000	0.000000	0.0483	0.000000	0.0588	0.0506
2023-11-30	TPB	0.0217	0.000000	0.000000	0.000000	0.000000	0.0588	0.0551
2023-11-30	STB	0.0367	0.000000	0.000000	0.0662	0.2945	0.0588	0.0484
2023-11-30	SHB	0.0215	0.000000	0.000000	0.000000	0.1055	0.0588	0.0484
2023-11-30	LPB	0.0238	0.1643	0.000000	0.000000	0.3000	0.0588	0.0499
2023-11-30	BAB	0.006964	0.000000	0.3000	0.000000	0.000000	0.0588	0.0715
2023-11-30	KLB	0.001812	0.000000	0.000273	0.000000	0.000000	0.0588	0.0407
2023-11-30	VBB	0.002997	0.000000	0.000000	0.0111	0.000000	0.0588	0.0465
2023-11-30	BVB	0.004025	0.000000	0.000000	0.000000	0.000000	0.0588	0.0419
2023-12-31	VCB	0.3277	0.3000	0.3000	0.000000	0.2275	0.0526	0.0794

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2023-12-31	BID	0.1554	0.000000	0.000000	0.3000	0.000000	0.0526	0.0549
2023-12-31	CTG	0.0675	0.000000	0.000000	0.1219	0.000000	0.0526	0.0563
2023-12-31	MBB	0.0484	0.000000	0.000000	0.3000	0.000000	0.0526	0.0592
2023-12-31	TCB	0.0672	0.000000	0.000000	0.000000	0.000000	0.0526	0.0493
2023-12-31	ACB	0.0551	0.1706	0.3000	0.000000	0.000000	0.0526	0.0803
2023-12-31	VPB	0.0960	0.2886	0.000000	0.1210	0.000000	0.0526	0.0502
2023-12-31	HDB	0.0272	0.000000	0.0534	0.000000	0.000000	0.0526	0.0652
2023-12-31	VIB	0.0251	0.000000	0.000000	0.0698	0.000000	0.0526	0.0468
2023-12-31	TPB	0.0225	0.000000	0.000000	0.000000	0.000000	0.0526	0.0493
2023-12-31	STB	0.0361	0.000000	0.000000	0.0742	0.2084	0.0526	0.0434
2023-12-31	SHB	0.0222	0.000000	0.000000	0.000000	0.000000	0.0526	0.0437
2023-12-31	LPB	0.0240	0.2408	0.000000	0.000000	0.2641	0.0526	0.0456
2023-12-31	NAB	0.007090	0.000000	0.000000	0.000000	0.3000	0.0526	0.0462
2023-12-31	BAB	0.006786	0.000000	0.3000	0.000000	0.000000	0.0526	0.0645
2023-12-31	KLB	0.001831	0.000000	0.000000	0.000000	0.000000	0.0526	0.0368
2023-12-31	VBB	0.003195	0.000000	0.000000	0.0131	0.000000	0.0526	0.0414
2023-12-31	SGB	0.002582	0.000000	0.0466	0.000000	0.000000	0.0526	0.0490
2023-12-31	BVB	0.004047	0.000000	0.000000	0.000000	0.000000	0.0526	0.0386
2024-01-31	VCB	0.3035	0.2269	0.3000	0.000000	0.3000	0.0526	0.0805
2024-01-31	BID	0.1696	0.000000	0.000000	0.3000	0.000000	0.0526	0.0542
2024-01-31	CTG	0.0680	0.000000	0.000000	0.0706	0.0373	0.0526	0.0569
2024-01-31	MBB	0.0495	0.000000	0.000000	0.2510	0.000000	0.0526	0.0597
2024-01-31	TCB	0.0708	0.000000	0.000000	0.000000	0.000000	0.0526	0.0492
2024-01-31	ACB	0.0587	0.1731	0.3000	0.000000	0.2127	0.0526	0.0821
2024-01-31	VPB	0.0942	0.3000	0.000000	0.1033	0.000000	0.0526	0.0502
2024-01-31	HDB	0.0300	0.000000	0.0877	0.000000	0.000000	0.0526	0.0665
2024-01-31	VIB	0.0257	0.000000	0.000000	0.1365	0.000000	0.0526	0.0461
2024-01-31	TPB	0.0225	0.000000	0.000000	0.000000	0.000000	0.0526	0.0500
2024-01-31	STB	0.0359	0.000000	0.000000	0.1055	0.1501	0.0526	0.0431
2024-01-31	SHB	0.0217	0.000000	0.000000	0.000000	0.000000	0.0526	0.0433

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-01-31	LPB	0.0242	0.3000	0.000000	0.007710	0.3000	0.0526	0.0451
2024-01-31	NAB	0.007629	0.000000	0.000000	0.000000	0.000000	0.0526	0.0459
2024-01-31	BAB	0.006630	0.000000	0.3000	0.000000	0.000000	0.0526	0.0637
2024-01-31	KLB	0.001771	0.000000	0.000000	0.000000	0.000000	0.0526	0.0365
2024-01-31	VBB	0.003253	0.000000	0.000000	0.0254	0.000000	0.0526	0.0406
2024-01-31	SGB	0.002483	0.000000	0.0123	0.000000	0.000000	0.0526	0.0483
2024-01-31	BVB	0.003992	0.000000	0.000000	0.000000	0.000000	0.0526	0.0381
2024-02-29	VCB	0.2994	0.0499	0.3000	0.000000	0.3000	0.0455	0.0692
2024-02-29	BID	0.1669	0.000000	0.000000	0.3000	0.000000	0.0455	0.0474
2024-02-29	CTG	0.0709	0.000000	0.000000	0.0950	0.000000	0.0455	0.0491
2024-02-29	MBB	0.0517	0.000000	0.000000	0.2137	0.000000	0.0455	0.0524
2024-02-29	TCB	0.0688	0.000000	0.000000	0.000000	0.000000	0.0455	0.0459
2024-02-29	ACB	0.0566	0.3000	0.3000	0.000000	0.2734	0.0455	0.0697
2024-02-29	VPB	0.0852	0.0585	0.000000	0.0462	0.000000	0.0455	0.0449
2024-02-29	HDB	0.0288	0.2596	0.0656	0.000000	0.000000	0.0455	0.0564
2024-02-29	VIB	0.0250	0.000000	0.000000	0.1540	0.000000	0.0455	0.0406
2024-02-29	TPB	0.0207	0.000000	0.000000	0.000000	0.000000	0.0455	0.0446
2024-02-29	STB	0.0344	0.000000	0.000000	0.1144	0.1266	0.0455	0.0376
2024-02-29	SHB	0.0208	0.000000	0.000000	0.000000	0.000000	0.0455	0.0373
2024-02-29	LPB	0.0242	0.3000	0.000000	0.0311	0.3000	0.0455	0.0390
2024-02-29	MSB	0.0138	0.000000	0.000000	0.000000	0.000000	0.0455	0.0525
2024-02-29	NAB	0.007003	0.0321	0.000000	0.000000	0.000000	0.0455	0.0394
2024-02-29	BAB	0.006278	0.000000	0.3000	0.000000	0.000000	0.0455	0.0591
2024-02-29	KLB	0.001657	0.000000	0.0344	0.000000	0.000000	0.0455	0.0348
2024-02-29	PGB	0.004209	0.000000	0.000000	0.000000	0.000000	0.0455	0.0264
2024-02-29	VBB	0.003016	0.000000	0.000000	0.0455	0.000000	0.0455	0.0350
2024-02-29	SGB	0.002328	0.000000	0.000000	0.000000	0.000000	0.0455	0.0418
2024-02-29	ABB	0.004669	0.000000	0.000000	0.000000	0.000000	0.0455	0.0433
2024-02-29	BVB	0.003607	0.000000	0.000000	0.000000	0.000000	0.0455	0.0337
2024-03-31	VCB	0.2959	0.2296	0.3000	0.003785	0.3000	0.0435	0.0646

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-03-31	BID	0.1667	0.000000	0.000000	0.2627	0.0459	0.0435	0.0468
2024-03-31	CTG	0.0717	0.000000	0.000000	0.1142	0.000000	0.0435	0.0462
2024-03-31	MBB	0.0516	0.000000	0.000000	0.3000	0.000000	0.0435	0.0480
2024-03-31	TCB	0.0756	0.000000	0.000000	0.000000	0.000000	0.0435	0.0430
2024-03-31	ACB	0.0553	0.3000	0.3000	0.000000	0.3000	0.0435	0.0659
2024-03-31	VPB	0.0786	0.000000	0.000000	0.0558	0.000000	0.0435	0.0429
2024-03-31	HDB	0.0277	0.0816	0.0390	0.000000	0.0541	0.0435	0.0536
2024-03-31	VIB	0.0237	0.000000	0.000000	0.1079	0.000000	0.0435	0.0382
2024-03-31	TPB	0.0207	0.000000	0.000000	0.000000	0.000000	0.0435	0.0425
2024-03-31	STB	0.0326	0.000000	0.000000	0.0921	0.000000	0.0435	0.0355
2024-03-31	SHB	0.0191	0.000000	0.000000	0.000000	0.000000	0.0435	0.0360
2024-03-31	LPB	0.0211	0.2550	0.000000	0.0315	0.3000	0.0435	0.0366
2024-03-31	MSB	0.0145	0.000000	0.000000	0.000000	0.000000	0.0435	0.0506
2024-03-31	OCB	0.0150	0.000000	0.0224	0.000000	0.000000	0.0435	0.0477
2024-03-31	NAB	0.006618	0.0426	0.000000	0.000000	0.000000	0.0435	0.0383
2024-03-31	BAB	0.005691	0.000000	0.3000	0.000000	0.000000	0.0435	0.0553
2024-03-31	KLB	0.001501	0.000000	0.0335	0.000000	0.000000	0.0435	0.0331
2024-03-31	PGB	0.004091	0.0911	0.000000	0.000000	0.000000	0.0435	0.0270
2024-03-31	VBB	0.002738	0.000000	0.000000	0.0320	0.000000	0.0435	0.0331
2024-03-31	SGB	0.002107	0.000000	0.005125	0.000000	0.000000	0.0435	0.0407
2024-03-31	ABB	0.004305	0.000000	0.000000	0.000000	0.000000	0.0435	0.0416
2024-03-31	BVB	0.003334	0.000000	0.000000	0.000000	0.000000	0.0435	0.0329
2024-04-30	VCB	0.2874	0.3000	0.3000	0.0189	0.3000	0.0435	0.0624
2024-04-30	BID	0.1632	0.000000	0.000000	0.2696	0.000000	0.0435	0.0458
2024-04-30	CTG	0.0715	0.000000	0.000000	0.1075	0.000000	0.0435	0.0470
2024-04-30	MBB	0.0540	0.000000	0.000000	0.3000	0.000000	0.0435	0.0482
2024-04-30	TCB	0.0848	0.000000	0.000000	0.0351	0.000000	0.0435	0.0423
2024-04-30	ACB	0.0561	0.3000	0.3000	0.000000	0.3000	0.0435	0.0723
2024-04-30	VPB	0.0777	0.000000	0.0891	0.000000	0.000000	0.0435	0.0469
2024-04-30	HDB	0.0285	0.0837	0.006275	0.000000	0.000000	0.0435	0.0536

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-04-30	VIB	0.0266	0.000000	0.000000	0.0931	0.000000	0.0435	0.0394
2024-04-30	TPB	0.0198	0.000000	0.000000	0.000000	0.000000	0.0435	0.0416
2024-04-30	STB	0.0325	0.000000	0.000000	0.1492	0.1000	0.0435	0.0350
2024-04-30	SHB	0.0184	0.000000	0.000000	0.000000	0.000000	0.0435	0.0355
2024-04-30	LPB	0.0216	0.1203	0.000000	0.000000	0.3000	0.0435	0.0364
2024-04-30	MSB	0.0132	0.000000	0.000000	0.000000	0.000000	0.0435	0.0486
2024-04-30	OCB	0.0147	0.000000	0.000000	0.000000	0.000000	0.0435	0.0471
2024-04-30	NAB	0.006653	0.0988	0.000000	0.000000	0.000000	0.0435	0.0378
2024-04-30	BAB	0.005669	0.000000	0.3000	0.000000	0.000000	0.0435	0.0545
2024-04-30	KLB	0.001483	0.000000	0.004632	0.000000	0.000000	0.0435	0.0325
2024-04-30	PGB	0.003821	0.0972	0.000000	0.000000	0.000000	0.0435	0.0266
2024-04-30	VBB	0.002623	0.000000	0.000000	0.0267	0.000000	0.0435	0.0332
2024-04-30	SGB	0.002241	0.000000	0.000000	0.000000	0.000000	0.0435	0.0401
2024-04-30	ABB	0.004181	0.000000	0.000000	0.000000	0.000000	0.0435	0.0410
2024-04-30	BVB	0.003291	0.000000	0.000000	0.000000	0.000000	0.0435	0.0323
2024-05-31	VCB	0.2814	0.3000	0.3000	0.000000	0.3000	0.0417	0.0579
2024-05-31	BID	0.1570	0.0159	0.000000	0.2660	0.3000	0.0417	0.0424
2024-05-31	CTG	0.0672	0.000000	0.000000	0.1182	0.000000	0.0417	0.0438
2024-05-31	MBB	0.0482	0.000000	0.000000	0.3000	0.000000	0.0417	0.0445
2024-05-31	TCB	0.0847	0.000000	0.000000	0.002089	0.000000	0.0417	0.0385
2024-05-31	ACB	0.0538	0.3000	0.3000	0.000000	0.3000	0.0417	0.0669
2024-05-31	VPB	0.0744	0.000000	0.0166	0.000000	0.000000	0.0417	0.0441
2024-05-31	HDB	0.0284	0.2308	0.000000	0.000000	0.000000	0.0417	0.0496
2024-05-31	VIB	0.0239	0.000000	0.000000	0.0574	0.000000	0.0417	0.0375
2024-05-31	TPB	0.0189	0.000000	0.000000	0.000000	0.000000	0.0417	0.0384
2024-05-31	STB	0.0296	0.000000	0.000000	0.1978	0.000000	0.0417	0.0329
2024-05-31	SHB	0.0188	0.000000	0.000000	0.000000	0.000000	0.0417	0.0429
2024-05-31	LPB	0.0262	0.000000	0.000000	0.000000	0.000000	0.0417	0.0344
2024-05-31	SSB	0.0307	0.002775	0.0834	0.000000	0.000000	0.0417	0.0430
2024-05-31	MSB	0.0126	0.000000	0.000000	0.000000	0.000000	0.0417	0.0460

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-05-31	OCB	0.0137	0.000000	0.000000	0.000000	0.1000	0.0417	0.0449
2024-05-31	NAB	0.006840	0.1493	0.000000	0.000000	0.000000	0.0417	0.0350
2024-05-31	BAB	0.005637	0.000000	0.3000	0.000000	0.000000	0.0417	0.0646
2024-05-31	KLB	0.001422	0.000000	0.000000	0.000000	0.000000	0.0417	0.0308
2024-05-31	PGB	0.003651	0.001171	0.000000	0.000000	0.000000	0.0417	0.0248
2024-05-31	VBB	0.002966	0.000000	0.000000	0.0585	0.000000	0.0417	0.0307
2024-05-31	SGB	0.002478	0.000000	0.000000	0.000000	0.000000	0.0417	0.0372
2024-05-31	ABB	0.004053	0.000000	0.000000	0.000000	0.000000	0.0417	0.0388
2024-05-31	BVB	0.003384	0.000000	0.000000	0.000000	0.000000	0.0417	0.0302
2024-06-30	VCB	0.2701	0.1988	0.3000	0.000000	0.3000	0.0417	0.0574
2024-06-30	BID	0.1509	0.1636	0.000000	0.2762	0.3000	0.0417	0.0420
2024-06-30	CTG	0.0656	0.000000	0.000000	0.0552	0.0390	0.0417	0.0430
2024-06-30	MBB	0.0484	0.000000	0.000000	0.3000	0.0225	0.0417	0.0430
2024-06-30	TCB	0.0884	0.000000	0.000000	0.000000	0.000000	0.0417	0.0382
2024-06-30	ACB	0.0591	0.1066	0.3000	0.000000	0.3000	0.0417	0.0654
2024-06-30	VPB	0.0763	0.000000	0.005910	0.000000	0.000000	0.0417	0.0497
2024-06-30	HDB	0.0281	0.2941	0.000000	0.000000	0.000000	0.0417	0.0492
2024-06-30	VIB	0.0247	0.000000	0.000000	0.0166	0.000000	0.0417	0.0373
2024-06-30	TPB	0.0187	0.000000	0.000000	0.000000	0.000000	0.0417	0.0379
2024-06-30	STB	0.0293	0.000000	0.000000	0.2875	0.000000	0.0417	0.0326
2024-06-30	SHB	0.0188	0.000000	0.000000	0.000000	0.000000	0.0417	0.0430
2024-06-30	LPB	0.0324	0.0648	0.000000	0.000000	0.0385	0.0417	0.0346
2024-06-30	SSB	0.0301	0.1147	0.0941	0.000000	0.000000	0.0417	0.0430
2024-06-30	MSB	0.0133	0.000000	0.000000	0.000000	0.000000	0.0417	0.0453
2024-06-30	OCB	0.0149	0.000000	0.000000	0.000000	0.000000	0.0417	0.0440
2024-06-30	NAB	0.006497	0.0411	0.000000	0.000000	0.000000	0.0417	0.0352
2024-06-30	BAB	0.005610	0.000000	0.3000	0.000000	0.000000	0.0417	0.0660
2024-06-30	KLB	0.001517	0.000000	0.000000	0.000000	0.000000	0.0417	0.0315
2024-06-30	PGB	0.003564	0.0164	0.000000	0.000000	0.000000	0.0417	0.0246
2024-06-30	VBB	0.002977	0.000000	0.000000	0.0645	0.000000	0.0417	0.0302

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-06-30	SGB	0.002326	0.000000	0.000000	0.000000	0.000000	0.0417	0.0367
2024-06-30	ABB	0.004541	0.000000	0.000000	0.000000	0.000000	0.0417	0.0398
2024-06-30	BVB	0.003888	0.000000	0.000000	0.000000	0.000000	0.0417	0.0300
2024-07-31	VCB	0.2690	0.3000	0.0962	0.000000	0.3000	0.0417	0.0493
2024-07-31	BID	0.1412	0.000000	0.000000	0.3000	0.1589	0.0417	0.0376
2024-07-31	CTG	0.0650	0.000000	0.000000	0.000000	0.000000	0.0417	0.0420
2024-07-31	MBB	0.0503	0.000000	0.000000	0.1000	0.000000	0.0417	0.0403
2024-07-31	TCB	0.0896	0.000000	0.000000	0.000000	0.000000	0.0417	0.0364
2024-07-31	ACB	0.0581	0.3000	0.2419	0.000000	0.3000	0.0417	0.0618
2024-07-31	VPB	0.0805	0.000000	0.000000	0.000000	0.000000	0.0417	0.0459
2024-07-31	HDB	0.0284	0.1000	0.000000	0.000000	0.000000	0.0417	0.0467
2024-07-31	VIB	0.0243	0.000000	0.000000	0.3000	0.000000	0.0417	0.0341
2024-07-31	TPB	0.0191	0.000000	0.000000	0.000000	0.000000	0.0417	0.0375
2024-07-31	STB	0.0309	0.000000	0.000000	0.3000	0.000000	0.0417	0.0319
2024-07-31	SHB	0.0191	0.000000	0.000000	0.000000	0.000000	0.0417	0.0400
2024-07-31	LPB	0.0361	0.3000	0.000000	0.000000	0.2411	0.0417	0.0320
2024-07-31	SSB	0.0296	0.000000	0.2649	0.000000	0.000000	0.0417	0.0547
2024-07-31	MSB	0.0136	0.000000	0.000000	0.000000	0.000000	0.0417	0.0434
2024-07-31	OCB	0.0145	0.000000	0.000000	0.000000	0.000000	0.0417	0.0406
2024-07-31	NAB	0.006645	0.000000	0.000000	0.000000	0.000000	0.0417	0.0375
2024-07-31	BAB	0.005717	0.000000	0.3000	0.000000	0.000000	0.0417	0.0687
2024-07-31	KLB	0.001559	0.000000	0.000000	0.000000	0.000000	0.0417	0.0284
2024-07-31	PGB	0.003590	0.000000	0.000000	0.000000	0.000000	0.0417	0.0238
2024-07-31	VBB	0.002790	0.000000	0.000000	0.000000	0.000000	0.0417	0.0337
2024-07-31	SGB	0.002388	0.000000	0.0970	0.000000	0.000000	0.0417	0.0501
2024-07-31	ABB	0.004358	0.000000	0.000000	0.000000	0.000000	0.0417	0.0436
2024-07-31	BVB	0.003712	0.000000	0.000000	0.000000	0.000000	0.0417	0.0400
2024-08-31	VCB	0.2685	0.000000	0.0659	0.000000	0.2137	0.0417	0.0519
2024-08-31	BID	0.1482	0.1558	0.000000	0.3000	0.000000	0.0417	0.0363
2024-08-31	CTG	0.0639	0.000000	0.000000	0.000000	0.000000	0.0417	0.0410

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-08-31	MBB	0.0527	0.000000	0.000000	0.000000	0.000000	0.0417	0.0406
2024-08-31	TCB	0.0850	0.000000	0.000000	0.1000	0.000000	0.0417	0.0356
2024-08-31	ACB	0.0572	0.2971	0.2390	0.000000	0.3000	0.0417	0.0611
2024-08-31	VPB	0.0782	0.000000	0.000000	0.000000	0.000000	0.0417	0.0447
2024-08-31	HDB	0.0322	0.0163	0.000000	0.000000	0.000000	0.0417	0.0458
2024-08-31	VIB	0.0234	0.000000	0.000000	0.3000	0.000000	0.0417	0.0332
2024-08-31	TPB	0.0192	0.000000	0.000000	0.000000	0.000000	0.0417	0.0366
2024-08-31	STB	0.0297	0.000000	0.000000	0.3000	0.0536	0.0417	0.0314
2024-08-31	SHB	0.0186	0.000000	0.000000	0.000000	0.000000	0.0417	0.0392
2024-08-31	LPB	0.0356	0.3000	0.000000	0.000000	0.3000	0.0417	0.0311
2024-08-31	SSB	0.0296	0.000000	0.2657	0.000000	0.000000	0.0417	0.0533
2024-08-31	MSB	0.0132	0.000000	0.000000	0.000000	0.000000	0.0417	0.0433
2024-08-31	OCB	0.0142	0.000000	0.000000	0.000000	0.1327	0.0417	0.0398
2024-08-31	NAB	0.007521	0.2308	0.000000	0.000000	0.000000	0.0417	0.0380
2024-08-31	BAB	0.005364	0.000000	0.3000	0.000000	0.000000	0.0417	0.0804
2024-08-31	KLB	0.001474	0.000000	0.000000	0.000000	0.000000	0.0417	0.0280
2024-08-31	PGB	0.003250	0.000000	0.000000	0.000000	0.000000	0.0417	0.0233
2024-08-31	VBB	0.002661	0.000000	0.000000	0.000000	0.000000	0.0417	0.0328
2024-08-31	SGB	0.002261	0.000000	0.1294	0.000000	0.000000	0.0417	0.0509
2024-08-31	ABB	0.004105	0.000000	0.000000	0.000000	0.000000	0.0417	0.0427
2024-08-31	BVB	0.003749	0.000000	0.000000	0.000000	0.000000	0.0417	0.0394
2024-09-30	VCB	0.2688	0.2479	0.1647	0.000000	0.2151	0.0400	0.0542
2024-09-30	BID	0.1483	0.0888	0.000000	0.3000	0.000000	0.0400	0.0349
2024-09-30	CTG	0.0682	0.000000	0.000000	0.000000	0.000000	0.0400	0.0408
2024-09-30	MBB	0.0523	0.000000	0.000000	0.2882	0.000000	0.0400	0.0384
2024-09-30	TCB	0.0831	0.000000	0.000000	0.1118	0.000000	0.0400	0.0334
2024-09-30	ACB	0.0562	0.000000	0.1337	0.000000	0.1849	0.0400	0.0571
2024-09-30	VPB	0.0759	0.000000	0.000000	0.000000	0.000000	0.0400	0.0429
2024-09-30	HDB	0.0330	0.0640	0.000000	0.000000	0.000000	0.0400	0.0406
2024-09-30	VIB	0.0231	0.000000	0.000000	0.000000	0.000000	0.0400	0.0324

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-09-30	TPB	0.0185	0.000000	0.000000	0.000000	0.000000	0.0400	0.0343
2024-09-30	STB	0.0304	0.000000	0.000000	0.3000	0.000000	0.0400	0.0294
2024-09-30	SHB	0.0172	0.000000	0.000000	0.000000	0.000000	0.0400	0.0367
2024-09-30	LPB	0.0374	0.2993	0.000000	0.000000	0.3000	0.0400	0.0293
2024-09-30	SSB	0.0289	0.000000	0.2424	0.000000	0.000000	0.0400	0.0497
2024-09-30	MSB	0.0133	0.000000	0.000000	0.000000	0.000000	0.0400	0.0406
2024-09-30	OCB	0.0136	0.000000	0.000000	0.000000	0.000000	0.0400	0.0373
2024-09-30	NAB	0.007976	0.3000	0.000000	0.000000	0.000000	0.0400	0.0352
2024-09-30	BAB	0.005182	0.000000	0.3000	0.000000	0.000000	0.0400	0.0839
2024-09-30	KLB	0.001567	0.000000	0.000000	0.000000	0.000000	0.0400	0.0270
2024-09-30	PGB	0.003106	0.000000	0.000000	0.000000	0.000000	0.0400	0.0219
2024-09-30	VAB	0.001697	0.000000	0.000000	0.000000	0.3000	0.0400	0.0389
2024-09-30	VBB	0.002516	0.000000	0.000000	0.000000	0.000000	0.0400	0.0312
2024-09-30	SGB	0.002185	0.000000	0.1592	0.000000	0.000000	0.0400	0.0502
2024-09-30	ABB	0.003948	0.000000	0.000000	0.000000	0.000000	0.0400	0.0410
2024-09-30	BVB	0.003696	0.000000	0.000000	0.000000	0.000000	0.0400	0.0388
2024-10-31	VCB	0.2626	0.0830	0.1477	0.000000	0.3000	0.0400	0.0538
2024-10-31	BID	0.1467	0.1812	0.000000	0.3000	0.1749	0.0400	0.0349
2024-10-31	CTG	0.0700	0.000000	0.000000	0.000000	0.000000	0.0400	0.0404
2024-10-31	MBB	0.0526	0.000000	0.000000	0.3000	0.000000	0.0400	0.0382
2024-10-31	TCB	0.0839	0.000000	0.000000	0.0373	0.000000	0.0400	0.0333
2024-10-31	ACB	0.0568	0.000000	0.1596	0.000000	0.1509	0.0400	0.0615
2024-10-31	VPB	0.0784	0.000000	0.000000	0.000000	0.000000	0.0400	0.0426
2024-10-31	HDB	0.0327	0.1794	0.000000	0.000000	0.000000	0.0400	0.0404
2024-10-31	VIB	0.0237	0.000000	0.000000	0.000000	0.000000	0.0400	0.0331
2024-10-31	TPB	0.0208	0.000000	0.000000	0.000000	0.000000	0.0400	0.0341
2024-10-31	STB	0.0323	0.000000	0.000000	0.3000	0.0958	0.0400	0.0294
2024-10-31	SHB	0.0174	0.000000	0.000000	0.0627	0.000000	0.0400	0.0363
2024-10-31	LPB	0.0370	0.3000	0.000000	0.000000	0.2784	0.0400	0.0295
2024-10-31	SSB	0.0250	0.000000	0.2382	0.000000	0.000000	0.0400	0.0494

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-10-31	MSB	0.0145	0.000000	0.000000	0.000000	0.000000	0.0400	0.0404
2024-10-31	OCB	0.0135	0.000000	0.000000	0.000000	0.000000	0.0400	0.0372
2024-10-31	NAB	0.007858	0.2565	0.000000	0.000000	0.000000	0.0400	0.0347
2024-10-31	BAB	0.005130	0.000000	0.3000	0.000000	0.000000	0.0400	0.0837
2024-10-31	KLB	0.001586	0.000000	0.000000	0.000000	0.000000	0.0400	0.0266
2024-10-31	PGB	0.003284	0.000000	0.000000	0.000000	0.000000	0.0400	0.0218
2024-10-31	VAB	0.001783	0.000000	0.000000	0.000000	0.000000	0.0400	0.0386
2024-10-31	VBB	0.002545	0.000000	0.000000	0.000000	0.000000	0.0400	0.0310
2024-10-31	SGB	0.002144	0.000000	0.1546	0.000000	0.000000	0.0400	0.0499
2024-10-31	ABB	0.003940	0.000000	0.000000	0.000000	0.000000	0.0400	0.0407
2024-10-31	BVB	0.003692	0.000000	0.000000	0.000000	0.000000	0.0400	0.0385
2024-11-30	VCB	0.2694	0.1767	0.1457	0.000000	0.3000	0.0400	0.0541
2024-11-30	BID	0.1421	0.0151	0.000000	0.3000	0.0728	0.0400	0.0350
2024-11-30	CTG	0.0682	0.000000	0.000000	0.000000	0.000000	0.0400	0.0406
2024-11-30	MBB	0.0517	0.000000	0.000000	0.3000	0.000000	0.0400	0.0383
2024-11-30	TCB	0.0835	0.000000	0.000000	0.0214	0.000000	0.0400	0.0333
2024-11-30	ACB	0.0565	0.1722	0.1694	0.000000	0.1626	0.0400	0.0615
2024-11-30	VPB	0.0805	0.000000	0.000000	0.000000	0.000000	0.0400	0.0428
2024-11-30	HDB	0.0315	0.1937	0.000000	0.000000	0.000000	0.0400	0.0406
2024-11-30	VIB	0.0234	0.000000	0.000000	0.000000	0.000000	0.0400	0.0331
2024-11-30	TPB	0.0210	0.000000	0.000000	0.000000	0.000000	0.0400	0.0353
2024-11-30	STB	0.0344	0.000000	0.000000	0.3000	0.0354	0.0400	0.0294
2024-11-30	SHB	0.0169	0.000000	0.000000	0.0786	0.000000	0.0400	0.0363
2024-11-30	LPB	0.0385	0.3000	0.000000	0.000000	0.3000	0.0400	0.0298
2024-11-30	SSB	0.0244	0.000000	0.2280	0.000000	0.1292	0.0400	0.0474
2024-11-30	MSB	0.0139	0.000000	0.000000	0.000000	0.000000	0.0400	0.0393
2024-11-30	OCB	0.0129	0.000000	0.000000	0.000000	0.000000	0.0400	0.0375
2024-11-30	NAB	0.007596	0.1423	0.000000	0.000000	0.000000	0.0400	0.0348
2024-11-30	BAB	0.005173	0.000000	0.3000	0.000000	0.000000	0.0400	0.0833
2024-11-30	KLB	0.001432	0.000000	0.000000	0.000000	0.000000	0.0400	0.0266

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-11-30	PGB	0.003032	0.000000	0.000000	0.000000	0.000000	0.0400	0.0218
2024-11-30	VAB	0.001722	0.000000	0.000000	0.000000	0.000000	0.0400	0.0381
2024-11-30	VBB	0.002518	0.000000	0.000000	0.000000	0.000000	0.0400	0.0310
2024-11-30	SGB	0.002096	0.000000	0.1570	0.000000	0.000000	0.0400	0.0501
2024-11-30	ABB	0.003779	0.000000	0.000000	0.000000	0.000000	0.0400	0.0409
2024-11-30	BVB	0.003658	0.000000	0.000000	0.000000	0.000000	0.0400	0.0392
2024-12-31	VCB	0.2740	0.3000	0.1346	0.000000	0.3000	0.0400	0.0539
2024-12-31	BID	0.1420	0.000000	0.000000	0.3000	0.0664	0.0400	0.0348
2024-12-31	CTG	0.0697	0.000000	0.000000	0.000000	0.000000	0.0400	0.0404
2024-12-31	MBB	0.0509	0.000000	0.000000	0.3000	0.000000	0.0400	0.0381
2024-12-31	TCB	0.0842	0.000000	0.000000	0.0414	0.000000	0.0400	0.0333
2024-12-31	ACB	0.0572	0.1035	0.1739	0.000000	0.1197	0.0400	0.0616
2024-12-31	VPB	0.0769	0.000000	0.000000	0.000000	0.000000	0.0400	0.0428
2024-12-31	HDB	0.0303	0.2789	0.000000	0.000000	0.000000	0.0400	0.0402
2024-12-31	VIB	0.0240	0.000000	0.000000	0.000000	0.000000	0.0400	0.0331
2024-12-31	TPB	0.0201	0.000000	0.000000	0.000000	0.000000	0.0400	0.0355
2024-12-31	STB	0.0332	0.000000	0.000000	0.3000	0.0812	0.0400	0.0293
2024-12-31	SHB	0.0168	0.000000	0.000000	0.0586	0.000000	0.0400	0.0363
2024-12-31	LPB	0.0384	0.3000	0.000000	0.000000	0.3000	0.0400	0.0299
2024-12-31	SSB	0.0256	0.000000	0.2412	0.000000	0.1327	0.0400	0.0486
2024-12-31	MSB	0.0132	0.000000	0.000000	0.000000	0.000000	0.0400	0.0390
2024-12-31	OCB	0.0123	0.000000	0.000000	0.000000	0.000000	0.0400	0.0381
2024-12-31	NAB	0.007628	0.0176	0.000000	0.000000	0.000000	0.0400	0.0346
2024-12-31	BAB	0.005101	0.000000	0.3000	0.000000	0.000000	0.0400	0.0829
2024-12-31	KLB	0.001450	0.000000	0.000000	0.000000	0.000000	0.0400	0.0273
2024-12-31	PGB	0.003035	0.000000	0.000000	0.000000	0.000000	0.0400	0.0217
2024-12-31	VAB	0.001720	0.000000	0.000000	0.000000	0.000000	0.0400	0.0380
2024-12-31	VBB	0.002925	0.000000	0.000000	0.000000	0.000000	0.0400	0.0309
2024-12-31	SGB	0.002037	0.000000	0.1503	0.000000	0.000000	0.0400	0.0498
2024-12-31	ABB	0.003654	0.000000	0.000000	0.000000	0.000000	0.0400	0.0408

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2024-12-31	BVB	0.003605	0.000000	0.000000	0.000000	0.000000	0.0400	0.0390
2025-01-31	VCB	0.2630	0.3000	0.1250	0.000000	0.0324	0.0400	0.0529
2025-01-31	BID	0.1355	0.000000	0.000000	0.3000	0.000000	0.0400	0.0342
2025-01-31	CTG	0.0724	0.000000	0.000000	0.000000	0.000000	0.0400	0.0399
2025-01-31	MBB	0.0520	0.000000	0.000000	0.2523	0.000000	0.0400	0.0373
2025-01-31	TCB	0.0864	0.000000	0.000000	0.1477	0.000000	0.0400	0.0327
2025-01-31	ACB	0.0576	0.3000	0.1504	0.000000	0.000000	0.0400	0.0604
2025-01-31	VPB	0.0757	0.000000	0.000000	0.000000	0.000000	0.0400	0.0417
2025-01-31	HDB	0.0359	0.000000	0.000000	0.000000	0.000000	0.0400	0.0420
2025-01-31	VIB	0.0244	0.000000	0.000000	0.000000	0.000000	0.0400	0.0333
2025-01-31	TPB	0.0203	0.000000	0.000000	0.000000	0.000000	0.0400	0.0346
2025-01-31	STB	0.0362	0.000000	0.000000	0.3000	0.3000	0.0400	0.0288
2025-01-31	SHB	0.0164	0.000000	0.000000	0.000000	0.000000	0.0400	0.0358
2025-01-31	LPB	0.0427	0.3000	0.000000	0.000000	0.3000	0.0400	0.0293
2025-01-31	SSB	0.0248	0.000000	0.2606	0.000000	0.1793	0.0400	0.0483
2025-01-31	MSB	0.0131	0.000000	0.000000	0.000000	0.000000	0.0400	0.0394
2025-01-31	OCB	0.0128	0.000000	0.000000	0.000000	0.1883	0.0400	0.0374
2025-01-31	NAB	0.007583	0.1000	0.000000	0.000000	0.000000	0.0400	0.0344
2025-01-31	BAB	0.005139	0.000000	0.3000	0.000000	0.000000	0.0400	0.0865
2025-01-31	KLB	0.001352	0.000000	0.000000	0.000000	0.000000	0.0400	0.0268
2025-01-31	PGB	0.002869	0.000000	0.004336	0.000000	0.000000	0.0400	0.0249
2025-01-31	VAB	0.001746	0.000000	0.000000	0.000000	0.000000	0.0400	0.0375
2025-01-31	VBB	0.002872	0.000000	0.000000	0.000000	0.000000	0.0400	0.0300
2025-01-31	SGB	0.002083	0.000000	0.1598	0.000000	0.000000	0.0400	0.0496
2025-01-31	ABB	0.003637	0.000000	0.000000	0.000000	0.000000	0.0400	0.0420
2025-01-31	BVB	0.003666	0.000000	0.000000	0.000000	0.000000	0.0400	0.0401
2025-02-28	VCB	0.2621	0.3000	0.1225	0.000000	0.2789	0.0400	0.0526
2025-02-28	BID	0.1417	0.000000	0.000000	0.3000	0.1199	0.0400	0.0341
2025-02-28	CTG	0.0714	0.000000	0.000000	0.000000	0.3000	0.0400	0.0396
2025-02-28	MBB	0.0523	0.000000	0.000000	0.2096	0.0186	0.0400	0.0370

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-02-28	TCB	0.0851	0.000000	0.000000	0.1904	0.000000	0.0400	0.0325
2025-02-28	ACB	0.0555	0.1563	0.1402	0.000000	0.000000	0.0400	0.0603
2025-02-28	VPB	0.0732	0.000000	0.000000	0.000000	0.000000	0.0400	0.0417
2025-02-28	HDB	0.0313	0.1868	0.000000	0.000000	0.1120	0.0400	0.0388
2025-02-28	VIB	0.0251	0.000000	0.000000	0.000000	0.000000	0.0400	0.0339
2025-02-28	TPB	0.0197	0.000000	0.000000	0.000000	0.0792	0.0400	0.0374
2025-02-28	STB	0.0356	0.000000	0.000000	0.3000	0.0915	0.0400	0.0286
2025-02-28	SHB	0.0162	0.000000	0.000000	0.000000	0.000000	0.0400	0.0358
2025-02-28	LPB	0.0481	0.3000	0.000000	0.000000	0.000000	0.0400	0.0289
2025-02-28	SSB	0.0272	0.000000	0.2958	0.000000	0.000000	0.0400	0.0523
2025-02-28	MSB	0.0123	0.000000	0.000000	0.000000	0.000000	0.0400	0.0405
2025-02-28	OCB	0.0124	0.000000	0.000000	0.000000	0.000000	0.0400	0.0370
2025-02-28	NAB	0.007891	0.0569	0.000000	0.000000	0.000000	0.0400	0.0344
2025-02-28	BAB	0.005044	0.000000	0.3000	0.000000	0.000000	0.0400	0.0850
2025-02-28	KLB	0.001327	0.000000	0.000000	0.000000	0.000000	0.0400	0.0269
2025-02-28	PGB	0.002778	0.000000	0.000000	0.000000	0.000000	0.0400	0.0251
2025-02-28	VAB	0.001658	0.000000	0.000000	0.000000	0.000000	0.0400	0.0372
2025-02-28	VBB	0.002728	0.000000	0.000000	0.000000	0.000000	0.0400	0.0299
2025-02-28	SGB	0.002141	0.000000	0.1415	0.000000	0.000000	0.0400	0.0490
2025-02-28	ABB	0.003427	0.000000	0.000000	0.000000	0.000000	0.0400	0.0418
2025-02-28	BVB	0.003876	0.000000	0.000000	0.000000	0.000000	0.0400	0.0398
2025-03-31	VCB	0.2569	0.1000	0.1160	0.000000	0.2838	0.0400	0.0552
2025-03-31	BID	0.1411	0.000000	0.000000	0.000000	0.1585	0.0400	0.0379
2025-03-31	CTG	0.0758	0.000000	0.000000	0.000000	0.000000	0.0400	0.0399
2025-03-31	MBB	0.0523	0.000000	0.000000	0.000000	0.000000	0.0400	0.0385
2025-03-31	TCB	0.0873	0.000000	0.000000	0.3000	0.000000	0.0400	0.0324
2025-03-31	ACB	0.0554	0.3000	0.1792	0.000000	0.000000	0.0400	0.0597
2025-03-31	VPB	0.0727	0.000000	0.000000	0.000000	0.000000	0.0400	0.0415
2025-03-31	HDB	0.0309	0.000000	0.000000	0.000000	0.000000	0.0400	0.0372
2025-03-31	VIB	0.0245	0.000000	0.000000	0.3000	0.000000	0.0400	0.0335

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-03-31	TPB	0.0194	0.000000	0.000000	0.000000	0.000000	0.0400	0.0372
2025-03-31	STB	0.0358	0.000000	0.000000	0.3000	0.2578	0.0400	0.0289
2025-03-31	SHB	0.0169	0.000000	0.000000	0.1000	0.000000	0.0400	0.0355
2025-03-31	LPB	0.0472	0.3000	0.000000	0.000000	0.3000	0.0400	0.0285
2025-03-31	SSB	0.0274	0.000000	0.2684	0.000000	0.000000	0.0400	0.0499
2025-03-31	MSB	0.0122	0.000000	0.000000	0.000000	0.000000	0.0400	0.0404
2025-03-31	OCB	0.0121	0.000000	0.000000	0.000000	0.000000	0.0400	0.0367
2025-03-31	NAB	0.008036	0.3000	0.000000	0.000000	0.000000	0.0400	0.0340
2025-03-31	BAB	0.005205	0.000000	0.3000	0.000000	0.000000	0.0400	0.0843
2025-03-31	KLB	0.001405	0.000000	0.000000	0.000000	0.000000	0.0400	0.0271
2025-03-31	PGB	0.002791	0.000000	0.000000	0.000000	0.000000	0.0400	0.0250
2025-03-31	VAB	0.001843	0.000000	0.000000	0.000000	0.000000	0.0400	0.0369
2025-03-31	VBB	0.002802	0.000000	0.000000	0.000000	0.000000	0.0400	0.0299
2025-03-31	SGB	0.002068	0.000000	0.1364	0.000000	0.000000	0.0400	0.0481
2025-03-31	ABB	0.003569	0.000000	0.000000	0.000000	0.000000	0.0400	0.0419
2025-03-31	BVB	0.004436	0.000000	0.000000	0.000000	0.000000	0.0400	0.0398
2025-04-30	VCB	0.2632	0.2207	0.1342	0.000000	0.2422	0.0400	0.0563
2025-04-30	BID	0.1334	0.000000	0.000000	0.000000	0.000000	0.0400	0.0388
2025-04-30	CTG	0.0757	0.000000	0.000000	0.000000	0.000000	0.0400	0.0407
2025-04-30	MBB	0.0547	0.000000	0.000000	0.1272	0.000000	0.0400	0.0386
2025-04-30	TCB	0.0919	0.000000	0.000000	0.3000	0.000000	0.0400	0.0326
2025-04-30	ACB	0.0553	0.3000	0.1596	0.000000	0.000000	0.0400	0.0600
2025-04-30	VPB	0.0715	0.000000	0.000000	0.000000	0.000000	0.0400	0.0418
2025-04-30	HDB	0.0297	0.0680	0.000000	0.000000	0.000000	0.0400	0.0379
2025-04-30	VIB	0.0235	0.000000	0.000000	0.000000	0.000000	0.0400	0.0336
2025-04-30	TPB	0.0166	0.000000	0.000000	0.000000	0.000000	0.0400	0.0373
2025-04-30	STB	0.0358	0.000000	0.000000	0.3000	0.3000	0.0400	0.0292
2025-04-30	SHB	0.0211	0.000000	0.000000	0.2728	0.000000	0.0400	0.0353
2025-04-30	LPB	0.0434	0.3000	0.000000	0.000000	0.3000	0.0400	0.0292
2025-04-30	SSB	0.0276	0.000000	0.2787	0.000000	0.1010	0.0400	0.0517

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-04-30	MSB	0.0131	0.000000	0.000000	0.000000	0.000000	0.0400	0.0405
2025-04-30	OCB	0.0120	0.000000	0.000000	0.000000	0.0568	0.0400	0.0370
2025-04-30	NAB	0.007553	0.1113	0.000000	0.000000	0.000000	0.0400	0.0341
2025-04-30	BAB	0.005201	0.000000	0.3000	0.000000	0.000000	0.0400	0.0805
2025-04-30	KLB	0.001324	0.000000	0.000000	0.000000	0.000000	0.0400	0.0269
2025-04-30	PGB	0.002843	0.000000	0.000000	0.000000	0.000000	0.0400	0.0253
2025-04-30	VAB	0.001858	0.000000	0.000000	0.000000	0.000000	0.0400	0.0356
2025-04-30	VBB	0.002949	0.000000	0.000000	0.000000	0.000000	0.0400	0.0299
2025-04-30	SGB	0.002081	0.000000	0.1276	0.000000	0.000000	0.0400	0.0483
2025-04-30	ABB	0.003659	0.000000	0.000000	0.000000	0.000000	0.0400	0.0412
2025-04-30	BVB	0.004161	0.000000	0.000000	0.000000	0.000000	0.0400	0.0376
2025-05-31	VCB	0.2553	0.3000	0.1265	0.000000	0.000000	0.0400	0.0565
2025-05-31	BID	0.1289	0.000000	0.000000	0.1323	0.2797	0.0400	0.0385
2025-05-31	CTG	0.0736	0.000000	0.000000	0.000000	0.000000	0.0400	0.0408
2025-05-31	MBB	0.0579	0.000000	0.000000	0.2015	0.000000	0.0400	0.0386
2025-05-31	TCB	0.0952	0.000000	0.000000	0.2505	0.000000	0.0400	0.0324
2025-05-31	ACB	0.0554	0.3000	0.1582	0.000000	0.000000	0.0400	0.0604
2025-05-31	VPB	0.0675	0.000000	0.000000	0.000000	0.000000	0.0400	0.0418
2025-05-31	HDB	0.0312	0.0917	0.000000	0.000000	0.000000	0.0400	0.0377
2025-05-31	VIB	0.0231	0.000000	0.000000	0.000000	0.000000	0.0400	0.0335
2025-05-31	TPB	0.0171	0.000000	0.000000	0.000000	0.000000	0.0400	0.0359
2025-05-31	STB	0.0396	0.000000	0.000000	0.3000	0.3000	0.0400	0.0293
2025-05-31	SHB	0.0231	0.000000	0.000000	0.1157	0.000000	0.0400	0.0324
2025-05-31	LPB	0.0460	0.3000	0.000000	0.000000	0.2987	0.0400	0.0290
2025-05-31	SSB	0.0281	0.000000	0.2880	0.000000	0.000000	0.0400	0.0529
2025-05-31	MSB	0.0130	0.000000	0.000000	0.000000	0.000000	0.0400	0.0400
2025-05-31	OCB	0.0121	0.000000	0.000000	0.000000	0.1216	0.0400	0.0370
2025-05-31	NAB	0.008131	0.008269	0.000000	0.000000	0.000000	0.0400	0.0338
2025-05-31	BAB	0.005300	0.000000	0.3000	0.000000	0.000000	0.0400	0.0804
2025-05-31	KLB	0.001495	0.000000	0.000000	0.000000	0.000000	0.0400	0.0312

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-05-31	PGB	0.002855	0.000000	0.000000	0.000000	0.000000	0.0400	0.0254
2025-05-31	VAB	0.002111	0.000000	0.000000	0.000000	0.000000	0.0400	0.0356
2025-05-31	VBB	0.003098	0.000000	0.000000	0.000000	0.000000	0.0400	0.0298
2025-05-31	SGB	0.002204	0.000000	0.1273	0.000000	0.000000	0.0400	0.0483
2025-05-31	ABB	0.003758	0.000000	0.000000	0.000000	0.000000	0.0400	0.0411
2025-05-31	BVB	0.004001	0.000000	0.000000	0.000000	0.000000	0.0400	0.0374
2025-06-30	VCB	0.2411	0.0570	0.0894	0.000000	0.2331	0.0400	0.0519
2025-06-30	BID	0.1288	0.000000	0.000000	0.1858	0.1302	0.0400	0.0385
2025-06-30	CTG	0.0727	0.1380	0.000000	0.000000	0.000000	0.0400	0.0412
2025-06-30	MBB	0.0575	0.000000	0.000000	0.0952	0.000000	0.0400	0.0386
2025-06-30	TCB	0.1058	0.000000	0.000000	0.3000	0.000000	0.0400	0.0323
2025-06-30	ACB	0.0559	0.000000	0.1338	0.000000	0.000000	0.0400	0.0568
2025-06-30	VPB	0.0721	0.000000	0.000000	0.000000	0.000000	0.0400	0.0391
2025-06-30	HDB	0.0303	0.2114	0.000000	0.000000	0.000000	0.0400	0.0379
2025-06-30	VIB	0.0229	0.000000	0.000000	0.1556	0.000000	0.0400	0.0326
2025-06-30	TPB	0.0172	0.000000	0.000000	0.000000	0.000000	0.0400	0.0369
2025-06-30	STB	0.0395	0.000000	0.000000	0.2634	0.2719	0.0400	0.0293
2025-06-30	SHB	0.0239	0.000000	0.000000	0.000000	0.000000	0.0400	0.0360
2025-06-30	LPB	0.0463	0.3000	0.000000	0.000000	0.1992	0.0400	0.0298
2025-06-30	SSB	0.0270	0.000000	0.2811	0.000000	0.000000	0.0400	0.0511
2025-06-30	MSB	0.0129	0.000000	0.000000	0.000000	0.000000	0.0400	0.0393
2025-06-30	OCB	0.0118	0.000000	0.000000	0.000000	0.1656	0.0400	0.0360
2025-06-30	NAB	0.007901	0.2936	0.000000	0.000000	0.000000	0.0400	0.0342
2025-06-30	BAB	0.005268	0.000000	0.3000	0.000000	0.000000	0.0400	0.0846
2025-06-30	KLB	0.002142	0.000000	0.000000	0.000000	0.000000	0.0400	0.0306
2025-06-30	PGB	0.002858	0.000000	0.000000	0.000000	0.000000	0.0400	0.0247
2025-06-30	VAB	0.002613	0.000000	0.000000	0.000000	0.000000	0.0400	0.0357
2025-06-30	VBB	0.003528	0.000000	0.000000	0.000000	0.000000	0.0400	0.0300
2025-06-30	SGB	0.002099	0.000000	0.1956	0.000000	0.000000	0.0400	0.0508
2025-06-30	ABB	0.003899	0.000000	0.000000	0.000000	0.000000	0.0400	0.0439

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-06-30	BVB	0.004159	0.000000	0.000000	0.000000	0.000000	0.0400	0.0381
2025-07-31	VCB	0.2346	0.0783	0.0820	0.000000	0.000000	0.0400	0.0516
2025-07-31	BID	0.1251	0.000000	0.000000	0.3000	0.000000	0.0400	0.0385
2025-07-31	CTG	0.0766	0.000000	0.000000	0.0780	0.000000	0.0400	0.0408
2025-07-31	MBB	0.0587	0.000000	0.000000	0.2953	0.000000	0.0400	0.0388
2025-07-31	TCB	0.1144	0.0108	0.000000	0.1825	0.000000	0.0400	0.0327
2025-07-31	ACB	0.0543	0.3000	0.1133	0.000000	0.000000	0.0400	0.0564
2025-07-31	VPB	0.0716	0.000000	0.000000	0.000000	0.000000	0.0400	0.0401
2025-07-31	HDB	0.0293	0.007236	0.000000	0.000000	0.000000	0.0400	0.0376
2025-07-31	VIB	0.0225	0.000000	0.000000	0.0155	0.000000	0.0400	0.0340
2025-07-31	TPB	0.0168	0.000000	0.000000	0.000000	0.000000	0.0400	0.0365
2025-07-31	STB	0.0437	0.000000	0.000000	0.1288	0.000000	0.0400	0.0312
2025-07-31	SHB	0.0227	0.000000	0.000000	0.000000	0.000000	0.0400	0.0362
2025-07-31	LPB	0.0452	0.3000	0.000000	0.000000	0.1119	0.0400	0.0302
2025-07-31	SSB	0.0256	0.000000	0.2882	0.000000	0.2864	0.0400	0.0520
2025-07-31	MSB	0.0129	0.000000	0.000000	0.000000	0.000000	0.0400	0.0400
2025-07-31	OCB	0.0126	0.000000	0.000883	0.000000	0.000000	0.0400	0.0382
2025-07-31	NAB	0.007607	0.3000	0.000000	0.000000	0.000000	0.0400	0.0354
2025-07-31	BAB	0.005334	0.000000	0.3000	0.000000	0.000000	0.0400	0.0826
2025-07-31	KLB	0.001959	0.000000	0.000000	0.000000	0.0876	0.0400	0.0254
2025-07-31	PGB	0.002637	0.000000	0.000000	0.000000	0.000000	0.0400	0.0256
2025-07-31	VAB	0.002693	0.003692	0.000000	0.000000	0.3000	0.0400	0.0326
2025-07-31	VBB	0.003497	0.000000	0.000000	0.000000	0.2141	0.0400	0.0290
2025-07-31	SGB	0.002006	0.000000	0.2157	0.000000	0.000000	0.0400	0.0527
2025-07-31	ABB	0.003944	0.000000	0.000000	0.000000	0.000000	0.0400	0.0433
2025-07-31	BVB	0.003912	0.000000	0.000000	0.000000	0.000000	0.0400	0.0383
2025-08-31	VCB	0.2181	0.1356	0.0875	0.000000	0.000000	0.0400	0.0507
2025-08-31	BID	0.1148	0.000000	0.000000	0.2961	0.000000	0.0400	0.0375
2025-08-31	CTG	0.0738	0.0706	0.000000	0.0888	0.000000	0.0400	0.0395
2025-08-31	MBB	0.0725	0.000000	0.000000	0.2336	0.000000	0.0400	0.0395

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-08-31	TCB	0.1001	0.000000	0.000000	0.1688	0.000000	0.0400	0.0313
2025-08-31	ACB	0.0516	0.0575	0.1187	0.000000	0.000000	0.0400	0.0556
2025-08-31	VPB	0.0878	0.000000	0.000000	0.0779	0.000000	0.0400	0.0392
2025-08-31	HDB	0.0313	0.000000	0.000000	0.000000	0.000000	0.0400	0.0372
2025-08-31	VIB	0.0273	0.000000	0.000000	0.000000	0.000000	0.0400	0.0364
2025-08-31	TPB	0.0172	0.000000	0.000000	0.000000	0.000000	0.0400	0.0377
2025-08-31	STB	0.0411	0.000000	0.000000	0.1145	0.000000	0.0400	0.0297
2025-08-31	SHB	0.0301	0.000000	0.000000	0.0202	0.1499	0.0400	0.0357
2025-08-31	LPB	0.0422	0.3000	0.000000	0.000000	0.000000	0.0400	0.0306
2025-08-31	SSB	0.0242	0.000000	0.2723	0.000000	0.1087	0.0400	0.0533
2025-08-31	MSB	0.0131	0.000000	0.000000	0.000000	0.000000	0.0400	0.0425
2025-08-31	OCB	0.0142	0.000000	0.000964	0.000000	0.000000	0.0400	0.0388
2025-08-31	NAB	0.009426	0.3000	0.000000	0.000000	0.000000	0.0400	0.0362
2025-08-31	BAB	0.005700	0.000000	0.3000	0.000000	0.000000	0.0400	0.0821
2025-08-31	KLB	0.003656	0.000000	0.000000	0.000000	0.3000	0.0400	0.0247
2025-08-31	PGB	0.002673	0.000000	0.000000	0.000000	0.000000	0.0400	0.0266
2025-08-31	VAB	0.003961	0.1363	0.000000	0.000000	0.3000	0.0400	0.0326
2025-08-31	VBB	0.003991	0.000000	0.000000	0.000000	0.0547	0.0400	0.0297
2025-08-31	SGB	0.001848	0.000000	0.2205	0.000000	0.0867	0.0400	0.0514
2025-08-31	ABB	0.005411	0.000000	0.000000	0.000000	0.000000	0.0400	0.0436
2025-08-31	BVB	0.003974	0.000000	0.000000	0.000000	0.000000	0.0400	0.0381
2025-09-30	VCB	0.2086	0.1416	0.1085	0.000000	0.000000	0.0400	0.0531
2025-09-30	BID	0.1091	0.000000	0.000000	0.2323	0.000000	0.0400	0.0404
2025-09-30	CTG	0.0693	0.0639	0.000000	0.0961	0.000000	0.0400	0.0412
2025-09-30	MBB	0.0819	0.000000	0.000000	0.1666	0.000000	0.0400	0.0416
2025-09-30	TCB	0.0978	0.000000	0.000000	0.1008	0.1625	0.0400	0.0331
2025-09-30	ACB	0.0523	0.008688	0.1055	0.000000	0.000000	0.0400	0.0575
2025-09-30	VPB	0.1000	0.000000	0.000000	0.1754	0.000000	0.0400	0.0333
2025-09-30	HDB	0.0332	0.0537	0.000000	0.000000	0.000000	0.0400	0.0367
2025-09-30	VIB	0.0267	0.000000	0.000000	0.0572	0.000000	0.0400	0.0393

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-09-30	TPB	0.0193	0.000000	0.000000	0.000000	0.000000	0.0400	0.0378
2025-09-30	STB	0.0384	0.000000	0.000000	0.0985	0.000000	0.0400	0.0319
2025-09-30	SHB	0.0312	0.000000	0.000000	0.0730	0.000000	0.0400	0.0331
2025-09-30	LPB	0.0466	0.3000	0.000000	0.000000	0.0390	0.0400	0.0329
2025-09-30	SSB	0.0223	0.000000	0.2534	0.000000	0.000000	0.0400	0.0548
2025-09-30	MSB	0.0141	0.000000	0.000000	0.000000	0.000000	0.0400	0.0435
2025-09-30	OCB	0.0133	0.000000	0.000000	0.000000	0.000000	0.0400	0.0389
2025-09-30	NAB	0.008408	0.3000	0.000000	0.000000	0.000000	0.0400	0.0376
2025-09-30	BAB	0.004990	0.000000	0.3000	0.000000	0.000000	0.0400	0.0709
2025-09-30	KLB	0.002975	0.000000	0.000000	0.000000	0.1985	0.0400	0.0241
2025-09-30	PGB	0.002382	0.000000	0.000000	0.000000	0.000000	0.0400	0.0289
2025-09-30	VAB	0.003845	0.1321	0.000000	0.000000	0.3000	0.0400	0.0340
2025-09-30	VBB	0.003494	0.000000	0.000000	0.000000	0.3000	0.0400	0.0309
2025-09-30	SGB	0.001668	0.000000	0.2326	0.000000	0.000000	0.0400	0.0539
2025-09-30	ABB	0.004370	0.000000	0.000000	0.000000	0.000000	0.0400	0.0311
2025-09-30	BVB	0.003758	0.000000	0.000000	0.000000	0.000000	0.0400	0.0394
2025-10-31	VCB	0.2009	0.000000	0.1742	0.000000	0.000000	0.0400	0.0549
2025-10-31	BID	0.1085	0.000000	0.000000	0.1936	0.000000	0.0400	0.0414
2025-10-31	CTG	0.0731	0.0741	0.000000	0.0797	0.000000	0.0400	0.0423
2025-10-31	MBB	0.0825	0.000000	0.000000	0.2315	0.000000	0.0400	0.0373
2025-10-31	TCB	0.1023	0.000000	0.000000	0.0801	0.000000	0.0400	0.0338
2025-10-31	ACB	0.0512	0.2085	0.000000	0.000000	0.000000	0.0400	0.0529
2025-10-31	VPB	0.0947	0.000000	0.000000	0.1638	0.000000	0.0400	0.0312
2025-10-31	HDB	0.0328	0.000000	0.000000	0.000000	0.000000	0.0400	0.0358
2025-10-31	VIB	0.0251	0.000000	0.000000	0.0513	0.000000	0.0400	0.0399
2025-10-31	TPB	0.0185	0.000000	0.000000	0.0454	0.000000	0.0400	0.0344
2025-10-31	STB	0.0419	0.000000	0.000000	0.0619	0.0253	0.0400	0.0331
2025-10-31	SHB	0.0299	0.000000	0.000000	0.0546	0.1429	0.0400	0.0329
2025-10-31	LPB	0.0546	0.3000	0.000000	0.0381	0.000000	0.0400	0.0319
2025-10-31	SSB	0.0215	0.000000	0.2589	0.000000	0.000000	0.0400	0.0552

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-10-31	MSB	0.0134	0.000000	0.000000	0.000000	0.000000	0.0400	0.0396
2025-10-31	OCB	0.0132	0.000000	0.000000	0.000000	0.000000	0.0400	0.0400
2025-10-31	NAB	0.008211	0.3000	0.000000	0.000000	0.000000	0.0400	0.0393
2025-10-31	BAB	0.004950	0.000000	0.3000	0.000000	0.000000	0.0400	0.0734
2025-10-31	KLB	0.003571	0.000000	0.000000	0.000000	0.2810	0.0400	0.0254
2025-10-31	PGB	0.002346	0.000000	0.000000	0.000000	0.000000	0.0400	0.0302
2025-10-31	VAB	0.003700	0.1174	0.000000	0.000000	0.2612	0.0400	0.0346
2025-10-31	VBB	0.003509	0.000000	0.000000	0.000000	0.000000	0.0400	0.0323
2025-10-31	SGB	0.001628	0.000000	0.2669	0.000000	0.000000	0.0400	0.0556
2025-10-31	ABB	0.004545	0.000000	0.000000	0.000000	0.2896	0.0400	0.0325
2025-10-31	BVB	0.003530	0.000000	0.000000	0.000000	0.000000	0.0400	0.0403
2025-11-30	VCB	0.1928	0.000000	0.2108	0.0166	0.000000	0.0400	0.0557
2025-11-30	BID	0.1017	0.000000	0.000000	0.1690	0.000000	0.0400	0.0429
2025-11-30	CTG	0.1019	0.2981	0.000000	0.0569	0.000000	0.0400	0.0469
2025-11-30	MBB	0.0736	0.000000	0.000000	0.1810	0.000000	0.0400	0.0388
2025-11-30	TCB	0.0943	0.000000	0.000000	0.0572	0.000000	0.0400	0.0351
2025-11-30	ACB	0.0507	0.000000	0.000000	0.000000	0.000000	0.0400	0.0523
2025-11-30	VPB	0.0866	0.000000	0.000000	0.1429	0.0998	0.0400	0.0309
2025-11-30	HDB	0.0478	0.000000	0.000000	0.000000	0.000000	0.0400	0.0353
2025-11-30	VIB	0.0232	0.000000	0.000000	0.1316	0.000000	0.0400	0.0388
2025-11-30	TPB	0.0179	0.000000	0.000000	0.0757	0.000000	0.0400	0.0337
2025-11-30	STB	0.0405	0.000000	0.000000	0.0492	0.000000	0.0400	0.0346
2025-11-30	SHB	0.0288	0.000395	0.000000	0.0102	0.2405	0.0400	0.0344
2025-11-30	LPB	0.0554	0.3000	0.0569	0.0188	0.000000	0.0400	0.0341
2025-11-30	SSB	0.0195	0.000000	0.2173	0.000000	0.000000	0.0400	0.0521
2025-11-30	MSB	0.0146	0.000000	0.000000	0.0879	0.000000	0.0400	0.0391
2025-11-30	OCB	0.0131	0.000000	0.000000	0.002773	0.0914	0.0400	0.0404
2025-11-30	NAB	0.007886	0.2890	0.000000	0.000000	0.000000	0.0400	0.0395
2025-11-30	BAB	0.004798	0.000000	0.3000	0.000000	0.000000	0.0400	0.0708
2025-11-30	KLB	0.003765	0.000000	0.0264	0.000000	0.2464	0.0400	0.0244

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-11-30	PGB	0.002790	0.000000	0.0164	0.000000	0.000000	0.0400	0.0298
2025-11-30	VAB	0.003588	0.0215	0.000000	0.000000	0.2146	0.0400	0.0334
2025-11-30	VBB	0.004721	0.0910	0.000000	0.000000	0.000000	0.0400	0.0331
2025-11-30	SGB	0.001650	0.000000	0.1721	0.000000	0.000000	0.0400	0.0526
2025-11-30	ABB	0.004762	0.000000	0.000000	0.000000	0.1073	0.0400	0.0324
2025-11-30	BVB	0.003375	0.000000	0.000000	0.000000	0.000000	0.0400	0.0390
2025-12-31	VCB	0.1895	0.000000	0.1768	0.007494	0.000000	0.0400	0.0532
2025-12-31	BID	0.1029	0.000000	0.000000	0.1853	0.000000	0.0400	0.0408
2025-12-31	CTG	0.1040	0.000000	0.000000	0.1372	0.000000	0.0400	0.0450
2025-12-31	MBB	0.0740	0.000000	0.000000	0.1768	0.1808	0.0400	0.0377
2025-12-31	TCB	0.0925	0.1705	0.000000	0.001376	0.0809	0.0400	0.0403
2025-12-31	ACB	0.0492	0.000000	0.000000	0.000000	0.000000	0.0400	0.0504
2025-12-31	VPB	0.0901	0.000000	0.000000	0.1493	0.000000	0.0400	0.0300
2025-12-31	HDB	0.0488	0.0177	0.000000	0.000000	0.000000	0.0400	0.0363
2025-12-31	VIB	0.0236	0.000000	0.000000	0.1342	0.000000	0.0400	0.0382
2025-12-31	TPB	0.0190	0.000000	0.000000	0.0579	0.000000	0.0400	0.0343
2025-12-31	STB	0.0362	0.1432	0.0512	0.0106	0.000000	0.0400	0.0385
2025-12-31	SHB	0.0301	0.000000	0.000000	0.005932	0.000000	0.0400	0.0337
2025-12-31	LPB	0.0542	0.3000	0.0737	0.0118	0.0674	0.0400	0.0341
2025-12-31	SSB	0.0194	0.000000	0.1851	0.000000	0.000000	0.0400	0.0491
2025-12-31	MSB	0.0162	0.000000	0.000000	0.1222	0.000000	0.0400	0.0370
2025-12-31	OCB	0.0129	0.000000	0.000000	0.000000	0.000000	0.0400	0.0416
2025-12-31	NAB	0.008080	0.2575	0.000000	0.000000	0.000000	0.0400	0.0394
2025-12-31	BAB	0.004598	0.000000	0.3000	0.000000	0.1171	0.0400	0.0737
2025-12-31	KLB	0.003519	0.000000	0.0329	0.000000	0.3000	0.0400	0.0237
2025-12-31	PGB	0.002760	0.0384	0.0345	0.000000	0.0396	0.0400	0.0287
2025-12-31	VAB	0.003500	0.006197	0.000000	0.000000	0.0103	0.0400	0.0337
2025-12-31	VBB	0.004468	0.0665	0.000000	0.000000	0.0484	0.0400	0.0319
2025-12-31	SGB	0.001672	0.000000	0.1458	0.000000	0.000000	0.0400	0.0507
2025-12-31	ABB	0.005382	0.000000	0.000000	0.000000	0.1556	0.0400	0.0333

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2025-12-31	BVB	0.003367	0.000000	0.000000	0.000000	0.000000	0.0400	0.0446
2026-01-31	VCB	0.1852	0.000000	0.1548	0.000000	0.000000	0.0400	0.0539
2026-01-31	BID	0.1053	0.000000	0.000000	0.1392	0.000000	0.0400	0.0435
2026-01-31	CTG	0.1070	0.000000	0.000000	0.1344	0.0550	0.0400	0.0453
2026-01-31	MBB	0.0785	0.000000	0.000000	0.2539	0.0339	0.0400	0.0373
2026-01-31	TCB	0.0933	0.1678	0.000000	0.009684	0.1365	0.0400	0.0394
2026-01-31	ACB	0.0475	0.000000	0.000000	0.0384	0.000000	0.0400	0.0490
2026-01-31	VPB	0.0861	0.000000	0.000000	0.1662	0.000000	0.0400	0.0296
2026-01-31	HDB	0.0573	0.0676	0.000000	0.006216	0.000000	0.0400	0.0362
2026-01-31	VIB	0.0221	0.000000	0.000000	0.1026	0.000000	0.0400	0.0377
2026-01-31	TPB	0.0183	0.000000	0.000000	0.0549	0.000000	0.0400	0.0340
2026-01-31	STB	0.0421	0.000000	0.000000	0.0318	0.000000	0.0400	0.0387
2026-01-31	SHB	0.0285	0.000000	0.000000	0.004903	0.000000	0.0400	0.0333
2026-01-31	LPB	0.0455	0.3000	0.0568	0.000185	0.0529	0.0400	0.0333
2026-01-31	SSB	0.0191	0.000000	0.2045	0.000000	0.000000	0.0400	0.0485
2026-01-31	MSB	0.0149	0.000000	0.000000	0.0384	0.000000	0.0400	0.0362
2026-01-31	OCB	0.0123	0.000000	0.000000	0.000000	0.000000	0.0400	0.0428
2026-01-31	NAB	0.007881	0.3000	0.000000	0.000000	0.0162	0.0400	0.0393
2026-01-31	BAB	0.004415	0.000000	0.3000	0.000000	0.1947	0.0400	0.0709
2026-01-31	KLB	0.003769	0.000000	0.0459	0.000000	0.3000	0.0400	0.0265
2026-01-31	PGB	0.002628	0.0583	0.0281	0.000000	0.0268	0.0400	0.0296
2026-01-31	VAB	0.003272	0.0488	0.000000	0.000000	0.000000	0.0400	0.0343
2026-01-31	VBB	0.004316	0.0307	0.000000	0.0192	0.0236	0.0400	0.0316
2026-01-31	SGB	0.001619	0.000000	0.2099	0.000000	0.000000	0.0400	0.0525
2026-01-31	ABB	0.005868	0.0268	0.000000	0.000000	0.1603	0.0400	0.0326
2026-01-31	BVB	0.003260	0.000000	0.000000	0.000000	0.000000	0.0400	0.0442
2026-02-28	VCB	0.2055	0.000000	0.1483	0.000000	0.000000	0.0400	0.0537
2026-02-28	BID	0.1369	0.000000	0.000000	0.1877	0.000000	0.0400	0.0436
2026-02-28	CTG	0.1050	0.000000	0.000000	0.0992	0.000000	0.0400	0.0449
2026-02-28	MBB	0.0764	0.000000	0.000000	0.3000	0.000000	0.0400	0.0369

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2026-02-28	TCB	0.0869	0.1823	0.000000	0.000000	0.000000	0.0400	0.0393
2026-02-28	ACB	0.0432	0.000000	0.000000	0.000000	0.000000	0.0400	0.0487
2026-02-28	VPB	0.0761	0.000000	0.000000	0.1441	0.1691	0.0400	0.0294
2026-02-28	HDB	0.0494	0.1863	0.000000	0.000000	0.1086	0.0400	0.0344
2026-02-28	VIB	0.0200	0.000000	0.000000	0.1475	0.000000	0.0400	0.0378
2026-02-28	TPB	0.0166	0.000000	0.000000	0.0771	0.1146	0.0400	0.0340
2026-02-28	STB	0.0414	0.000000	0.007043	0.000000	0.000000	0.0400	0.0372
2026-02-28	SHB	0.0253	0.000000	0.000000	0.000000	0.0717	0.0400	0.0330
2026-02-28	LPB	0.0410	0.3000	0.0450	0.000000	0.1230	0.0400	0.0313
2026-02-28	SSB	0.0170	0.000000	0.2138	0.000000	0.000000	0.0400	0.0498
2026-02-28	MSB	0.0135	0.000000	0.000000	0.0444	0.1968	0.0400	0.0358
2026-02-28	OCB	0.0110	0.000000	0.000000	0.000000	0.000000	0.0400	0.0455
2026-02-28	NAB	0.007057	0.2409	0.000000	0.000000	0.000000	0.0400	0.0391
2026-02-28	BAB	0.004515	0.000000	0.3000	0.000000	0.000000	0.0400	0.0711
2026-02-28	KLB	0.003087	0.0332	0.0467	0.000000	0.000593	0.0400	0.0284
2026-02-28	PGB	0.002949	0.000000	0.0240	0.000000	0.000000	0.0400	0.0293
2026-02-28	VAB	0.003218	0.000000	0.000000	0.000000	0.000000	0.0400	0.0362
2026-02-28	VBB	0.003983	0.0121	0.000000	0.000000	0.000000	0.0400	0.0316
2026-02-28	SGB	0.001586	0.000000	0.2152	0.000000	0.000000	0.0400	0.0523
2026-02-28	ABB	0.005019	0.0452	0.000000	0.000000	0.2156	0.0400	0.0320
2026-02-28	BVB	0.003130	0.000000	0.000000	0.000000	0.000000	0.0400	0.0445
2026-03-31	VCB	0.1937	0.000000	0.0691	0.000000	0.000000	0.0400	0.0483
2026-03-31	BID	0.1245	0.000000	0.000000	0.2781	0.000000	0.0400	0.0353
2026-03-31	CTG	0.1061	0.000000	0.000000	0.1180	0.000000	0.0400	0.0449
2026-03-31	MBB	0.0820	0.000000	0.000000	0.3000	0.000000	0.0400	0.0372
2026-03-31	TCB	0.0898	0.1787	0.000000	0.000000	0.000000	0.0400	0.0397
2026-03-31	ACB	0.0450	0.000000	0.002997	0.000000	0.000000	0.0400	0.0534
2026-03-31	VPB	0.0806	0.000000	0.000000	0.2163	0.1624	0.0400	0.0291
2026-03-31	HDB	0.0500	0.1382	0.000000	0.000000	0.2752	0.0400	0.0345
2026-03-31	VIB	0.0199	0.000000	0.000000	0.000000	0.000000	0.0400	0.0412

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2026-03-31	TPB	0.0182	0.000000	0.000000	0.0206	0.000000	0.0400	0.0351
2026-03-31	STB	0.0441	0.0277	0.0607	0.000000	0.000000	0.0400	0.0387
2026-03-31	SHB	0.0258	0.000000	0.000000	0.000000	0.000000	0.0400	0.0327
2026-03-31	LPB	0.0434	0.3000	0.0659	0.000000	0.000000	0.0400	0.0310
2026-03-31	SSB	0.0172	0.000000	0.2217	0.000000	0.000000	0.0400	0.0493
2026-03-31	MSB	0.0136	0.000000	0.000000	0.0670	0.000000	0.0400	0.0356
2026-03-31	OCB	0.0111	0.000000	0.000000	0.000000	0.000000	0.0400	0.0449
2026-03-31	NAB	0.007150	0.2626	0.0454	0.000000	0.000000	0.0400	0.0436
2026-03-31	BAB	0.004594	0.000000	0.3000	0.000000	0.000000	0.0400	0.0695
2026-03-31	KLB	0.003170	0.000000	0.008083	0.000000	0.3000	0.0400	0.0275
2026-03-31	PGB	0.003115	0.000000	0.0257	0.000000	0.000000	0.0400	0.0289
2026-03-31	VAB	0.003148	0.000000	0.000000	0.000000	0.000000	0.0400	0.0355
2026-03-31	VBB	0.004038	0.0670	0.0689	0.000000	0.000000	0.0400	0.0365
2026-03-31	SGB	0.001624	0.000000	0.1315	0.000000	0.000000	0.0400	0.0514
2026-03-31	ABB	0.005138	0.0258	0.000000	0.000000	0.2624	0.0400	0.0315
2026-03-31	BVB	0.003089	0.000000	0.000000	0.000000	0.000000	0.0400	0.0445
2026-04-30	VCB	0.1913	0.000000	0.0341	0.000000	0.0787	0.0400	0.0457
2026-04-30	BID	0.1131	0.000000	0.000000	0.2930	0.3000	0.0400	0.0334
2026-04-30	CTG	0.1059	0.000000	0.000000	0.1002	0.0926	0.0400	0.0462
2026-04-30	MBB	0.0840	0.000000	0.000000	0.2782	0.1853	0.0400	0.0383
2026-04-30	TCB	0.0840	0.1922	0.000000	0.000000	0.000000	0.0400	0.0405
2026-04-30	ACB	0.0477	0.000000	0.0422	0.000000	0.000000	0.0400	0.0536
2026-04-30	VPB	0.0820	0.000000	0.000000	0.2005	0.000000	0.0400	0.0293
2026-04-30	HDB	0.0502	0.0704	0.000000	0.000000	0.0435	0.0400	0.0345
2026-04-30	VIB	0.0216	0.000000	0.000000	0.0351	0.000000	0.0400	0.0412
2026-04-30	TPB	0.0178	0.000000	0.000000	0.008271	0.000000	0.0400	0.0345
2026-04-30	STB	0.0462	0.1009	0.0722	0.000000	0.3000	0.0400	0.0402
2026-04-30	SHB	0.0278	0.000000	0.000000	0.000000	0.000000	0.0400	0.0330
2026-04-30	LPB	0.0466	0.3000	0.0558	0.000000	0.000000	0.0400	0.0308
2026-04-30	SSB	0.0190	0.000000	0.2315	0.000000	0.000000	0.0400	0.0486

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2026-04-30	MSB	0.0142	0.000000	0.000000	0.0848	0.000000	0.0400	0.0358
2026-04-30	OCB	0.0117	0.000000	0.000000	0.000000	0.000000	0.0400	0.0477
2026-04-30	NAB	0.007350	0.2584	0.0514	0.000000	0.000000	0.0400	0.0446
2026-04-30	BAB	0.004691	0.000000	0.3000	0.000000	0.000000	0.0400	0.0677
2026-04-30	KLB	0.002994	0.000000	0.0237	0.000000	0.000000	0.0400	0.0275
2026-04-30	PGB	0.003224	0.000000	0.0121	0.000000	0.000000	0.0400	0.0283
2026-04-30	VAB	0.003250	0.000000	0.000000	0.000000	0.000000	0.0400	0.0360
2026-04-30	VBB	0.004499	0.000000	0.0279	0.000000	0.000000	0.0400	0.0372
2026-04-30	SGB	0.001749	0.000000	0.1492	0.000000	0.000000	0.0400	0.0506
2026-04-30	ABB	0.006038	0.0782	0.000000	0.000000	0.000000	0.0400	0.0316
2026-04-30	BVB	0.003132	0.000000	0.000000	0.000000	0.000000	0.0400	0.0431
2026-05-31	VCB	0.1913	0.000000	0.000000	0.000000	0.0173	0.0400	0.0442
2026-05-31	BID	0.1118	0.000000	0.000000	0.3000	0.1471	0.0400	0.0314
2026-05-31	CTG	0.1039	0.000000	0.000000	0.0814	0.3000	0.0400	0.0446
2026-05-31	MBB	0.0803	0.1202	0.000000	0.2827	0.2355	0.0400	0.0378
2026-05-31	TCB	0.0899	0.0144	0.000000	0.0362	0.000000	0.0400	0.0373
2026-05-31	ACB	0.0462	0.000000	0.0520	0.000000	0.000000	0.0400	0.0534
2026-05-31	VPB	0.0791	0.000000	0.000000	0.2275	0.000000	0.0400	0.0310
2026-05-31	HDB	0.0510	0.000000	0.000000	0.000000	0.000000	0.0400	0.0344
2026-05-31	VIB	0.0212	0.000000	0.000000	0.000000	0.000000	0.0400	0.0416
2026-05-31	TPB	0.0173	0.000000	0.000000	0.0383	0.000000	0.0400	0.0338
2026-05-31	STB	0.0487	0.1316	0.0771	0.000000	0.3000	0.0400	0.0405
2026-05-31	SHB	0.0256	0.000000	0.000000	0.000000	0.000000	0.0400	0.0335
2026-05-31	LPB	0.0511	0.3000	0.0453	0.000000	0.000000	0.0400	0.0310
2026-05-31	SSB	0.0182	0.000000	0.2653	0.000000	0.000000	0.0400	0.0491
2026-05-31	MSB	0.0149	0.000000	0.000000	0.0339	0.000000	0.0400	0.0359
2026-05-31	OCB	0.0115	0.000000	0.000000	0.000000	0.000000	0.0400	0.0476
2026-05-31	NAB	0.007717	0.2599	0.0162	0.000000	0.000000	0.0400	0.0436
2026-05-31	BAB	0.004638	0.000000	0.3000	0.000000	0.000000	0.0400	0.0653
2026-05-31	KLB	0.003265	0.000000	0.000000	0.000000	0.000000	0.0400	0.0268

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_BL_KIO	w_EW	w_RP
2026-05-31	PGB	0.003053	0.000000	0.0280	0.000000	0.000000	0.0400	0.0315
2026-05-31	VAB	0.003203	0.000000	0.000000	0.000000	0.000000	0.0400	0.0359
2026-05-31	VBB	0.005524	0.000000	0.009129	0.000000	0.000000	0.0400	0.0400
2026-05-31	SGB	0.001685	0.000000	0.1952	0.000000	0.000000	0.0400	0.0539
2026-05-31	ABB	0.005904	0.1739	0.0117	0.000000	0.000000	0.0400	0.0318
2026-05-31	BVB	0.003042	0.000000	0.000000	0.000000	0.000000	0.0400	0.0443

Phụ lục A – Phương pháp luận Chi tiết

Nguồn dữ liệu: output/backtest_v2.py

Dữ liệu	25 cổ phiếu ngân hàng niêm yết HOSE/HNX. Giá đóng cửa tháng và vốn hoá thị trường (bn VND). Giai đoạn 06/2014–05/2026 (144 tháng). NaN với cổ phiếu chưa niêm yết xử lý bằng dynamic universe.
Rolling window	Lookback T = 36 tháng. Tại mỗi bước i, training = [i–36, i–1], OOS = tháng i+1. Rebalancing hàng tháng → 106 bước OOS (08/2017–05/2026).
Tín hiệu K-means	Features: (1) idiosyncratic momentum = mean(log_ret_idio[–6:–1]) × 12, với idio = ret – ew_market_ret; (2) –ann_vol = –std(log_ret[–36:]) × √12. Chuẩn hoá StandardScaler, K-means k=4, random_state=42, n_init=10. Composite score = idio_ret – ann_vol → xếp hạng best/worst cluster.
Black-Litterman gốc (BL)	Không có active views (P=∅, q=∅). Posterior đơn giản hóa: $\mu_{BL} = \Pi$ (equilibrium), $\Sigma_{BL} = (1+\tau)\cdot\Sigma$. Tối ưu Tangency trên $(\Pi, (1+\tau)\Sigma)$. Đây là BL chuẩn Black & Litterman (1992) không có quan điểm chủ quan.
BL + K-means Signal (BL_KIO)	Prior: $\Pi = \delta\cdot\Sigma\cdot w_{mkt}$ (Reverse-CAPM). $\delta = (w_{mkt}\cdot\mu - rf) / (w_{mkt}\cdot\Sigma\cdot w_{mkt})$, clipped [0.5, 10]. View uncertainty: $\Omega = \tau\cdot P\cdot\Sigma\cdot P^T + \varepsilon\cdot I$, $\tau = 1/36$. Posterior: $M = (\tau\Sigma)^{-1} + P^T\cdot\Omega^{-1}\cdot P$; $\mu_{BL_KIO} = M^{-1}[(\tau\Sigma)^{-1}\Pi + P^T\cdot\Omega^{-1}\cdot q]$; $\Sigma_{BL_KIO} = M^{-1} + \Sigma$. q = idiosyncratic momentum spread (best cluster – worst cluster).
Tối ưu hoá	SLSQP maximise Sharpe(μ, Σ_{post}) subject to $\Sigma w_i=1, 0\leq w_i\leq 0.30$. 10 điểm xuất phát ngẫu nhiên (Dirichlet), chọn nghiệm tốt nhất.
Monte Carlo	2,000 mô phỏng nhiễu Gaussian trên q (delta ∈ {–10%,–5%,0%,+5%,+10%}). Khai thác $\mu_{BL}(q') = \text{base_vec} + A_mat @ q'$ (tuyến tính) để tránh tối ưu lại.
Kiểm định	Paired t-test (scipy.stats.ttest_rel); Jobson-Korkie (1981) z-test; Jarque-Bera (scipy.stats.jarque_bera); Ljung-Box lag 5/10/20 (statsmodels).

Phụ lục B – Lãi suất Phi rủi ro VN10Y theo Năm

Nguồn dữ liệu: output/backtest_v2.py (VN10Y_RATES dict)

Lãi suất trái phiếu chính phủ Việt Nam kỳ hạn 10 năm (VN10Y). RF tháng = VN10Y_năm / 100 / 12.

Năm	VN10Y (%/năm)	RF tháng (%)
2014	6.5	0.5417
2015	6.5	0.5417
2016	6.2	0.5167

2017	5.8	0.4833
2018	4.58	0.3817
2019	4.58	0.3817
2020	3.0	0.2500
2021	2.53	0.2108
2022	4.0	0.3333
2023	3.5	0.2917
2024	3.06	0.2550
2025	3.1	0.2583
2026	4.35	0.3625